NONSOFIC GROUPS EXIST

ANONYMOUS

Abstract. We prove that nonsofic groups exist: for every $m \geq 1$, the group $\mathrm{EL}_{m+1}(L_{\mathbb{F}_2}(1,2))$ of elementary matrices over the binary Leavitt algebra is infinite, finitely generated, has Kazhdan's property (T), admits no nontrivial homomorphism to a finite group, and is not sofic. The proof is engineered for verification: the two auxiliary matrices on which the construction turns are explicit products of elementary matrices, the remaining memberships reduce to one Whitehead identity and exhibited commutators of units, and no K-theory is used.

The engine is a compression–centralizer criterion. If a Kazhdan group is generated by an infinite Kazhdan subgroup Γ together with finitely many elements that conjugate Γ into itself, then in a hypothetical sofic approximation the expander decompositions supplied by Kun's theorem, for Γ and for the ambient group, can be matched almost bijectively: a median normalization of component sizes is almost monotone along compressor arcs, exact conservation under permutations makes the estimate two-sided, and an ℓ^1 co-area inequality pins the normalized sizes at their median. Localizing to one matched component yields a sofic approximation of $K \times J$ whose K-edges expand, and the Kun–Thom obstruction forces the commuting subgroup J to be locally embeddable into finite groups (LEF). The Leavitt relations realize the scheme with J a two-generator subgroup of a corner copy of Thompson's group V that is not LEF for a finite reason: in a finite group the two F-relators force commutation, and the witness pair does not commute.

Self-similarity spreads the conclusion: for each fixed finite field k, the unit group of $L_k(1,2)$, every GL_r, and every EL_r with $r \geq 2$ are mutually isomorphic Kazhdan nonsofic groups. Every finitely generated nonsofic group is a quotient of an infinite finitely presented nonsofic group, Kazhdan whenever the original is; so not every finitely presented group is sofic, and neither soficity, local embeddability, nor residual finiteness is closed under quotients. Soficity is consequently a Markov property of finitely presented groups, so by Adian–Rabin no algorithm decides it from a presentation. The arXiv proofs of the two theorems that carry the approximation-theoretic weight—Kun's expander decomposition and the Kun–Thom centralizer obstruction—are audited: two local gaps are repaired, and a spectral implication asserted in the former is refuted by counterexample. An appendix gives an elementary proof that $K_1(L_k(1,2)) = 0$ over every field, from the Leavitt relations and one division lemma imported from pure infinite simplicity.

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Contents

1. Introduction

A countable discrete group is *sofic* if every finite portion of its multiplication table admits arbitrarily accurate and asymptotically faithful models by permutations of finite sets. The notion arose in Gromov's work on symbolic dynamics [14]; Weiss introduced the term and asked whether every countable group is sofic [33]. The class contains all amenable and all residually finite groups, and it is stable under a long list of standard constructions; see [11, 27]. Whether every group is sofic had remained a central open problem of modern group theory; recent work examines the structure a minimal counterexample would have, conditional on existence [12].

We answer the question in the negative. Two consequences follow at once: not every finitely presented group is sofic (Theorem C), and none of the classes of sofic, LEF, or residually finite groups is closed under quotients (Remark 10.16).

Endpoint/ManuscriptStatements
`theoremA_exact`

Main results. Let k be a field and let

$$L = L_k(1,2) = k\langle s_0, s_1, t_0, t_1 \mid t_i s_j = \delta_{ij},\ s_0t_0 + s_1t_1 = 1\rangle \tag{1}$$

be the binary Leavitt algebra [24]. For a unital ring A and $r \geq 2$, write

$$x_{ij}(a) = I_r + aE_{ij} \in \mathrm{GL}_r(A), \qquad a \in A, \quad i \neq j,$$

and let $\mathrm{EL}_r(A) \leq \mathrm{GL}_r(A)$ be the subgroup generated by these elementary matrices.

Endpoint/MainResults
`universalLeavittEL4_not_isSofic`
`universalLeavitt_profile`
`ambient_profile`
`ambient_full_profile`
`countable_nonsofic_groups_exist`

Theorem A. *Let $R = L_{\mathbb{F}_2}(1,2)$. For every $m \geq 1$, the group*

$$\mathrm{EL}_{m+1}(R)$$

is infinite, finitely generated, has property (T), and is not sofic. In particular $G = \mathrm{EL}_4(R)$ is not sofic, and not every countable discrete group is sofic.

Endpoint/ManuscriptStatements
`theoremB_exact`
`theoremB_Theta`
`theoremB_Theta_apply`

The proof of Theorem A is engineered for audit. The two auxiliary matrices of the construction are exhibited as explicit products of elementary matrices; every remaining elementary membership follows from one explicit Whitehead identity together with explicit commutator certificates—every cylinder transposition is a single commutator of units; the ranks below four are reached by an explicit isomorphism between the elementary groups of any two ranks ≥ 2 over R (Proposition 5.8); and no K-theory is used.

Endpoint/MainResults
`binaryLeavitt_finiteField_profile`
`binaryLeavittUnits_not_isSofic`
`binaryLeavittGL_not_isSofic`
Leavitt/UnitsGLProfile
`binaryLeavittUnits_profile`
`binaryLeavittGL_profile`

Theorem B. *Let k be a finite field and let $L = L_k(1,2)$.*

(i) *The unit group $L^\times$ is infinite, finitely generated, has property (T), and is not sofic.*
(ii) *For every $r \geq 1$, every ordered complete binary leaf set $\mathcal{C} = (\alpha_1, \ldots, \alpha_r)$ induces an explicit group isomorphism*

$$\Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \xrightarrow{\cong} L^\times, \qquad (a_{ij}) \longmapsto \sum_{i,j=1}^{r} \alpha_i a_{ij} \alpha_j^*. \tag{2}$$

For $r \geq 2$ one has $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$. Hence all the groups $\mathrm{GL}_r(L)$, $r \geq 1$, and $\mathrm{EL}_r(L)$, $r \geq 2$, are mutually isomorphic, and each is an infinite finitely generated property (T) nonsofic group.

Endpoint/ManuscriptStatements
`theoremC_first`
`theoremC_second`
`theoremC_exact`
`theoremC_covers_theoremB_groups`

Theorem C. *Every finitely generated nonsofic group is a quotient of an infinite finitely presented nonsofic group. Every nonsofic group with property (T) is a quotient of an infinite finitely presented nonsofic group with property (T). In particular, there is an infinite finitely presented property (T) group H that is not sofic; for each fixed finite field k of Theorem B, H may be chosen to surject onto $L^\times$, and hence to have a quotient isomorphic to every group listed there.*

Endpoint/MainResults
`exists_finitelyPresented_nonsofic_group`
`exists_infinite_finitelyPresented_nonsofic_ambient_cover`

Strategy. The proof has three independent layers.

The first layer is approximation-theoretic.

Covers/KazhdanCover
`exists_kazhdan_finitelyPresented_cover_of_not_isSofic`
Endpoint/ManuscriptStatements
`theoremD_subgroups`
Criterion/CriterionAssembly
`isLEF_of_isSofic`
`not_isSofic_of_not_isLEF`
Finite generation and countability are derived from (T), not assumed.

Theorem D (Kazhdan compression–centralizer criterion)**.** *Let G be a finitely generated property-(T) group, and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup. Suppose there is a finite set $Q \subseteq G$ with*

$$G = \langle \Gamma, Q \rangle \qquad \text{and} \qquad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$, put $K = q_0 \Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$[K, J] = 1, \qquad K \cap J = \{1\}.$$

If G is sofic, then J is LEF.

Finite generation of G is listed for readability only: it follows from the generation hypothesis, Γ being finitely generated and Q finite, and the formal statement omits it.

The point of departure is the tension between the two theorems that govern sofic approximations of Kazhdan groups. Kun's theorem [21] proves Bowen's conjecture: every sofic approximation of an infinite property-(T) group is essentially a vertex-disjoint union of uniformly expanding components. The Kun–Thom theorem [22] requires more: if the restriction of a sofic approximation of $\Lambda \times \Delta$ to a Kazhdan factor Λ is essentially *one* expander, then Δ is locally embeddable into finite groups (LEF). A disjoint union of equal-sized expanders is not enough, because the commuting factor may permute the components—this is exactly what happens in a product approximation.

Compression closes the gap. A hypothetical sofic approximation of G yields a Kun decomposition for Γ and another for G. Transporting the Γ-components by a compressor q gives one-sided inclusions into other Γ-components. Write $s(y)$ for the size of the Γ-component containing y; on each ambient G-component, normalize by a vertex-weighted median m of the sizes and set

$$f(y) = \frac{s(y)}{s(y)+m}. \tag{3}$$

The one-sided inclusions make f almost monotone along compressor arcs. But a compressor acts by a permutation, so total increase equals total decrease *exactly*. The resulting small ℓ^1 edge variation, fed into an elementary co-area inequality on the ambient expanders, pins f at its median value $1/2$. Hence transported and target components have asymptotically equal sizes, and one-sided containment becomes almost-bijective matching. A weighted diagonal selection extracts a single matched component on which every multiplication, faithfulness, invariance, conjugacy, and graph-edit error is negligible. Restricting and completing the permutations yields a sofic approximation of $K \times J$ whose K-edges form an essential expander sequence, and the Kun–Thom obstruction forces J to be LEF. Theorem D is proved in Section 4 as a corollary of a local statement, Theorem 4.1, which isolates the exact graph-theoretic hypotheses and records precisely what property (T) is used for. Remarks 4.6 and 4.7 test the criterion against sofic examples and show that each hypothesis earns its place.

The second layer is algebraic, and it is where the Leavitt relations enter. Fix $m \geq 2$ and a nontrivial unital ring A carrying a binary Leavitt family—four elements satisfying the two Leavitt relations; no copy of L is assumed. A left-comb binary partition with $m+1$ leaves produces a compressor $u \in \mathrm{GL}_{m+1}(A)$ that conjugates the block subgroup $\Gamma \cong \mathrm{EL}_m(A)$ into the corner $a \mapsto s_0 a t_0$ of itself. A block involution z centralizes the compressed image K, so $v = zu$ is a second compressor with the same image, and $vu^{-1} = z$ generates every elementary matrix involving the last coordinate; whenever u and z lie in $\mathrm{EL}_{m+1}(A)$, it follows that $G = \mathrm{EL}_{m+1}(A) = \langle \Gamma, u, v\rangle$. The complementary corner $a \mapsto p_0 + s_1 a t_1$ carries a commuting copy of Thompson's group V, and inside it the witness: the two-generator standard copy of Thompson's F, which is not LEF for a finite reason—in a finite group the two F-relators force their two elements to commute, and the witness pair does not commute. The witness is elementary over every ambient ring: its two generators are products of explicitly exhibited commutators of units—every cylinder transposition is a single commutator—and the Whitehead lemma places diagonal products of commutators of units in EL_2. The only membership hypotheses of the general theorem therefore concern u and z. Over $\mathbb{F}_2$ they are discharged by explicit four-term elementary factorizations, and an explicit rank equivalence between the elementary groups of any two ranks ≥ 2—block flattening along the self-similarity of the Leavitt relations—then spreads the conclusion from rank four to every rank, which gives Theorem A with no K-theory. Over every finite field the memberships also follow from $K_1(L) = 0$ and the GE property of purely infinite simple rings; the K_1 input itself admits an elementary proof, written out in full in Appendix A. Matrix self-similarity $M_r(L) \cong L$ then spreads nonsoficity and property (T) to the unit group and to all GL_r, and a separate two-by-two realization shows that the scheme already closes in rank two.

The third layer is a finite multiplication-table construction. Every finitely generated nonsofic group G has a finite table fragment that admits no sufficiently accurate permutation model. Imposing precisely that finite table as relators gives a finitely presented group mapping onto G; the map proves infinitude and preserves distinctness of the named elements, while the forbidden finite table proves nonsoficity directly. When G is Kazhdan, Shalom's finite-presentation theorem first supplies a finitely presented Kazhdan cover, and imposing the same table inside its kernel produces a finitely presented Kazhdan nonsofic cover. The covers are proved nonsofic directly, from the forbidden table itself; quotient closure of soficity—which in fact fails outright once nonsofic groups exist (Remark 10.16)—plays no role.

Figure 1 charts how the three layers assemble.

External inputs, and how to verify this paper. The proof is organized so that its trust surface is explicit and a skeptical reader can audit it efficiently.

Theorems A and D rest on the following external inputs: Kun's expander decomposition theorem [21], cited in one-way form and with two proof repairs recorded in Remark 2.11; the Kun–Thom centralizer obstruction [22], with the repair recorded there as well; property (T) at rank three [13], stated and used in the finite-field case; and, for context only, the classical facts about Thompson's group V [17, 8, 7]—the main line replaces them by the two-relator obstruction of Lemma 5.12 and the explicit commutator certificates of Lemma 6.6. Everything else in their proofs is self-contained; the concluding remarks of Section 10 draw on further classical results, cited where they are used. In the proof of Theorem A, the two auxiliary matrices u and z are factored explicitly into elementary matrices, and every remaining elementary membership follows from the explicit Whitehead identity of Lemma 5.5 together with the commutator certificates of Lemma 6.6. The analytic sections use no spectral theory: the only analytic inequality proved and used in the matching argument is the ℓ^1 co-area estimate of Lemma 2.7, established in a few lines from the layer-cake formula, while the approximation-theoretic depth is confined to the cited theorems of Kun and Kun–Thom.

Theorem B additionally needs $K_1(L) = 0$, the collapse $\mathrm{GL}_r = \mathrm{EL}_r$, and the perfectness of $L^\times$. Appendix A proves all three—K_1 by the two-exit elimination, the collapse at every rank, and perfectness through the three-leaf self-similarity—from the Leavitt relations together with exactly one external input: strong division (Lemma A.3), imported from pure infinite simplicity through the graph criteria of [1]. The classical K-theoretic computations

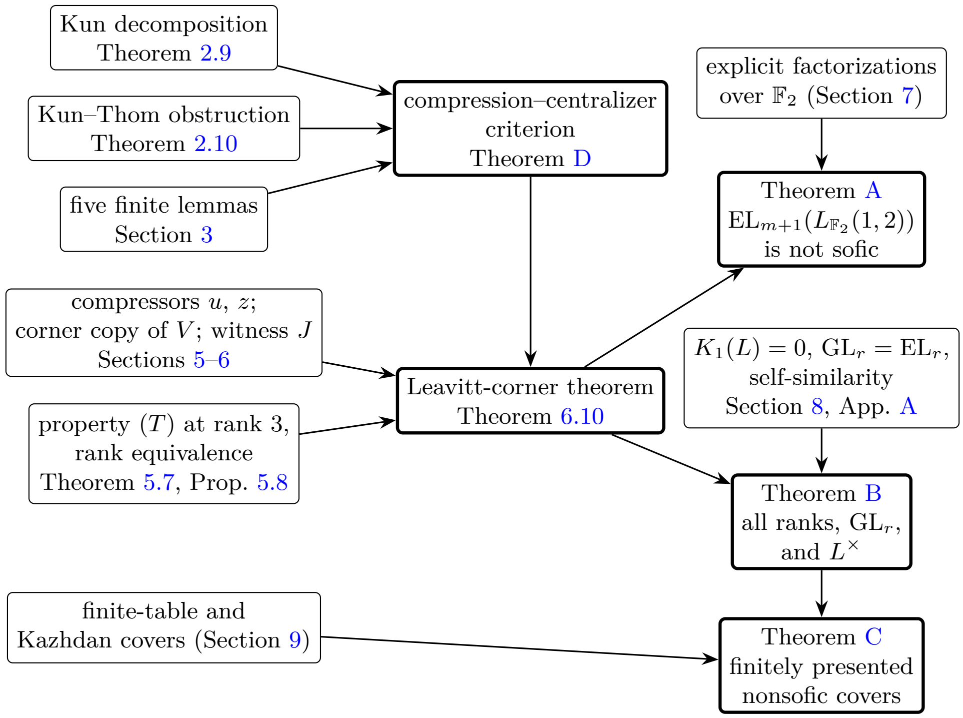


FIGURE 1. The architecture of the proof. The approximation-theoretic layer (top left) produces the criterion; the algebraic layer (middle left) realizes its hypotheses over any ring carrying a binary Leavitt family; the memberships are discharged twice, explicitly in characteristic two and K-theoretically over every finite field; and the finite-table layer converts nonsoficity into finitely presented covers.

[5, 4] are retained as independent confirmation only; the accompanying formal development proves strong division internally, so the machine-checked route has no external K-theoretic input.

Theorem C is self-contained apart from Shalom's finitely presented Kazhdan covers [30], used only for the Kazhdan refinement.

A reader who wants to check the headline claim should verify, in order: the five finite lemmas of Section 3, each elementary and a few lines long; the proof of Theorem 4.1 in Section 4, which assembles them and invokes Kun's theorem twice and the Kun–Thom theorem once, in exactly the forms stated in Section 2.5; the matrix identities of Sections 5 and 6, each a finite computation from the two Leavitt relations; and the two explicit factorizations of Section 7. That path proves Theorem A in full. Sections 8 and 9 lie downstream and can be audited separately.

Organization. Section 2 fixes the sofic, expander, and LEF vocabulary and states the theorems of Kun and Kun–Thom in the forms used. Section 3 isolates the five elementary lemmas of the matching mechanism. Section 4 proves the local compression–centralizer theorem and deduces Theorem D. Section 5 develops Leavitt leaf calculus, matrix self-similarity, the elementary rank equivalence, the Whitehead lemma, the characteristic-two block factorizations, and the corner realization of V. Section 6 presents the compression scheme in adjacent ranks over an arbitrary ring carrying a binary Leavitt family and proves the Leavitt-corner theorem. Section 7 discharges all memberships explicitly over $\mathbb{F}_2$ and proves Theorem A. Section 8 proves the two-by-two theorem, the equality $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$, and Theorem B. Section 9 proves Theorem C. Section 10 collects concluding remarks. Appendix A proves $K_1(L_k(1,2)) = 0$ elementarily.

2. Sofic approximations, expanders, and LEF groups

2.1. **Sofic approximations.** For a nonempty finite set Y and $\sigma, \tau \in \mathrm{Sym}(Y)$, let

$$d_{\mathrm{H}}(\sigma, \tau) = \frac{1}{|Y|} \left|\{y \in Y \mid \sigma(y) \neq \tau(y)\}\right|$$

denote the normalized Hamming distance.

Sofic/Sofic
`SoficApproximation`
`IsSofic`
`isSofic_of_soficApproximation`
Sofic/SoficSequential
`isSofic_iff_nonempty_soficApproximation`

Definition 2.1. Let G be a countable discrete group. A *sofic approximation* of G is a sequence of maps $\sigma_n : G \to \mathrm{Sym}(Y_n)$, with $|Y_n| \to \infty$, such that

(a) $d_{\mathrm{H}}(\sigma_n(g)\sigma_n(h), \sigma_n(gh)) \to 0$ for all $g, h \in G$;
(b) $d_{\mathrm{H}}(\sigma_n(g), \mathrm{id}) \to 1$ for all $g \in G \setminus \{1\}$.

The group G is *sofic* if it admits a sofic approximation.

The maps σ_n are not assumed to be homomorphisms. It is standard that (a) and (b) force $d_{\mathrm{H}}(\sigma_n(1), \mathrm{id}) \to 0$; the exact normalization used in the sequel is the following.

Sofic/Normalization
`normalize`
`normalize_map_one`
`normalize_map_inv`
`normalize_close`
`exists_normalization`
`perturb`
`involutionNormalize`
`halfOrbit`
Sofic/Sofic
`map_one_close`
Matching/InverseNormalization
`inverseError_negligible`
σ' is `normalize`, on the same finite models.

Lemma 2.2 (Finite normalization)**.** *Let $\sigma_n : G \to \mathrm{Sym}(Y_n)$ be a sofic approximation and let $F \subseteq G$ be a finite symmetric set. Then there is a sofic approximation $\sigma'_n : G \to \mathrm{Sym}(Y_n)$ with $d_{\mathrm{H}}(\sigma_n, \sigma'_n) \to 0$ pointwise, such that for all n:*

$$\sigma'_n(1) = \mathrm{id}, \qquad \sigma'_n(g^{-1}) = \sigma'_n(g)^{-1} \quad (g \in F).$$

Proof. Set $\sigma'_n(1) = \mathrm{id}$, and note that $d_{\mathrm{H}}(\sigma_n(1), \mathrm{id}) \to 0$: applying (a) with $g = h = 1$ gives $d_{\mathrm{H}}(\sigma_n(1)^2, \sigma_n(1)) \to 0$, and composing with the permutation $\sigma_n(1)^{-1}$ preserves d_{H}.

Choose one representative from each pair $\{g, g^{-1}\} \subseteq F \setminus \{1\}$ with $g \neq g^{-1}$; put $\sigma'_n(g) = \sigma_n(g)$ and $\sigma'_n(g^{-1}) = \sigma_n(g)^{-1}$. Then $d_{\mathrm{H}}(\sigma_n(g^{-1}), \sigma_n(g)^{-1}) \to 0$, because $\sigma_n(g^{-1})\sigma_n(g)$ is d_{H}-close to $\sigma_n(1)$, hence to id, and d_{H} is invariant under composition with the permutation $\sigma_n(g)^{-1}$ on the right.

If $g = g^{-1} \in F \setminus \{1\}$ is an involution, let $A_n = \{y \mid \sigma_n(g)^2 y = y\}$. Since $\sigma_n(g)^2$ is d_{H}-close to $\sigma_n(g^2) = \sigma_n(1)$, hence to id, we have $|Y_n \setminus A_n| = o(|Y_n|)$. The set A_n is $\sigma_n(g)$-invariant—if $\sigma_n(g)^2 y = y$, then applying $\sigma_n(g)$ gives $\sigma_n(g)^2(\sigma_n(g)y) = \sigma_n(g)y$—and $\sigma_n(g)$ restricts to an involution of A_n. Define

$\sigma'_n(g)$ to be $\sigma_n(g)$ on A_n and the identity on $Y_n \setminus A_n$; this is an involution of Y_n differing from $\sigma_n(g)$ on $o(|Y_n|)$ points.

Outside $F \cup \{1\}$ put $\sigma'_n = \sigma_n$. Each $\sigma'_n(g)$ differs from $\sigma_n(g)$ on $o(|Y_n|)$ points, so properties (a) and (b) persist. □

The next lemma is the bridge, used many times below, between statements about products of letter permutations and statements about the permutation of the product.

Lemma 2.3 (Word approximation)**.** *Let $\sigma_n : G \to \mathrm{Sym}(Y_n)$ be a sofic approximation, let $g \in G$, and let $g = t_1 t_2 \cdots t_\ell$ be any fixed factorization with $t_1, \ldots, t_\ell \in G$. Then*

Sofic/Sofic
`word_close`
Matching/BlockWordCrossing
`wordDisagreement_negligible`
`all_wordCrossing_negligible`

$$\big|\{y \in Y_n \mid \sigma_n(t_1)\sigma_n(t_2)\cdots\sigma_n(t_\ell)\, y \neq \sigma_n(g)y\}\big| = o(|Y_n|).$$

Proof. Induction on ℓ, the case $\ell = 1$ being trivial. For $\ell \geq 2$ put $h = t_2 \cdots t_\ell$. Composing with the permutation $\sigma_n(t_1)$ on the left does not change disagreement sets, so $\sigma_n(t_1)\cdots\sigma_n(t_\ell)$ agrees with $\sigma_n(t_1)\sigma_n(h)$ outside the $o(|Y_n|)$ points supplied by the inductive hypothesis; and $\sigma_n(t_1)\sigma_n(h)$ agrees with $\sigma_n(t_1 h) = \sigma_n(g)$ outside $o(|Y_n|)$ points by Definition 2.1(a). The union of the two sets has size $o(|Y_n|)$. □

Whenever an element of a subgroup is written below as a word in a generating set and a property is verified letter by letter, Lemma 2.3 is what transfers the conclusion from the product of the letter permutations to the permutation of the element itself; it is cited explicitly at each such point.

2.2. **Generator graphs and edit distance.** Let $T \subseteq G$ be a finite set and $\sigma : G \to \mathrm{Sym}(Y)$ a map. The *T-generator graph* $X(Y, \sigma, T)$ is the multigraph on the vertex set Y with one edge $\{y, \sigma(t)y\}$ for each pair $(y, t) \in Y \times T$ with $\sigma(t)y \neq y$; parallel edges are retained, fixed points contribute no edge, and every vertex has degree at most $2|T|$. An *arc* of $t \in T$ is an ordered pair $(y, \sigma(t)y)$.

Conventions. All graphs in this paper are finite multigraphs without loops, and all edge counts—degrees, boundaries, edit distances, and sums over $E(X)$—count edges with multiplicity. For multigraphs X, X' on the same vertex set Y, the *edit distance* $\mathrm{ed}(X, X')$ is the minimal number of edge insertions and deletions, with multiplicity, transforming X into X'. A sequence of graphs X_n on Y_n is *edit-equivalent* to a sequence X'_n if $\mathrm{ed}(X_n, X'_n) = o(|Y_n|)$.

For a finite multigraph X and $\emptyset \neq U \subseteq V(X)$, write ∂U for the multiset of edges with exactly one endpoint in U, and

$$h(X) = \min\left\{ \frac{|\partial U|}{|U|} \;\middle|\; \emptyset \neq U \subseteq V(X),\ |U| \leq \tfrac{1}{2}|V(X)| \right\}$$

for the Cheeger constant; for a one-vertex graph the minimum runs over the empty set, and we set $h(X) = +\infty$. A sequence of bounded-degree graphs X_n with $|V(X_n)| \to \infty$ is an *expander sequence* if $\inf_n h(X_n) > 0$.

Remark 2.4 (Robustness of the conventions)**.** The conventions are robust in exactly the two directions in which they interface with the cited external theorems. Loops are immaterial: a loop lies in no boundary ∂U, so inserting or deleting loops changes no Cheeger constant, no component, and no estimate below; arcs with $\sigma(t)y = y$ are accordingly discarded once and for all in the definition above. Multiplicities are controlled by a uniform bound: in a T-generator graph every unordered pair carries at most $2|T|$ parallel edges, one per arc. For multigraphs of multiplicity at most M: forgetting multiplicities preserves components and connectivity; boundary sizes change by a factor between 1 and M, so a componentwise Cheeger bound h in one convention yields at least h/M in the other; and edit distances transfer with the same factor, one direction at a time: forgetting multiplicities is 1-Lipschitz for edit distance, since each multigraph insertion or deletion changes the underlying simple graph by at most one edge; and conversely, from a multigraph X of multiplicity at most M and a simple-graph edit path of length e from the simplification of X, one obtains a multigraph X' lifting its endpoint—keep the multiplicities of retained edges, delete all copies of deleted edges, insert new edges with multiplicity one—at multigraph edit distance at most Me from X. Hence $o(|Y_n|)$ edit-distance statements transfer in both directions against these chosen lifts. We do not assert any comparison for arbitrary multigraphs—two multigraphs can share an underlying simple graph and differ by linearly many parallel copies; the transfer above uses the multiplicity bound, which holds for every graph in this paper. Consequently every statement below, and each of the two external theorems of Section 2.5, may be read in either convention with constants adjusted; we fix the multiplicity-counting convention throughout.

Criterion/Criterion
`ExpanderDecomposition`
Stronger than printed: also records the block structure, an edge-edit witness, and (6).

Definition 2.5. Let (X_n) be a sequence of graphs of uniformly bounded degree on vertex sets Y_n with $|Y_n| \to \infty$. An *expander decomposition* of (X_n) consists of graphs $\widetilde{X}_n$ on Y_n, of uniformly bounded degree, with $\mathrm{ed}(X_n, \widetilde{X}_n) = o(|Y_n|)$, together with a constant $h > 0$ such that every connected component of every $\widetilde{X}_n$ has Cheeger constant at least h.

Criterion/Criterion
`IsEssentialExpanderSequence`
`MatchingCertificate`
`isEssentialExpanderSequence`
Matching/EssentialExpander Repair
`induced_expands_eventually`
`repairedActualGraph_expands_eventually`

Definition 2.6. A sequence of bounded-degree graphs (X_n) is an *essential expander sequence* if it is edit-equivalent to an expander sequence.

2.3. **Expansion and an ℓ^1 co-area estimate.**

The single analytic inequality used in this paper is the following co-area estimate.

Matching/FiniteGraph
`coarea`
`coarea_mul`
Matching/DirectedCoarea
`nonnegative_coarea`
Holds for every finite multigraph.

Lemma 2.7 (ℓ^1 co-area estimate)**.** *Let X be a finite connected graph and $f : V(X) \to \mathbb{R}$. Let $c \in \mathbb{R}$ be such that*

$$|\{x \mid f(x) > c\}| \le \tfrac{1}{2}|V(X)| \qquad \text{and} \qquad |\{x \mid f(x) < c\}| \le \tfrac{1}{2}|V(X)|.$$

Then

$$\sum_{x \in V(X)} |f(x) - c| \;\le\; \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f(x) - f(y)|.$$

Proof. If X has a single vertex, the median hypotheses force $f \equiv c$ and both sides vanish (with $1/h(X) = 0$ by convention); assume $|V(X)| \geq 2$. Replacing f by $f - c$ we may assume $c = 0$. Write $f = f_+ - f_-$ with $f_\pm \geq 0$. For $t > 0$, the level set $U_t = \{x \mid f_+(x) > t\}$ satisfies $|U_t| \leq \frac{1}{2}|V(X)|$, so $|\partial U_t| \geq h(X)|U_t|$ whenever $U_t \neq \emptyset$. By the layer-cake formula,

$$\sum_x f_+(x) = \int_0^\infty |U_t| \,\mathrm{d}t \leq \tfrac{1}{h(X)} \int_0^\infty |\partial U_t| \,\mathrm{d}t \leq \tfrac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f_+(x) - f_+(y)|,$$

since an edge $\{x, y\}$ lies in ∂U_t exactly for t in an interval of length $|f_+(x) - f_+(y)|$. The same bound holds for f_-, and $|f_+(x) - f_+(y)| + |f_-(x) - f_-(y)| \leq |f(x) - f(y)|$ pointwise on edges, because f_+ and f_- have disjoint supports. Adding the two estimates gives the claim. ☐

2.4. **LEF groups.** A group J is *LEF* (locally embeddable into finite groups) [32] if for every finite subset $F \subseteq J$ there are a finite group P and a map $\varphi : F \to P$ that is injective and satisfies $\varphi(g)\varphi(h) = \varphi(gh)$ whenever $g, h, gh \in F$. Every LEF group is sofic; finite and residually finite groups are LEF.

Lemma 2.8. *A finitely presented LEF group is residually finite.*

Sofic/FinitelyPresentedLEF
`finitelyPresented_isLEF_residuallyFinite`
Any generator set, finitely many relators, explicit symmetric-group quotient.

Proof. Let $J = \langle T \mid W \rangle$ with T, W finite and T symmetric, and let $g \in J \backslash \{1\}$; fix a word w_g in T representing g. Let $F \subseteq J$ be the finite set consisting of 1, the elements of T, all partial products $t_1 \cdots t_i$ of every relator $t_1 \cdots t_\ell \in W$, and all partial products of w_g; in particular $g \in F$. Let $\varphi : F \to P$ be a local embedding into a finite group. First, $\varphi(1)\varphi(1) = \varphi(1)$ gives $\varphi(1) = 1_P$. Next, the assignment $t \mapsto \varphi(t)$ kills every relator: its partial products $p_i = t_1 \cdots t_i$ lie in F along with the letters, so $\varphi(p_{i+1}) = \varphi(p_i)\varphi(t_{i+1})$ by multiplicativity, and induction gives $\varphi(t_1) \cdots \varphi(t_\ell) = \varphi(p_\ell) = \varphi(1) = 1_P$. Hence $t \mapsto \varphi(t)$ extends to a homomorphism $\psi : J \to P$. The same induction along the partial products of w_g, all of which lie in F, gives $\psi(g) = \varphi(g)$, and $\varphi(g) \neq \varphi(1) = 1_P$ by injectivity of φ. Thus g survives in a finite quotient, and J is residually finite. ☐

2.5. **The theorems of Kun and Kun–Thom.** We use two structure theorems for sofic approximations of Kazhdan groups.

Theorem 2.9 (Kun [21]). *Let Γ be an infinite finitely generated property-(T) group with a finite symmetric generating set S, and let $\sigma_n : \Gamma \to \mathrm{Sym}(Y_n)$ be a sofic approximation. Then the S-generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition. Moreover, the modified graphs $\widetilde{X}_n$ may be chosen of degree at most d, with the Cheeger constants of all components bounded below by a single constant $h > 0$; the constants d and h depend only on (Γ, S).*

Kun/KunDecomposition
`exists_expanderDecomposition`
Kun/KunFixedDecomposition
`expanderDecomposition`
Proved internally; generating set normalized to contain 1, Γ infinite as printed.

Kun's theorem proves Bowen's conjecture: sofic approximations of infinite Kazhdan groups are essentially disjoint unions of expanders. Theorem 2.9 is deliberately a *one-way* statement: the four-way equivalence of [21, Theorem 3] (v5, the current version) contains a spectral implication that fails as written and is not imported anywhere in this paper; Remark 2.11 records this precisely, together with two local repairs to the arXiv proofs of the theorems of this subsection. The companion result of Kun and Thom is an obstruction for centralizers.

KunThom/KunThomTheorem
`isLEF_of_exactProductExpansion`
KunThom/KunThomEssential
`isLEF_of_matchingCertificate`
Proved internally; generating set normalized to contain 1.

Theorem 2.10 (Kun–Thom [22])**.** *Let Λ be an infinite finitely generated property-(T) group with finite symmetric generating set S_Λ, and let Δ be a finitely generated group. Suppose $\Lambda \times \Delta$ admits a sofic approximation $\pi_n : \Lambda \times \Delta \to \mathrm{Sym}(D_n)$ such that the S_Λ-generator graphs $X(D_n, \pi_n|_\Lambda, S_\Lambda)$ form an essential expander sequence. Then Δ is LEF.*

Theorem 2.10 states the theorem of [22] in its *essential* form, which is exactly how it is invoked in Section 4. The statement in [22] assumes that the Λ-graphs form a genuine expander sequence; the essential form is nonetheless the preprint's own: its proof of Theorem 1.1 opens by assuming that the S_Λ-labelled subgraphs form "an essentially expander sequence" and is carried out under that hypothesis throughout, and its Remark 1.2—already present in the audited v2, and restated in sharper wording in v3—states the distinction between the two hypotheses explicitly, asking of the theorem that the S_Λ-graphs be *(essentially)* one expander rather than a disjoint union of expanders. The heuristic content: a genuine expander is essentially connected, so the commuting factor Δ cannot shuffle components and is forced to act almost by automorphisms of finite structures, which yields LEF approximations.

Interface. The proof in [22] is carried out for *regularly labeled* graphs: each generator label has exactly one outgoing directed edge at every vertex. An approximation normalized as in Lemma 2.2 supplies this structure. Every generator acts by a genuine permutation, and an inverse pair of generators acts by an inverse pair of permutations, so the two directed arcs of a pair combine into one undirected labeled edge; identity loops are discarded by the conventions of Section 2.2. Distinct labels rarely collide: for generators g, h with $h \neq g^{\pm 1}$, a collision $\sigma_n(g)y = \sigma_n(h)y$ makes y a preimage under the permutation $\sigma_n(h)$ of a fixed point of $\sigma_n(h)^{-1}\sigma_n(g)$, which agrees with $\sigma_n(h^{-1}g)$ outside a vanishing fraction of points by approximate multiplicativity and has a vanishing fraction of fixed points by asymptotic faithfulness. These properties are verified for the localized approximation at the end of the proof of Theorem 4.1.

formalized in part
Kun/KunSpectralCounterexample
`switched`
`hasCheegerLowerBound_one`
`not_isBipartite`
`rayleigh_testVector`
`no_uniform_spectral_gap`
`repeated_component_expands`
`rayleigh_repeatedTestVector`
`repetition_of_components`
The graph theory is formalized: the switched family, its uniform expansion, its non-bipartiteness, the Rayleigh value, and the repetition. Not formalized: the reading into the numbered conditions of [21, Theorem 3]*, a citation of that paper's text, and the elementary eigenvector computation that displays the defect.*

Remark 2.11 (Audit of the two cited proofs)**.** Because the two theorems above carry the entire approximation-theoretic weight of this paper, their proofs—in the arXiv versions cited, v5 of [21] and v2 of [22]—were audited line by line, with three findings; we record them here both to delimit exactly what is imported and to credit the arguments of [21, 22], which survive intact.

First, Theorem 3 of [21] is stated as a four-way equivalence, and its spectral implication (4) ⇒ (1) fails as written: it invokes the Cheeger–Alon–Milman bound, which separates the nonconstant Markov spectrum from

$+1$ but not from -1. The counterexample is explicit. In $K_{n,n}$ with sides $\{l_i\}$, $\{r_i\}$, replace the two edges $\{l_0, r_0\}$, $\{l_1, r_1\}$ by $\{l_0, l_1\}$, $\{r_0, r_1\}$; this preserves degrees, creates a triangle (so the graph is genuinely non-bipartite for $n \geq 3$), and keeps $h \geq 1$ for $n \geq 6$: a set U with $2|U| \leq 2n$ and a, b vertices on the two sides has $K_{n,n}$-boundary $n(a+b) - 2ab \geq n|U|/2$, since $4ab \leq (a+b)^2 \leq n(a+b)$, and the switch removes at most two boundary occurrences. Meanwhile the vector equal to $+1$ on one side and -1 on the other has normalized Markov Rayleigh quotient $-1 + 4/n^2$, so the least eigenvalue runs to -1 while the expansion stays uniform—non-bipartiteness alone does not repair the implication. Repeating each switched graph enough times gives a sequence satisfying (2)–(4) while violating (1). The repetition is what makes the violation possible, and it is worth seeing why. Condition (1) of [21, Theorem 3] asks that $\|M_n^4 f - M_n^2 f\| \leq (1 - \varepsilon)\|M_n^2 f - f\| + \alpha |V(G_n)|^{1/2} \|f\|_\infty$ for all f, with ε fixed first and $\alpha > 0$ arbitrary afterwards; applied to a bottom eigenvector, $M_n f = \lambda f$, it reads $\lambda^2(1-\lambda^2)\|f\| \leq (1-\varepsilon)(1-\lambda^2)\|f\| + \alpha |V(G_n)|^{1/2} \|f\|_\infty$, so the quantity to be beaten is $(\lambda^2 - 1 + \varepsilon)(1-\lambda^2)\|f\|$. Along the switched family itself that defect is of order εn^{-2} and tends to 0, while the additive slack stays proportional to $\|f\|$: letting n grow hands the inequality to any fixed α. Disjoint copies of one switched graph leave both λ and the ratio $|V|^{1/2}\|f\|_\infty / \|f\|$ unchanged, so one fixes n with $1 - \lambda^2 < \varepsilon/2$ after ε is given—the least eigenvalue is not -1 exactly, by non-bipartiteness, so this defect is positive—then takes α below it and lets the number of copies grow. Expansion in (2)–(4) is asked of each component, which is the only form available here since a disjoint union has global Cheeger constant 0, and every copy still expands at rate 1. This does not affect the present paper, nor Theorem 1 of [21]: the decomposition theorem needs only the forward implications, and Theorem 2.9 above is exactly that one-way statement—the only form used here.

Second, the proof of the forward direction in [21] normalizes a Markov defect (Lemma 10) without first excluding the degenerate case in which that defect vanishes; the repair separates the zero-defect case, where the displacement estimate holds for homogeneous reasons, before dividing.

Third, the centralizer argument of [22] constructs a correspondence between finite models and later treats it as the graph of a permutation, which it need not yet be; the repair measures the row and column fibers of the correspondence, extracts its injective matching core, and completes that core to a genuine permutation, after which the argument proceeds as written.

Both repairs are local to the proofs; a reader auditing this paper by hand should read [21, 22] with the three points above in mind. The record is version-specific: the arXiv comments of [21] (v5) already note that earlier versions did not sufficiently separate one-sided from two-sided expansion. The essential-versus-single-expander distinction discussed after Theorem 2.10 is not a later correction: it is stated in Remark 1.2 of [22] in the audited v2 itself, and again, in sharper wording, in v3 (August 2026), which appeared as this paper was completed and which records that the preprint was never submitted for publication. Neither source is a journal publication.

3. Conservative component matching: the toolkit

This section isolates five elementary lemmas about finite graphs, permutations, and sequences. They are stated once, in the exact quantitative forms used in Section 4; the only genuinely group-theoretic input of the paper appears there.

3.1. **Expander refinement.**

Matching/Refinement
`exists_dominant_cell`
`le_crossing_of_cell`
Matching/ComponentRefinement
`refineComponent`
Stated per component.

Lemma 3.1 (Expander refinement)**.** *Let X be a finite graph, every connected component of which has Cheeger constant at least $h > 0$. Let $V(X) = A_1 \sqcup \cdots \sqcup A_r$ be a partition, and let e be the number of edges of X whose endpoints lie in different parts. Then there is an assignment $C \mapsto i(C)$ of a part to each component of X, with $|C \cap A_{i(C)}|$ maximal, such that*

$$\sum_C |C \setminus A_{i(C)}| \;\le\; \frac{4e}{h}.$$

Proof. Every edge of X lies inside a component, so the crossing edges distribute among the components; let e_C be the number inside C, with $\sum_C e_C = e$. Fix a component C, let $a_1 \ge a_2 \ge \cdots \ge a_k$ be the sizes of its nonempty intersections with the parts, and let $A_{i(C)}$ realize a_1. We claim

$$|C| - a_1 \;\le\; \frac{4e_C}{h},$$

which proves the lemma upon summation. If C meets only one part, then $C \setminus A_{i(C)} = \emptyset$ and the claim is trivial; assume C meets at least two parts. If a subset $W \subseteq C$ is a union of part-intersections with $|W| \le \frac{1}{2}|C|$, then every edge of $\partial_{X[C]} W$ joins different parts, so $e_C \ge |\partial_{X[C]} W| \ge h|W|$.

If $a_1 \ge \frac{1}{2}|C|$, apply this to $W = C \setminus A_{i(C)}$, a union of part-intersections of size $|C| - a_1 \le \frac{1}{2}|C|$; this gives $|C| - a_1 \le e_C/h$.

If $a_1 < \frac{1}{2}|C|$, accumulate part-intersections in any order until the running total first reaches at least $\frac{1}{4}|C|$; the resulting union W satisfies $\frac{1}{4}|C| \le |W| < \frac{1}{4}|C| + a_1 < \frac{3}{4}|C|$, so $\min\bigl(|W|, |C \setminus W|\bigr) \ge \frac{1}{4}|C|$, and whichever of W, $C \setminus W$ has size at most $\frac{1}{2}|C|$ is again a union of part-intersections. Applying the boundary bound to it gives $e_C \ge \frac{h}{4}|C| \ge \frac{h}{4}(|C| - a_1)$. $\square$

3.2. **Permutation conservation.**

Matching/PermutationConservation
`permutation_conservation_full`

Lemma 3.2 (Permutation conservation)**.** *Let Y be a finite set, $\pi \in \mathrm{Sym}(Y)$, and $f : Y \to \mathbb{R}$. Then*

$$\sum_{y \in Y} \bigl(f(y) - f(\pi y)\bigr)_+ \;=\; \sum_{y \in Y} \bigl(f(\pi y) - f(y)\bigr)_+ \;=\; \tfrac{1}{2} \sum_{y \in Y} |f(\pi y) - f(y)|.$$

Proof. Both $\sum_y f(y)$ and $\sum_y f(\pi y)$ equal the same number, so $\sum_y \bigl(f(y) - f(\pi y)\bigr) = 0$; split into positive and negative parts. $\square$

This exact identity is what converts a one-sided (monotonicity) estimate along the arcs of a permutation into a two-sided ℓ^1 estimate.

3.3. **Median pinning.**

Lemma 3.3 (Median pinning)**.** *Let Z be a finite graph in which every connected component B satisfies $h(Z[B]) \geq h > 0$. Let $f : V(Z) \to \mathbb{R}$, and suppose that on every component at most half of the vertices satisfy $f > \frac{1}{2}$ and at most half satisfy $f < \frac{1}{2}$. Then*

Matching/Pinning
median_pinning
Matching/ComponentPinning
normalized_pinning_global

$$\sum_{x \in V(Z)} \left| f(x) - \tfrac{1}{2} \right| \;\leq\; \frac{1}{h} \sum_{\{x,y\} \in E(Z)} |f(x) - f(y)|.$$

Proof. Every edge of Z lies within a component. Apply Lemma 2.7 with $c = \frac{1}{2}$ on each component and sum. □

3.4. **Weighted selection.**

Lemma 3.4 (Weighted selection)**.** *For each n, let $\{D_{n,i}\}_{i \in I_n}$ be disjoint nonempty subsets of a finite set Y_n, $|Y_n| = N_n \to \infty$, with $\sum_{i \in I_n} |D_{n,i}| \geq (1 - o(1))N_n$. Suppose that for every fixed M,*

Matching/Selection
exists_selection
exists_selection_diverging
diagonalLevel_diverges
diagonalLevel_error

$$\sum_{i \colon |D_{n,i}| \leq M} |D_{n,i}| = o(N_n).$$

Let, for each $r \in \mathbb{N}$, nonnegative error weights $E_{n,i}^{(r)}$ be given with

$$\sum_{i \in I_n} E_{n,i}^{(r)} = o(N_n) \qquad \text{for every fixed } r.$$

Then there exist indices $i(n) \in I_n$, for all n in a cofinite set, such that $|D_{n,i(n)}| \to \infty$ and, for every fixed r,

$$E_{n,i(n)}^{(r)} = o\big(|D_{n,i(n)}|\big).$$

Proof. Replacing $E_{n,i}^{(r)}$ by $\max_{r' \leq r} E_{n,i}^{(r')}$—which only increases the weights, preserves the hypothesis for each fixed r because the maximum of finitely many families, each of total weight $o(N_n)$, has total weight at most their sum, and only strengthens the conclusion—we may and do assume that $E_{n,i}^{(r)}$ is nondecreasing in r.

For integers $k \geq 1$, call an index $i \in I_n$ *k-bad* if $|D_{n,i}| \leq k$ or $E_{n,i}^{(k)} > \frac{1}{k}|D_{n,i}|$, and k-good otherwise; let

$$q_k(n) = \frac{1}{N_n} \sum_{i \;\; k\text{-bad}} |D_{n,i}|.$$

For each fixed k,

$$q_k(n) \leq \frac{1}{N_n}\Big(\sum_{i \colon |D_{n,i}| \leq k} |D_{n,i}| \;+\; k \sum_{i \in I_n} E_{n,i}^{(k)} \Big) \xrightarrow[n \to \infty]{} 0,$$

the second summand bounding the mass of the second kind of badness by Markov's inequality, and both summands being $o(N_n)$ by hypothesis. Moreover the set of k-bad indices is nondecreasing in k: $|D_{n,i}| \le k$ implies $|D_{n,i}| \le k+1$, and $E^{(k)}_{n,i} > \frac{1}{k}|D_{n,i}|$ implies $E^{(k+1)}_{n,i} \ge E^{(k)}_{n,i} > \frac{1}{k+1}|D_{n,i}|$, using monotonicity in r. Hence $q_k(n)$ is nondecreasing in k for each fixed n.

Now make one simultaneous choice. Since $q_k(n) \to 0$ as $n \to \infty$ for each fixed k, there are integers $n_1 < n_2 < \cdots$ with $q_k(n) \le \frac{1}{k}$ for all $n \ge n_k$. For $n \ge n_1$ put $k_n = \max\{k \,|\, n_k \le n\}$, a finite maximum since the n_k strictly increase; then $k_n \to \infty$ and, by the definition of n_{k_n}, $q_{k_n}(n) \le \frac{1}{k_n} \to 0$.

For n large, the k_n-bad indices carry mass $q_{k_n}(n)N_n \le N_n/k_n = o(N_n)$, while all indices together carry mass at least $(1-o(1))N_n$; hence some k_n-good index $i(n)$ exists. Then $|D_{n,i(n)}| > k_n \to \infty$, and for every fixed r: as soon as $k_n \ge r$, monotonicity in r gives

$$E^{(r)}_{n,i(n)} \le E^{(k_n)}_{n,i(n)} \le \tfrac{1}{k_n}|D_{n,i(n)}| = o\big(|D_{n,i(n)}|\big). \qquad \square$$

3.5. **Localization by completion.**

Matching/Localization
`exists_completion`
`involutiveCompletion`
Criterion/LocalizedApproximation
`toSoficApproximation`

Lemma 3.5 (Localization by completion)**.** *Let H be a countable group and $\sigma_n : H \to \mathrm{Sym}(Y_n)$ maps. Suppose $D_n \subseteq Y_n$ are subsets with $|D_n| \to \infty$ such that, for every fixed $g, h \in H$:*

(i) $|\{y \in D_n \,|\, \sigma_n(g)y \notin D_n\}| = o(|D_n|)$;
(ii) $|\{y \in D_n \,|\, \sigma_n(g)\sigma_n(h)y \ne \sigma_n(gh)y\}| = o(|D_n|)$;
(iii) $|\{y \in D_n \,|\, \sigma_n(g)y = y\}| = o(|D_n|)$ *whenever* $g \ne 1$.

For $g \in H$ define $\pi_n(g) \in \mathrm{Sym}(D_n)$ by letting $\pi_n(g)$ agree with $\sigma_n(g)$ on $\{y \in D_n \,|\, \sigma_n(g)y \in D_n,\ \sigma_n(g)y \ne \sigma_n(g)y' \ \forall y' \ne y \in D_n\}$ and extending the resulting partial injection of D_n into itself to a permutation of D_n arbitrarily. Then $\pi_n : H \to \mathrm{Sym}(D_n)$ is a sofic approximation of H. Moreover, if $g, g^{-1} \in H$ satisfy $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$, the completions may be chosen with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$, and then

$$\big|\{y \in D_n \,|\, \pi_n(g)y \ne \sigma_n(g)y\}\big| = o(|D_n|).$$

Proof. Since $\sigma_n(g)$ is injective on Y_n, the described set consists precisely of the points of D_n sent into D_n by $\sigma_n(g)$, and its complement in D_n has size $o(|D_n|)$ by (i); the partial map is an injection, and any extension to a permutation of D_n differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n. Then (ii) transfers: for fixed g, h, the points of D_n where $\pi_n(g)\pi_n(h)y \ne \pi_n(gh)y$ are contained in the union of the $o(|D_n|)$-sized exceptional sets for g, h, gh (points where $\pi \ne \sigma$, including the $\sigma_n(h)$-preimage of the exceptional set of g) and the set in (ii); similarly (iii) transfers faithfulness. When $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$ and $g \ne g^{-1}$, the two partial injections attached to g and to g^{-1} are mutually inverse, so a completion of one can be transported to the other. When $g = g^{-1}$, so that $\sigma_n(g)^2 = \mathrm{id}$, the set $A_{n,g} = \{y \in D_n \,|\, \sigma_n(g)y \in D_n\}$ is $\sigma_n(g)$-invariant—if y and $\sigma_n(g)y$ both lie in D_n, then so do $\sigma_n(g)y$ and $\sigma_n(g)^2y = y$—so $\sigma_n(g)$ restricts to an involution of $A_{n,g}$; extending by the identity on $D_n \setminus A_{n,g}$

yields an involutive completion $\pi_n(g) = \pi_n(g)^{-1}$ that differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n by (i). $\square$

4. The compression–centralizer criterion

This section proves the analytic core of the paper. We first state a local version that isolates the exact graph-theoretic hypotheses; property (T) of the ambient group is a sufficient condition for the second hypothesis, and the local statement records precisely what it is used for.

4.1. Statement.

Criterion/LocalCriterion
`LocalCriterionData`
Criterion/CriterionAssembly
`isLEF_of_soficApproximation`

Theorem 4.1 (Local compression–centralizer theorem)**.** *Let G be a finitely generated group and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup with finite symmetric generating set S. Let $Q \subseteq G$ be a finite set with*

$$G = \langle \Gamma, Q \rangle \qquad \text{and} \qquad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q). \tag{4}$$

Fix $q_0 \in Q$, put $K = q_0 \Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$[K, J] = 1, \qquad K \cap J = \{1\}. \tag{5}$$

Put $A = S \cup Q \cup Q^{-1}$. Suppose G admits a sofic approximation $\sigma_n : G \to \mathrm{Sym}(Y_n)$ such that

(i) *the S-generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition, and*
(ii) *the A-generator graphs $X(Y_n, \sigma_n, A)$ admit an expander decomposition.*

Then J is LEF.

Criterion/CriterionAssembly
`isLEF_of_isSofic`
`not_isSofic_of_not_isLEF`

Corollary 4.2 (Kazhdan compression–centralizer criterion; Theorem D)**.** *In the setting of (4) and (5), suppose in addition that G and Γ have property (T). If G is sofic, then J is LEF. Equivalently: if J is not LEF, then G is not sofic.*

Proof. Suppose G is sofic and fix a sofic approximation σ_n. By (4) the finite symmetric set $A = S \cup Q \cup Q^{-1}$ generates G, and G is infinite, containing the infinite subgroup Γ. The restriction $\sigma_n|_\Gamma$ is a sofic approximation of the infinite Kazhdan group Γ, so Theorem 2.9 yields hypothesis (i); applying Theorem 2.9 to G itself with the generating set A yields hypothesis (ii). Theorem 4.1 concludes. $\square$

Remark 4.3. Only the componentwise ℓ^1 median inequality of Lemma 3.3 is extracted from hypothesis (ii); that hypothesis may accordingly be weakened to any decomposition of the A-generator graphs, up to $o(|Y_n|)$ edge edits, into components satisfying a uniform ℓ^1 Poincaré inequality around medians. This separates the approximation-theoretic input, which property (T) supplies through Kun's theorem, from the compression mechanism, which is combinatorial.

The remainder of the section proves Theorem 4.1. Write $N_n = |Y_n|$. All $o(N_n)$ estimates are uniform over the fixed finite data (S, Q, q_0) and the decomposition constants; the set Q itself is never symmetrized inside the compression steps, since the compression hypothesis $q\Gamma q^{-1} \le \Gamma$ is assumed only for $q \in Q$.

4.2. **The two component partitions.** Put $S_K = q_0 S q_0^{-1}$, a finite symmetric generating set of K, and fix a finite symmetric generating set S_J of the finitely generated group J. By Lemma 2.2 we may and do assume the exact identities

$$\sigma_n(1) = \mathrm{id}, \qquad \sigma_n(g^{-1}) = \sigma_n(g)^{-1} \quad (g \in A \cup S_K \cup S_J),$$

after a pointwise $o(N_n)$ modification; each relevant generator graph changes by $o(N_n)$ edges, so hypotheses (i) and (ii) persist.

By hypothesis (i), fix an expander decomposition of the S-generator graphs: graphs $\widetilde{X}_n$ on Y_n of degree at most d_Γ, with

$$\mathrm{ed}\big(X(Y_n, \sigma_n, S), \widetilde{X}_n\big) = o(N_n),$$

such that every component of $\widetilde{X}_n$ has Cheeger constant at least $h_\Gamma > 0$. Let $\mathcal{P}_n$ denote the partition of Y_n into the components of $\widetilde{X}_n$, and write $P_n(y)$ for the block containing y. We refer to the blocks of $\mathcal{P}_n$ as Γ-*components*.

By hypothesis (ii), fix likewise an expander decomposition $\widetilde{Z}_n$ of the A-generator graphs, with all components of Cheeger constant at least $h_G > 0$ and uniformly bounded degree; let $\mathcal{B}_n$ be the partition of Y_n into the components of $\widetilde{Z}_n$, with blocks $B_n(y)$, the G-*components*.

Both partitions are almost invariant in the following sense. For every $s \in S$,

$$\big|\{y \in Y_n \mid P_n(\sigma_n(s)y) \neq P_n(y)\}\big| = o(N_n), \tag{6}$$

because apart from the $o(N_n)$ edge edits, every arc $(y, \sigma_n(s)y)$ is an edge of $\widetilde{X}_n$ or a loop, and edges of $\widetilde{X}_n$ do not cross $\mathcal{P}_n$. For every $a \in A$,

$$\big|\{y \in Y_n \mid B_n(\sigma_n(a)y) \neq B_n(y)\}\big| = o(N_n), \tag{7}$$

for the same reason applied to $\widetilde{Z}_n$.

Both partitions stay fixed from here on: $\mathcal{P}_n$ and $\mathcal{B}_n$ are the exact component partitions of $\widetilde{X}_n$ and $\widetilde{Z}_n$, and every block keeps its connectivity and its Cheeger constant. This is deliberate—the final step of the proof selects a block of $\mathcal{P}_n$ and uses precisely that it is a genuine component of $\widetilde{X}_n$. The two partitions are related only pointwise, in two ways: through the almost-invariance estimates (6) and (7), and through the median normalization of Section 4.4, which compares a point with its compressor image exactly when the two lie in the same G-component; the $o(N_n)$ points where this fails are carried as explicit exceptional sets through every estimate below. In particular, neither partition is assumed to refine the other.

4.3. **One-sided refinement along a compressor.** Fix $q \in Q$. The group identity $qs = (qsq^{-1})q$ and approximate multiplicativity give, for each $s \in S$, the *conjugacy defect* bound

$$|\Big\{y \in Y_n \,\Big|\, \sigma_n(q)\sigma_n(s)y \neq \sigma_n(qsq^{-1})\sigma_n(q)y\Big\}| = o(N_n). \tag{8}$$

Indeed, $\sigma_n(q)\sigma_n(s)$ agrees with $\sigma_n(qs)$ outside $o(N_n)$ points, and $\sigma_n(qs) = \sigma_n((qsq^{-1})q)$ agrees with $\sigma_n(qsq^{-1})\sigma_n(q)$ outside $o(N_n)$ points, both by Definition 2.1(a); here qsq^{-1} belongs to the fixed finite set $qSq^{-1} \subseteq \Gamma$, by the compression hypothesis (4).

Transport the graphs $\widetilde{X}_n$ by the permutation $\sigma_n(q)$: the transported graph $\sigma_n(q)\widetilde{X}_n$ has vertex set Y_n and one edge $\{\sigma_n(q)x, \sigma_n(q)y\}$ for each edge $\{x, y\}$ of $\widetilde{X}_n$, multiplicities preserved; its components are $\{\sigma_n(q)C \,|\, C \in \mathcal{P}_n\}$, with the same component Cheeger constants $\geq h_\Gamma$ and degree bound d_Γ. By (8) and the edit bounds, apart from $o(N_n)$ edge occurrences the edges of the transported graph are of the form $\{\sigma_n(q)y, \sigma_n(\gamma)\sigma_n(q)y\}$ with $\gamma \in qSq^{-1} \subseteq \Gamma$; for each of the finitely many $\gamma \in qSq^{-1} \subseteq \Gamma$, fix a word in S representing it; by Lemma 2.3, $\sigma_n(\gamma)$ agrees with the product of the letter permutations outside $o(N_n)$ points, and by (6) applied to each letter in turn, that product preserves the block $P_n(\cdot)$ outside $o(N_n)$ points. Hence all but $o(N_n)$ of the transported edges lie inside blocks of $\mathcal{P}_n$.

Apply Lemma 3.1 to the transported graph with the partition $\mathcal{P}_n$. This produces a map

$$\Phi_{q,n} : \mathcal{P}_n \to \mathcal{P}_n, \quad \Phi_{q,n}(C) = \text{the block meeting } \sigma_n(q)C \text{ in maximal size,}$$

with total leakage

$$\sum_{C \in \mathcal{P}_n} e_{q,n}(C) = o(N_n), \qquad \text{where} \qquad e_{q,n}(C) := \big|\sigma_n(q)C \setminus \Phi_{q,n}(C)\big|. \tag{9}$$

Thus each transported Γ-component is almost contained in a single target Γ-component. The containment is a priori one-sided: a large target could still absorb several small transported components. Ruling this out is the business of the median normalization.

4.4. **Median normalization and conservation.** For $y \in Y_n$, let $s_n(y) = |P_n(y)|$ be the size of its Γ-component. For a G-component $B \in \mathcal{B}_n$, let $m_n(B) \geq 1$ be a *vertex-weighted median* of s_n on B: a value such that

$$|\{y \in B \,|\, s_n(y) > m_n(B)\}| \leq \tfrac{1}{2}|B| \text{ and } |\{y \in B \,|\, s_n(y) < m_n(B)\}| \leq \tfrac{1}{2}|B|.$$

Define the *median normalization*

$$f_n(y) = \frac{s_n(y)}{s_n(y) + m_n(B_n(y))} \;\in\; (0, 1). \tag{10}$$

Since $t \mapsto t/(t + m)$ is strictly increasing, on every G-component B at most half the vertices satisfy $f_n > \frac{1}{2}$ and at most half satisfy $f_n < \frac{1}{2}$; that is, $\frac{1}{2}$ is a median of f_n on every component of $\widetilde{Z}_n$.

Variation along S. For $s \in S$,

$$\sum_{y \in Y_n} |f_n(\sigma_n(s)y) - f_n(y)| = o(N_n), \tag{11}$$

since the summand vanishes unless $P_n(\sigma_n(s)y) \neq P_n(y)$ or $B_n(\sigma_n(s)y) \neq B_n(y)$, a set of size $o(N_n)$ by (6) and (7), and $0 < f_n < 1$.

One-sided variation along $q \in Q$. Call $y \in Y_n$ a *good point for q* if

$$\sigma_n(q)y \in \Phi_{q,n}(P_n(y)) \qquad \text{and} \qquad B_n(\sigma_n(q)y) = B_n(y). \tag{12}$$

Let y be good, $C = P_n(y)$, $D = \Phi_{q,n}(C)$, and $e = e_{q,n}(C)$; then

$$s_n(\sigma_n(q)y) = |D| \geq |\sigma_n(q)C \cap D| = |C| - e = s_n(y) - e.$$

Since both y and $\sigma_n(q)y$ lie in the same G-component B, they are normalized by the same median $m = m_n(B) \geq 1$. For $a \geq (1-\delta)a' \geq 0$ and $m \geq 1$,

$$\frac{a'}{a'+m} - \frac{a}{a+m} \;\leq\; \frac{a'}{a'+m} - \frac{(1-\delta)a'}{(1-\delta)a'+m} \;\leq\; \delta, \tag{13}$$

where the first inequality is monotonicity of $t \mapsto t/(t+m)$ and the second is a direct computation: the difference equals $\delta a' m/((a'+m)((1-\delta)a'+m)) \leq \delta$. Applying (13) with $a' = |C|$, $a = |D| \geq |C| - e$, $\delta = e/|C|$ gives, for every good point y,

$$\big(f_n(y) - f_n(\sigma_n(q)y)\big)_+ \leq \frac{e_{q,n}(P_n(y))}{|P_n(y)|}.$$

Summing over Y_n—good points of a block C contribute at most $e_{q,n}(C)$ in total, points with $\sigma_n(q)y \notin \Phi_{q,n}(P_n(y))$ number $e_{q,n}(C)$ per block and contribute at most 1 each, and points whose arc crosses $\mathcal{B}_n$ contribute at most 1 each—we obtain

(14)
$$\begin{aligned}\sum_{y \in Y_n} \big(f_n(y) - f_n(\sigma_n(q)y)\big)_+ \;&\leq\; 2\sum_{C \in \mathcal{P}_n} e_{q,n}(C) + \big|\{y \mid B_n(\sigma_n(q)y) \neq B_n(y)\}\big| \\ &=\; o(N_n)\end{aligned}$$

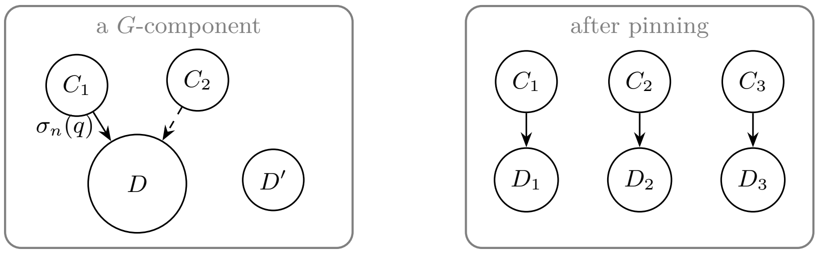
by (9) and (7). The compressed image can only grow—but growth is globally impossible:

Conservation. Since $\sigma_n(q)$ is a permutation of Y_n, Lemma 3.2 upgrades (14) to the two-sided estimate

(15)
$$\sum_{y \in Y_n} |f_n(\sigma_n(q)y) - f_n(y)| = 2\sum_{y \in Y_n} \big(f_n(y) - f_n(\sigma_n(q)y)\big)_+ = o(N_n) \qquad (q \in Q),$$

and the change of variables $y = \sigma_n(q)x$, legitimate because $\sigma_n(q^{-1}) = \sigma_n(q)^{-1}$, extends (15) to all $q \in Q \cup Q^{-1}$.

Pinning. Every edge of the A-generator graph $X(Y_n, \sigma_n, A)$ arises from an arc $(y, \sigma_n(a)y)$ with $a \in A = S \cup Q \cup Q^{-1}$, and each edge arises from at



(a) one-sided refinement (b) almost-bijective matching

FIGURE 2. Conservative component matching. (a) Transporting a Γ-component by a compressor almost lands it inside a single target component, but a priori a large target could absorb several small sources (dashed). (b) Exact conservation and median pinning force transported and target sizes to agree asymptotically, so the assignment $\Phi_{q,n}$ becomes injective on almost all mass (Proposition 4.4).

most $2|A|$ arcs. Hence (11) and (15) give

$$\sum_{\{x,y\}\in E(X(Y_n,\sigma_n,A))} |f_n(x) - f_n(y)| = o(N_n).$$

Since $0 < f_n < 1$, editing $o(N_n)$ edges changes this sum by at most $o(N_n)$; therefore the same bound holds over $E(\widetilde{Z}_n)$. As $\frac{1}{2}$ is a median of f_n on every component of $\widetilde{Z}_n$ and all component Cheeger constants are at least h_G, Lemma 3.3 yields the concentration estimate

$$\sum_{y\in Y_n} \left| f_n(y) - \tfrac{1}{2} \right| = o(N_n). \tag{16}$$

In words: after median normalization, almost every Γ-component has size asymptotically equal to the median size on its G-component, because $s/(s+m) \to \frac{1}{2}$ exactly when $s/m \to 1$.

4.5. **Almost-bijective matching.** The mechanism of this subsection is drawn in Figure 2.

Proposition 4.4 (Matching)**.** *Fix $q \in Q$. There are subcollections $\mathcal{G}_{q,n} \subseteq \mathcal{P}_n$ such that:*

(i) $\sum_{C\in\mathcal{P}_n\setminus\mathcal{G}_{q,n}} |C| = o(N_n)$;
(ii) *for every $C \in \mathcal{G}_{q,n}$, with $D = \Phi_{q,n}(C)$,*

$$\big|\sigma_n(q)C \,\triangle\, D\big| = o(1)\cdot|C| \qquad \textit{uniformly over } C \in \mathcal{G}_{q,n}; \tag{17}$$

(iii) *for n large, $\Phi_{q,n}$ is injective on $\mathcal{G}_{q,n}$;*
(iv) $\sum_{C\in\mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1-o(1))\,N_n$.

Matching/MedianNormalization
`medianNormalize`
Matching/MatchingPreparation
`acceptable_symmDiff_le`
Criterion/Criterion
`symmDiff_le_of_pinned`
`matching_injective`
Stated per component.

Proof. Choose $\eta_n \downarrow 0$ so slowly that the three global errors—in (7) for the element q, in (9), and in (16)—are all $o(\eta_n^2 N_n)$. Let $W_n \subseteq Y_n$ be the union of the four sets

$$\left\{y \,\middle|\, |f_n(y) - \tfrac{1}{2}| > \eta_n\right\}, \quad \sigma_n(q)^{-1}\left\{y \,\middle|\, |f_n(y) - \tfrac{1}{2}| > \eta_n\right\},$$
$$\{y \mid B_n(\sigma_n(q)y) \neq B_n(y)\}, \quad \{y \mid \sigma_n(q)y \notin \Phi_{q,n}(P_n(y))\}.$$

Markov's inequality applied to (16) bounds the first set by $\eta_n^{-1} \cdot o(\eta_n^2 N_n) = o(\eta_n N_n)$; the second is a bijective preimage of the first; the last two have sizes $o(\eta_n^2 N_n)$ by the choice of η_n. Hence $|W_n| = o(\eta_n N_n)$.

Discard from $\mathcal{P}_n$ every block C with $e_{q,n}(C) > \eta_n|C|$ or $|C \cap W_n| > \eta_n|C|$, and call the remaining collection $\mathcal{G}_{q,n}$. The discarded mass is at most $\eta_n^{-1}\sum_C e_{q,n}(C) + \eta_n^{-1}|W_n| = o(N_n)$, proving (i).

Let $C \in \mathcal{G}_{q,n}$ and pick $y \in C \setminus W_n$, possible since $\eta_n < 1$. Put $D = \Phi_{q,n}(C)$ and $B = B_n(y) = B_n(\sigma_n(q)y)$, and write $m = m_n(B)$. Since $y \notin W_n$, we have $\sigma_n(q)y \in D$, so $s_n(\sigma_n(q)y) = |D|$, and

$$\Big|\frac{|C|}{|C| + m} - \frac{1}{2}\Big| \leq \eta_n, \qquad \Big|\frac{|D|}{|D| + m} - \frac{1}{2}\Big| \leq \eta_n.$$

Solving $|\frac{a}{a+m} - \frac{1}{2}| \leq \eta$ gives $\frac{1-2\eta}{1+2\eta}\, m \leq a \leq \frac{1+2\eta}{1-2\eta}\, m$; applying this to $a = |C|$ and to $a = |D|$ and dividing,

$$|D| = (1 + O(\eta_n))\,|C| \qquad \text{uniformly over } C \in \mathcal{G}_{q,n}. \tag{18}$$

With $e = e_{q,n}(C) \leq \eta_n|C|$,

$$\begin{aligned}|\sigma_n(q)C \mathbin{\triangle} D| &= |\sigma_n(q)C \setminus D| + |D \setminus \sigma_n(q)C| \\ &\leq e + (|D| - |C| + e) \leq 2e + \big||D| - |C|\big| = O(\eta_n)|C|,\end{aligned}$$

proving (ii).

For (iii): if $C \neq C'$ in $\mathcal{G}_{q,n}$ had $\Phi_{q,n}(C) = \Phi_{q,n}(C') = D$, the disjoint sets $\sigma_n(q)C \cap D$ and $\sigma_n(q)C' \cap D$ would, by (17) and (18), each occupy a fraction $1 - o(1)$ of D, uniformly—impossible for n large.

For (iv): by (i), (ii), and (18),

$$\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1+o(1)) \sum_{C \in \mathcal{G}_{q,n}} |C| = (1+o(1))(1-o(1))N_n = (1-o(1))N_n.$$

□

Thus one-sided refinement, conservation, and pinning together force the compressor to permute the Γ-components almost bijectively, with asymptotically equal sizes. One more observation is needed before selecting a single matched pair.

Matching/ComponentDivergence

`SoficApproximation.smallBlockVertices_negligible`

`ExpanderDecomposition.smallBlockVertices_negligible`

Lemma 4.5. *For every fixed M, $\sum\{|C| : C \in \mathcal{P}_n,\ |C| \leq M\} = o(N_n)$.*

Proof. Let $g_1, \ldots, g_{M+1} \in \Gamma$ be distinct elements; they exist since Γ is infinite. For all but $o(N_n)$ points y the points $\sigma_n(g_1)y, \ldots, \sigma_n(g_{M+1})y$ are pairwise distinct: for $i \neq j$, $\sigma_n(g_i)$ agrees with $\sigma_n(g_i g_j^{-1})\sigma_n(g_j)$ outside $o(N_n)$ points by Definition 2.1(a), and $\sigma_n(g_i g_j^{-1})$ moves $1 - o(1)$ of all points by Definition 2.1(b). Moreover, fixing a word in S representing each g_i, Lemma 2.3 together with (6) applied along its letters shows that for all but $o(N_n)$ points y all these points lie in $P_n(y)$. A point y satisfying both conditions has $|P_n(y)| \geq M+1$; hence the mass of blocks of size at most M is $o(N_n)$. $\square$

4.6. **Selection, localization, and the proof of Theorem 4.1.**

Proof of Theorem 4.1. Apply Proposition 4.4 with $q = q_0$. For n large this gives a collection of *matched pairs*

$$\big\{(C,D) : C \in \mathcal{G}_{q_0,n},\ D = \Phi_{q_0,n}(C)\big\}, \tag{19}$$

with pairwise distinct sources, pairwise distinct targets, target mass $(1-o(1))N_n$, and $|\sigma_n(q_0)C \triangle D| = o(1)|C|$ uniformly.

Fix an exhaustion $L_1 \subseteq L_2 \subseteq \cdots$ of the countable group $KJ \leq \Gamma$ by finite subsets with $\bigcup_r L_r = KJ$, each L_r containing 1, the finite sets S_K and S_J, and closed under inverses, and such that $L_r \cdot L_r \subseteq L_{r+1}$. To each matched pair (C,D) and each r attach the error weight $E^{(r)}(C,D)$, the sum of:

(a) $|\sigma_n(q_0)C \triangle D|$;
(b) the number of edges of $\widetilde{X}_n$ with an endpoint in C or D that are edited relative to the S-generator graph, plus the number of edges of the transported graph $\sigma_n(q_0)\widetilde{X}_n$ with an endpoint in $D \cup \sigma_n(q_0)C$ that are edited relative to the transported S-generator graph;
(c) $|\Big\{y \in C \,\Big|\, \sigma_n(q_0)\sigma_n(s)y \neq \sigma_n(q_0 s q_0^{-1})\sigma_n(q_0)y \text{ for some } s \in S\Big\}|$;
(d) $|\{y \in D \,|\, \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y \text{ for some } g,h \in L_r\}|$;
(e) $|\{y \in D \,|\, \sigma_n(g)y = y \text{ for some } g \in L_r \setminus \{1\}\}|$;
(f) $|\{y \in D \,|\, P_n(\sigma_n(g)y) \neq P_n(y) \text{ for some } g \in L_r\}|$.

For every fixed r, the sum of $E^{(r)}$ over the matched pairs is $o(N_n)$: (a) by Proposition 4.4(ii) and (i); (b): the sources are pairwise disjoint, the targets are pairwise disjoint, and a fixed block can occur at most once as a source and at most once as a target; hence each endpoint of an edge lies in at most one source and at most one target, and, an edge having two endpoints, each edited edge is counted for at most four matched pairs—likewise in the transported graph, where the sets $\sigma_n(q_0)C$ are pairwise disjoint images of the disjoint sources—while the global edit counts are $o(N_n)$; (c) by (8) over the finite set S, the sources being disjoint; (d) and (e) by soficity of σ_n over the finite set $L_r \cup L_r^2$, the targets being disjoint; (f) by Lemma 2.3 together with (6) applied along the letters of a fixed word in S representing each $g \in L_r \subseteq \Gamma$, the targets being disjoint.

The targets D are pairwise disjoint blocks of $\mathcal{P}_n$ with total mass $(1-o(1))N_n$, and by Lemma 4.5 the total mass of targets of size at most M is $o(N_n)$ for every fixed M. Lemma 3.4 therefore produces matched pairs (C_n, D_n) with

$$|D_n| \to \infty, \qquad E^{(r)}(C_n,D_n) = o(|D_n|) \quad \text{for every fixed } r. \tag{20}$$

Put $U_n = \sigma_n(q_0)C_n$, so $|U_n \triangle D_n| = o(|D_n|)$ and $|U_n| = |C_n| = (1+o(1))|D_n|$. Define

$$D'_n = U_n, \tag{21}$$

a set of size $|C_n| \to \infty$ that agrees with D_n up to $o(|D_n|)$ points. Every error weight in (20) transfers from the pair (C_n, D_n) to D'_n at a cost of $o(|D'_n|)$, by separate accounting: the point-counted errors (d), (e), (f) are counted over D_n, and replacing D_n by D'_n changes each count by at most $|D_n \triangle D'_n| = o(|D'_n|)$; the error (c) is counted over C_n, which is unchanged; the edge-counted errors (b) concern graphs of degree at most $\max(d_\Gamma, 2|S|)$, so replacing D_n by D'_n in the incidence condition changes those counts by at most that degree bound times $|D_n \triangle D'_n| = o(|D'_n|)$; and the quantity (a) does not change.

Localization. Consider the countable group $H = KJ \leq \Gamma$. By (5), K and J commute and intersect trivially, so

$$H = KJ \;\cong\; K \times J.$$

For every fixed $g \in H$, the transferred error (f) shows that $\sigma_n(g)$ maps all but $o(|D'_n|)$ points of D'_n into D_n. Injectivity upgrades D_n to D'_n: since $\sigma_n(g)$ is a permutation of Y_n, at most $|D_n \setminus D'_n| = o(|D'_n|)$ points of D'_n can land in $D_n \setminus D'_n$, so $\sigma_n(g)$ maps all but $o(|D'_n|)$ points of D'_n into D'_n itself. The transferred errors (d) and (e) give multiplicativity and faithfulness on D'_n for every fixed pair $g, h \in H$, by choosing r with $g, h \in L_r$. Lemma 3.5, with completions on the symmetric set $S_K \cup S_J$ chosen compatibly with the exact inverse identities of Section 4.2, yields a sofic approximation

$$\pi_n : K \times J \longrightarrow \mathrm{Sym}(D'_n)$$

with $|\{y \in D'_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D'_n|)$ for every fixed g, and $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K \cup S_J$.

The K-edges expand. Restrict $\widetilde{X}_n$ to C_n and transport by $\sigma_n(q_0)$: the resulting graph Ξ_n on $D'_n = \sigma_n(q_0)C_n$ is connected with Cheeger constant at least h_Γ and degree at most d_Γ, because C_n is a component of $\widetilde{X}_n$. We claim that Ξ_n and the S_K-generator graph $X(D'_n, \pi_n, S_K)$ differ by $o(|D'_n|)$ edges. Indeed, an edge of either graph that fails to be an edge of the other must involve one of the following, each affecting $o(|D'_n|)$ edge occurrences by (20):

1. an edge of $\widetilde{X}_n[C_n]$ edited relative to the S-generator graph on C_n, or conversely (error (b));
2. an arc $(y, \sigma_n(s)y)$, $y \in C_n$, whose $\sigma_n(q_0)$-transport disagrees with the $\sigma_n(q_0 s q_0^{-1})$-arc at $\sigma_n(q_0)y$ (error (c));
3. a point of D'_n where $\pi_n(g) \neq \sigma_n(g)$ for some $g \in S_K$ (localization defect);
4. an arc of some $g \in S_K$ leaving D'_n (transferred error (f) and the symmetric difference (a)).

Hence the S_K-generator graphs of $\pi_n|_K$ form an essential expander sequence in the sense of Definition 2.6.

Conclusion. The group $K = q_0 \Gamma q_0^{-1} \cong \Gamma$ is an infinite finitely generated property-(T) group with symmetric generating set S_K, the group J is finitely generated, and π_n is a sofic approximation of $K \times J$ whose S_K-generator graphs form an essential expander sequence. The regular-labeling requirements recorded after Theorem 2.10 hold for π_n as well: each $\pi_n(g)$ is a genuine permutation of D'_n, with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K \cup S_J$; and for $g, h \in S_K \cup S_J$ with $h \neq g^{\pm 1}$, the collision set $\{y \in D'_n \mid \pi_n(g)y = \pi_n(h)y\}$ has size $o(|D'_n|)$, by the transferred errors (d) and (e) applied to the pair (h^{-1}, g) and the element $h^{-1}g \in L_2 \setminus \{1\}$, together with the $o(|D'_n|)$ agreement between π_n and σ_n on $S_K \cup S_J \cup \{h^{-1}g\}$. Theorem 2.10 applies and shows that J is LEF. ☐

4.7. **Two remarks on the mechanism.**

Remark 4.6 (The product obstruction)**.** The compression hypothesis cannot be deleted from Theorem D. For a direct product $\Lambda \times \Delta$ of an infinite Kazhdan group with an infinite residually finite Kazhdan group, product sofic approximations exist whenever Λ is sofic—a sofic approximation of Λ times the permutation action of Δ on large finite quotients—and in them the Δ-factor genuinely permutes the Λ-components: the Λ-component partition has all blocks of equal size, and no single component is almost invariant under the second factor. The role of the compressors $q \in Q$, through (4), is to generate the whole ambient group while conjugating Γ *into* itself, so that the ambient expansion of hypothesis (ii) couples the component structure of Γ to itself across all of G and forbids exactly this shuffling.

Remark 4.7 (The hypothesis $J \leq \Gamma$ is essential)**.** The subgroup J is required to lie in Γ, and this is used twice: through (6) in error (f), which keeps the J-arcs inside the matched component, and through the localization step, which restricts σ_n on elements of $KJ \leq \Gamma$. The hypothesis cannot be dropped in favor of $J \leq G$. Indeed, sofic groups that are not LEF exist: $J = \mathrm{BS}(2,3) = \langle a, t \mid ta^2t^{-1} = a^3 \rangle$ is finitely presented, residually solvable [20]—hence residually amenable and sofic—and non-Hopfian [6], therefore not residually finite, therefore not LEF by Lemma 2.8. We do not construct an ambient group satisfying the remaining hypotheses around such a J; the assertion is only that the proof uses $J \leq \Gamma$ at the two steps quoted and does not extend merely from $J \leq G$. The criterion is calibrated exactly so that the commuting subgroup rides inside the compressed expander, and this is what the Leavitt corner construction of Sections 6–7 arranges algebraically.

5. Leavitt algebras, elementary groups, and Thompson's group V

Throughout, k is a field and $L = L_k(1,2)$ is the binary Leavitt algebra (1), with generators s_0, s_1, t_0, t_1 and relations $t_i s_j = \delta_{ij}$, $s_0 t_0 + s_1 t_1 = 1$. Everything in this section is a consequence of the two relations alone.

Leavitt/Leavitt

`LeavittFamily`

Definition 5.1. A *binary Leavitt family* in a unital ring A is a quadruple (s_0, s_1, t_0, t_1) of elements of A satisfying

$$t_i s_j = \delta_{ij} \quad (i, j \in \{0, 1\}) \qquad \text{and} \qquad s_0 t_0 + s_1 t_1 = 1.$$

The generators of L form a binary Leavitt family in L, and a unital homomorphic image of L carries one; but a family is strictly less than a copy of L—no injectivity is assumed—and the statements below that mention only a family are correspondingly stronger than their specializations to L. Fix a unital ring A carrying a binary Leavitt family, and write

$$p_0 = s_0 t_0, \qquad p_1 = s_1 t_1,$$

a pair of orthogonal idempotents summing to 1: $p_i p_j = \delta_{ij} p_i$, since $p_0 p_1 = s_0(t_0 s_1)t_1 = 0$ and $p_i^2 = s_i(t_i s_i)t_i = p_i$. For a binary word $w = w_1 \cdots w_\ell \in \{0, 1\}^*$, put $s_w = s_{w_1} \cdots s_{w_\ell}$ and $s_w^* = t_{w_\ell} \cdots t_{w_1}$, so that $s_w^* s_{w'} = \delta_{w,w'}$ whenever $|w| = |w'|$, and more generally $s_w^* s_{w'}$ vanishes unless one of w, w' is a prefix of the other.

Two remarks on size. The algebra L itself is infinite-dimensional over k: the elements $s_{0^i} s_1$, $i \geq 0$, are killed on the left by mutually orthogonal projections and are linearly independent. And a family forces infinitude with no reference to L:

Leavitt/Leavitt
`infinite`
`s0_pow_injective`

Lemma 5.2. *A nontrivial unital ring A carrying a binary Leavitt family is infinite.*

Proof. The idempotents $e_i = s_0^i t_0^i$, $i \geq 0$, are pairwise distinct. Indeed, suppose $e_i = e_j$ with $i < j$. Multiplying on the left by t_0^i and on the right by s_0^i, and using $t_0^i s_0^i = 1$, gives $1 = s_0^d t_0^d$ with $d = j - i \geq 1$. Then $t_1 = t_1 \cdot 1 = (t_1 s_0)\, s_0^{d-1} t_0^d = 0$, whence $1 = t_1 s_1 = 0$, contradicting nontriviality. ☐

Leavitt/LeavittWords
`wordT_mul_wordS_self`
`wordT_mul_wordS_of_incomparable`
`cylinder_split`
Leavitt/MatrixSelfSimilarity
`ringEquivMatrix`
Leavitt/LeavittSelfSimilarity
`binaryMatrixRingEquiv`
Leavitt/PrefixCode
`prefixRingEquiv`
`prefixRingEquiv_apply`
`prefixUnitsEquiv`
Leavitt/SelfSimilarityAlgebra
`matrixAlgEquiv`
`prefixAlgEquiv`
Every positive rank, with explicit inverses.

5.1. **Leaf sets and matrix self-similarity.** A finite set of binary words $\mathcal{C} = (\alpha_1, \ldots, \alpha_r)$ (we use the same symbols for the words and for the corresponding elements $s_{\alpha_i} \in A$) is a *complete leaf set* if the words are the leaves of a finite full rooted binary tree; equivalently, if they are pairwise prefix-incomparable and the cylinders they define partition the boundary $\{0, 1\}^{\mathbb{N}}$.

Lemma 5.3 (Leaf identities)**.** *Let $\mathcal{C} = (\alpha_1, \ldots, \alpha_r)$ be a complete leaf set. Then in every unital ring A carrying a binary Leavitt family:*

$$\alpha_i^* \alpha_j = \delta_{ij} \qquad \textit{and} \qquad \sum_{i=1}^{r} \alpha_i \alpha_i^* = 1.$$

Proof. Prefix-incomparability and the relations $t_i s_j = \delta_{ij}$ give the first identity. The second holds by induction on the tree: for the trivial tree it reads $1 = 1$; replacing a leaf α by its two children $\alpha 0, \alpha 1$ changes the sum by

$$s_{\alpha 0} s_{\alpha 0}^* + s_{\alpha 1} s_{\alpha 1}^* - s_\alpha s_\alpha^* = s_\alpha \big(s_0 t_0 + s_1 t_1 - 1\big) s_\alpha^* = 0. \qquad ☐$$

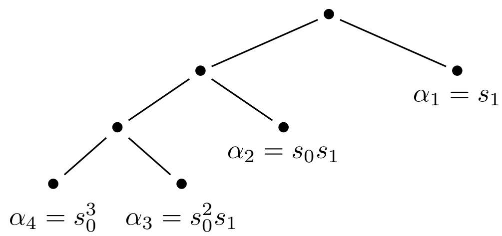


FIGURE 3. The left comb $\mathcal{C}_4$: every left child is subdivided. Reading left edges as s_0 and right edges as s_1, each leaf spells its word along the path from the root; the ordering lists the leaves by depth, with the leftmost branch last.

Proposition 5.4 (Matrix self-similarity)**.** *Let A be a unital ring carrying a binary Leavitt family and let $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ be an ordered complete leaf set. Then*

$$\Theta_{\mathcal{C}} : M_r(A) \longrightarrow A, \qquad (a_{ij}) \longmapsto \sum_{i,j=1}^{r} \alpha_i \, a_{ij} \, \alpha_j^*,$$

is an isomorphism of unital rings—of unital k-algebras when A is a k-algebra—with inverse $x \mapsto (\alpha_i^* x \alpha_j)_{ij}$. It restricts to group isomorphisms $\mathrm{GL}_r(A) \cong A^\times$ for all $r \geq 1$. In particular $M_r(L) \cong L$ for every $r \geq 1$.

Proof. The map $\Theta_{\mathcal{C}}$ is additive—and k-linear when A is a k-algebra—sends the identity matrix to $\sum_i \alpha_i \alpha_i^* = 1$ by Lemma 5.3, and is multiplicative:

$$\Theta_{\mathcal{C}}(a)\,\Theta_{\mathcal{C}}(b) = \sum_{i,j,l,m} \alpha_i a_{ij} \, \alpha_j^* \alpha_l \, b_{lm} \alpha_m^* = \sum_{i,m} \alpha_i \Big(\sum_j a_{ij} b_{jm} \Big) \alpha_m^* = \Theta_{\mathcal{C}}(ab),$$

using $\alpha_j^* \alpha_l = \delta_{jl}$. The displayed inverse is two-sided by the same identities: $\alpha_i^* \Theta_{\mathcal{C}}(a) \alpha_j = a_{ij}$, and $\sum_{i,j} \alpha_i (\alpha_i^* x \alpha_j) \alpha_j^* = 1 \cdot x \cdot 1 = x$. An algebra isomorphism restricts to an isomorphism of unit groups, and $\mathrm{GL}_r(L) = M_r(L)^\times$. $\square$

For every $r \geq 2$ there is a distinguished choice, the *left comb*

$$\mathcal{C}_r = \big(s_1,\ s_0 s_1,\ s_0^2 s_1,\ \dots,\ s_0^{r-2} s_1,\ s_0^{r-1}\big), \tag{22}$$

the leaf set of the tree in which every left child is subdivided; the ordering lists the leaves by depth with the leftmost branch last (Figure 3).

5.2. **Elementary groups over rings.** Let A be a unital ring. Recall $x_{ij}(a) = I_r + aE_{ij}$ for $i \neq j$, the relations $x_{ij}(a) x_{ij}(b) = x_{ij}(a+b)$, and the Steinberg commutator relation

$$[x_{il}(a), x_{lj}(b)] = x_{ij}(ab) \qquad (i, j, l \text{ distinct}),$$

where $[g, h] = ghg^{-1}h^{-1}$.

Leavitt/Whitehead
`whitehead_commutator`
`whitehead_diagonal`
Leavitt/MatrixDiagonalization
`exists_elementary_whitehead`
Leavitt/DiagonalElementary
`firstDiagonalUnit_mem_of_mem_commutator`

Lemma 5.5 (Whitehead commutator lemma)**.** *Let A be a unital ring and $a, b \in A^\times$. Put*

$$w(a) = x_{12}(a)\, x_{21}(-a^{-1})\, x_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}.$$

Then

$$\mathrm{diag}(a, a^{-1}) = w(a)\, w(-1) \in \mathrm{EL}_2(A),$$

and

$$\mathrm{diag}(aba^{-1}b^{-1},\, 1) = \mathrm{diag}(a, a^{-1})\ \mathrm{diag}(b, b^{-1})\ \mathrm{diag}\big((ba)^{-1}, ba\big) \in \mathrm{EL}_2(A).$$

Consequently, for every $m \geq 2$, every product of commutators of units $c \in [A^\times, A^\times]$, and every diagonal position, the matrix with c at that position and 1 elsewhere on the diagonal lies in $\mathrm{EL}_m(A)$.

Proof. First, $x_{12}(a)x_{21}(-a^{-1}) = \left(\begin{smallmatrix} 1-aa^{-1} & a \\ -a^{-1} & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & a \\ -a^{-1} & 1 \end{smallmatrix}\right)$, and multiplying by $x_{12}(a)$ on the right adds the first column, multiplied on the right by a, to the second, giving $w(a) = \left(\begin{smallmatrix} 0 & a \\ -a^{-1} & 0 \end{smallmatrix}\right)$. Then

$$w(a)w(-1) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The displayed product of three diagonal matrices has first entry $ab(ba)^{-1} = aba^{-1}b^{-1}$ and second entry $a^{-1}b^{-1}(ba) = 1$. For the last statement, embed $\mathrm{EL}_2(A) \hookrightarrow \mathrm{EL}_m(A)$ in any chosen pair of coordinates and multiply the resulting matrices for the commutator factors of c. ☐

Leavitt/IntegralGeneration
`elementaryGroup_eq_closure_of_adjoin_int`
`elementaryGroup_finitelyGenerated_int`
Leavitt/ElementaryGroup
`elementaryGroup_finitelyGenerated`
Any finitely generated unital ring, with the printed generating set.

Lemma 5.6. *Let A be a finitely generated unital ring and $r \geq 3$. Then $\mathrm{EL}_r(A)$ is finitely generated: if A is generated as a unital ring by the finite set T, then $\mathrm{EL}_r(A)$ is generated by the finitely many matrices $x_{ij}(1)$ and $x_{ij}(t)$, $t \in T$, $i \neq j$.*

Proof. Let E be the subgroup generated by the listed matrices, and let $A_0 \subseteq A$ be the set of a with $x_{ij}(a) \in E$ for all $i \neq j$. By the addition relation, A_0 is an additive subgroup containing 1 and T; by the Steinberg relation with a third index $l \notin \{i, j\}$ (here $r \geq 3$ is used), $a, b \in A_0$ implies $ab \in A_0$, since $x_{ij}(ab) = [x_{il}(a), x_{lj}(b)]$ and $x_{il}(a), x_{lj}(b) \in E$. Hence A_0 is a unital subring containing T, so $A_0 = A$ and $E = \mathrm{EL}_r(A)$. ☐

PropertyT/FreeElementaryPropertyT
`controlSet_isKazhdanPair`
PropertyT/FiniteFieldElementaryPropertyT
`finiteFieldElementaryThree_hasKazhdanPropertyT`
Proved internally, with an explicit Kazhdan pair.

Theorem 5.7 (Property (T) at rank three; Ershov–Jaikin-Zapirain [13])**.** *Let A be a finite-type algebra over a finite field. Then $\mathrm{EL}_3(A)$ has Kazhdan's property (T).*

This is the case of the Ershov–Jaikin-Zapirain theorem that this paper consumes: [13] proves property (T) for EL_r of *every* finitely generated associative unital ring and every $r \geq 3$—their statement is for the universal rings $\mathbb{Z}\langle x_1, \ldots, x_d\rangle$, and the general case follows since EL_r of a quotient ring is a quotient group of EL_r of the ring and property (T) passes to quotients. We deliberately state only the finite-field, rank-three case: over rings carrying a binary Leavitt family it is all that is ever needed, because the rank equivalence below spreads property (T) from rank three to every rank ≥ 2. Where a statement over an arbitrary finitely generated ring is wanted, the full theorem of [13] may be substituted; we flag each such use.

Over a ring carrying a binary Leavitt family, the rank of an elementary group is immaterial: matrix self-similarity collapses all ranks at the level of the *elementary* groups themselves, with no K-theory and no GE input. This is the engine that will spread every property of one rank to all ranks.

Proposition 5.8 (Elementary rank equivalence)**.** *Let A be a unital ring carrying a binary Leavitt family and let $n, m \geq 2$. Then there is an explicit group isomorphism*

Leavitt/LeavittRankEquivalence
`rankSuccEquiv`
`rankSuccToProduct`
`rankSucc_propertyT_of_rankSucc`

$$\mathrm{EL}_n(A) \;\cong\; \mathrm{EL}_m(A),$$

the composite

$$\mathrm{EL}_n(A) \;\cong\; \mathrm{EL}_n\big(M_m(A)\big) \;=\; \mathrm{EL}_{nm}(A) \;=\; \mathrm{EL}_m\big(M_n(A)\big) \;\cong\; \mathrm{EL}_m(A),$$

where the outer isomorphisms apply $\Theta_{\mathcal{C}}^{-1}$ and $\Theta_{\mathcal{C}'}$ of Proposition 5.4 to each matrix coefficient, and the middle equalities are block flattening. In particular finite generation, infinitude, property (T), and (non)soficity hold at one rank ≥ 2 if and only if they hold at every rank ≥ 2.

Proof. Coefficient transport. A unital ring isomorphism $\varphi : B \to B'$ carries $x_{ij}(b)$ to $x_{ij}(\varphi b)$, hence induces an isomorphism $\mathrm{EL}_p(B) \cong \mathrm{EL}_p(B')$ for every p. Apply this with $\varphi = \Theta_{\mathcal{C}_m}^{-1} : A \to M_m(A)$, which exists by Proposition 5.4.

Block flattening. Identify $M_p(M_q(A)) = M_{pq}(A)$ by the index pairing (i, c), $1 \leq i \leq p$, $1 \leq c \leq q$. We claim $\mathrm{EL}_p(M_q(A)) = \mathrm{EL}_{pq}(A)$ as subgroups of $\mathrm{GL}_{pq}(A)$, for every $p \geq 2$ and $q \geq 1$. One inclusion is direct: a block transvection $I + aE_{ij}$, $i \neq j$, $a \in M_q(A)$, equals the product $\prod_{c,d} x_{(i,c),(j,d)}(a_{cd})$ of q^2 commuting entry transvections. For the other inclusion, an entry transvection $x_{(i,c),(j,d)}(r)$ with $i \neq j$ is a block transvection with the single-entry coefficient rE_{cd}; and one with $i = j$ (so $c \neq d$) is produced by the Steinberg relation through any third block index $l \neq i$, which exists because $p \geq 2$:

$$x_{(i,c),(i,d)}(r) = \big[\, x_{(i,c),(l,c)}(r),\; x_{(l,c),(i,d)}(1) \,\big],$$

both factors being cross-block transvections, hence in $\mathrm{EL}_p(M_q(A))$ by the previous sentence. Applying block flattening with $(p, q) = (n, m)$ and—after reindexing along the bijection $(i, c) \mapsto (c, i)$, which conjugates by a permutation matrix and matches entry transvections to entry transvections—with $(p, q) = (m, n)$, gives the middle equalities. The final claim holds because all four properties are isomorphism-invariant. $\square$

Leavitt/Leavitt
`characteristicTwo_involution`
`characteristicTwo_compressor`
`compressor_factorization`
The factorization is proved in every characteristic.

5.3. **Characteristic-two block factorizations.** The following two identities, valid in characteristic two, are the engine of the explicit spine.

Lemma 5.9. *Let A be a unital ring of characteristic 2 carrying a binary Leavitt family. Then, in $M_2(A)$:*

(a) $x_{12}(s_1)\, x_{21}(t_1)\, x_{12}(s_1) = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}$;

(b) $x_{21}(1+t_0)\, x_{12}(1)\, x_{21}(1+s_0)\, x_{12}(t_0) = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}$.

Proof. (a) Since $1 + p_1 = p_0$ in characteristic 2,

$$\begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 + s_1 t_1 & s_1 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix},$$

and then

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & p_0 s_1 + s_1 \\ t_1 & t_1 s_1 + 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix},$$

using $p_0 s_1 = s_0(t_0 s_1) = 0$ and $t_1 s_1 + 1 = 1 + 1 = 0$.

(b) Compute the product of the first two factors:

$$\begin{pmatrix} 1 & 0 \\ 1 + t_0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix}$$

(using $1 + t_0 + 1 = t_0$ in characteristic 2). Multiplying by the third factor,

$$\begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 + s_0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 1 + t_0 + t_0 + t_0 s_0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix},$$

where the $(1,1)$-entry is $1 + 1 + s_0 = s_0$ and the $(2,1)$-entry is $(1 + t_0) + t_0(1 + s_0) = 1 + t_0 + t_0 + t_0 s_0 = 1 + 1 = 0$, using $t_0 s_0 = 1$. Finally,

$$\begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & t_0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & s_0 t_0 + 1 \\ 0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix},$$

since $s_0 t_0 + 1 = p_0 + 1 = p_1 = s_1 t_1$ in characteristic 2. □

5.4. **Thompson's group V in a Leavitt corner.** Thompson's group V is the group of homeomorphisms of the boundary $\{0,1\}^{\mathbb{N}}$ given by *tree tables*: a pair of ordered complete leaf sets $(\beta_1, \dots, \beta_\ell)$, $(\alpha_1, \dots, \alpha_\ell)$ with the same number ℓ of leaves, acting by the prefix substitutions $\beta_i x \mapsto \alpha_i x$; each such map carries the cylinder of β_i bijectively onto the cylinder of α_i and is a homeomorphism of the Cantor set. Two tables define the same element exactly when they have a common refinement, refinement being simultaneous subdivision of a source leaf and its target leaf.

Leavitt/FamilyVEmbedding
`vEmbeddingOfFamily`
`vEmbeddingOfFamily_injective`
`vEmbeddingOfFamily_tableEquiv`
`tableSum_eq_of_mapsCylinder`
`tableSum_mul`
`wordS_injective`

Proposition 5.10 (Tree-table embedding)**.** *Let A be a nontrivial unital ring carrying a binary Leavitt family. For $g \in V$ given by a table $((\beta_i), (\alpha_i))$, put*

Leavitt/ThompsonVEmbedding
`vEmbedding`
`vEmbedding_injective`
`isComplete_of_covers`
Leavitt/ThompsonV
`tableEquiv`
Over any nontrivial ring carrying a binary family.

$$U_g = \sum_{i=1}^{\ell} \alpha_i\, \beta_i^* \ \in\ A. \tag{23}$$

Then U_g is well defined (independent of the table), $U_g U_h = U_{gh}$, $U_1 = 1$, each U_g is a unit with $U_g^{-1} = U_{g^{-1}}$, and $g \mapsto U_g$ is an injective homomorphism $V \hookrightarrow A^\times$.

Proof. Refining a table replaces a summand $\alpha\beta^*$ by $\alpha 0\,(\beta 0)^* + \alpha 1\,(\beta 1)^* = \alpha(s_0t_0 + s_1t_1)\beta^* = \alpha\beta^*$; since any two tables of the same element admit a common refinement, U_g is well defined. For multiplicativity, choose tables of g and h such that the source leaf set of g equals the target leaf set of h (refine both accordingly); then

$$U_gU_h = \sum_{i,j} \alpha_i\beta_i^*\,\gamma_j\delta_j^* = \sum_i \alpha_i\delta_i^* = U_{gh},$$

using $\beta_i^*\gamma_j = \delta_{ij}$ for the common leaf set, and the composed table represents gh. Also $U_1 = \sum_i \alpha_i\alpha_i^* = 1$ by Lemma 5.3, so each U_g is a unit with inverse $U_{g^{-1}}$. For injectivity, suppose $U_g = 1$ with a table $\big((\beta_i),(\alpha_i)\big)$. Multiplying $\sum_i \alpha_i\beta_i^* = 1$ on the right by β_j gives $s_{\alpha_j} = s_{\beta_j}$ in A for each j—the products $\beta_i^*\beta_j = \delta_{ij}$ collapse the sum. Equality of these elements forces equality of the binary words. If neither word is a prefix of the other, multiplying $s_{\alpha_j} = s_{\beta_j}$ on the left by $s_{\alpha_j}^*$ gives $1 = s_{\alpha_j}^* s_{\beta_j} = 0$, contradicting the nontriviality of A. If, say, $\beta_j = \alpha_j\gamma$ with γ nonempty, the same multiplication gives $1 = s_\gamma$; choosing $i \in \{0,1\}$ different from the first letter of γ then gives $t_i = t_i\,s_\gamma = 0$, whence $1 = t_is_i = 0$, again a contradiction. The case $\alpha_j = \beta_j\gamma$ is symmetric. Hence $\alpha_j = \beta_j$ as words for every j, and g is the identity substitution. This is the standard realization of the Higman–Thompson groups inside Leavitt algebras and their Cuntz-algebra completions; compare [25, 26]. □

Lemma 5.11 (Corner embedding)**.** *Let A be a nontrivial unital ring carrying a binary Leavitt family. For $g \in V$ put* *Leavitt/LeavittCorner* `cornerHom`

$$\rho(g) = p_0 + s_1U_gt_1 \;\in\; A.$$

`cornerHom_s0` `t0_cornerHom`

Then $\rho : V \to A^\times$ is an injective group homomorphism, and for all $g \in V$: `t1_cornerHom_s1`

$$t_1\rho(g)s_1 = U_g, \qquad \rho(g)\,s_0 = s_0, \qquad t_0\,\rho(g) = t_0.$$

Proof. All computations take place in the subring generated by the Leavitt family, using $t_1s_0 = 0$, $t_0s_1 = 0$, $t_1s_1 = t_0s_0 = 1$, $p_0s_0 = s_0$, $t_0p_0 = t_0$, $p_0s_1 = 0$, $t_1p_0 = 0$. Multiplicativity:

$$\begin{aligned}\rho(g)\rho(h) &= (p_0 + s_1U_gt_1)(p_0 + s_1U_ht_1)\\ &= p_0^2 + p_0s_1U_ht_1 + s_1U_gt_1p_0 + s_1U_gt_1s_1U_ht_1\\ &= p_0 + s_1U_gU_ht_1 = \rho(gh),\end{aligned}$$

by Proposition 5.10. Also $\rho(1) = p_0 + p_1 = 1$, so $\rho(g) \in A^\times$ with inverse $\rho(g^{-1})$. The three displayed identities are immediate: $t_1\rho(g)s_1 = t_1p_0s_1 + t_1s_1U_gt_1s_1 = U_g$; $\rho(g)s_0 = p_0s_0 + s_1U_gt_1s_0 = s_0$; $t_0\rho(g) = t_0p_0 + t_0s_1U_gt_1 = t_0$. Injectivity: $\rho(g) = 1$ forces $U_g = t_1\rho(g)s_1 = t_1s_1 = 1$, hence $g = 1$ by Proposition 5.10. □

Thus ρ realizes V inside the corner $p_0 + s_1(\cdot)t_1$, acting as the identity on the complementary corner $s_0(\cdot)t_0$: this complementarity is the algebraic seed of the centralizer relation $[K, J] = 1$.

The non-LEF input of the whole paper is the following finite obstruction. Its two relators are the standard defining relators of Thompson's group F, used here purely as relators.

Sofic/ThompsonFObstruction
`not_isLEF_of_two_relations`
`finite_commute_of_two_relations`
`conjugacy_relation_all`

Lemma 5.12 (Two-relator obstruction)**.** *Let a, b be elements of a group satisfying*

$$[\,ab^{-1},\, a^{-1}ba\,] = 1 \qquad \text{and} \qquad [\,ab^{-1},\, a^{-2}ba^2\,] = 1. \tag{24}$$

If a and b do not commute, the group is not LEF.

Proof. We first show that in a *finite* group the relations (24) force $[a, b] = 1$. Put $c_n = a^{-n}ba^n$, so $c_0 = b$ and $a^{-1}c_na = c_{n+1}$. A relation $[ab^{-1}, c_n] = 1$ rearranges: $ab^{-1}c_nba^{-1} = c_n$ is exactly

$$b^{-1}c_nb = a^{-1}c_na = c_{n+1}.$$

Thus the two relations (24) are the cases $n = 1, 2$ of the family (C_n): $b^{-1}c_nb = c_{n+1}$. Conjugating (C_1) by a^{-n} gives, since $a^{-n}ba^n = c_n$,

$$(A_n): \qquad c_n^{-1}\, c_{n+1}\, c_n = c_{n+2} \qquad (n \geq 1).$$

Now (C_n) holds for every $n \geq 1$, by strong induction with base $(C_1), (C_2)$: using (A_{n-1}), then (C_{n-1}) and (C_n), then (A_n),

$$b^{-1}c_{n+1}b = b^{-1}(c_{n-1}^{-1}c_nc_{n-1})b = c_n^{-1}\, c_{n+1}\, c_n = c_{n+2}.$$

In a finite group $a^N = 1$ for some $N \geq 1$, so $c_N = c_0 = b$, and (C_N) reads $b^{-1}bb = c_{N+1} = c_1$; that is, $b = a^{-1}ba$, so $[a, b] = 1$.

Now suppose the ambient group were LEF. Take F to consist of 1, $[a, b]$, and every product of at most twenty letters drawn from $\{a^{\pm 1}, b^{\pm 1}\}$—a finite set containing a, b, and all partial products needed to evaluate both relators and $[a, b]$ letter by letter. A local embedding $\varphi : F \to P$ into a finite group is multiplicative on these partial products, so $\varphi(a), \varphi(b)$ satisfy (24) in P, whence $[\varphi(a), \varphi(b)] = 1_P$ by the previous paragraph; but $\varphi([a, b]) = [\varphi(a), \varphi(b)] = 1_P = \varphi(1)$ and injectivity force $[a, b] = 1$, a contradiction. $\square$

Leavitt/ThompsonVWitness
`thompsonV_not_isLEF`
Leavitt/ThompsonWitness
`not_isLEF_cornerWitnessSubgroup`
Criterion/Scheme
`not_isLEF_cornerSubgroup`
Proved for V itself, with no simplicity or presentation input.

Proposition 5.13. *Thompson's group V is not LEF.*

Proof. Let $x_0, x_1 \in V$ be the tree tables

$$x_0: \ \{\, 00 \mapsto 0,\ 01 \mapsto 10,\ 1 \mapsto 11\,\},$$
$$x_1: \ \{\, 0 \mapsto 0,\ 100 \mapsto 10,\ 101 \mapsto 110,\ 11 \mapsto 111\,\}$$

(x_1 acts as x_0 inside the cylinder 1 and as the identity outside it; Figure 4 draws the pair of trees of x_0). Both relators (24) hold for $(a, b) = (x_0, x_1)$: each side is a composition of finitely many prefix substitutions, and refining all tables to a common leaf depth and comparing leaf images is a finite computation. The pair does not commute: on the point 011^∞ of the Cantor set,

$$x_0x_1(011^\infty) = x_0(011^\infty) = 101^\infty, \qquad x_1x_0(011^\infty) = x_1(101^\infty) = 1101^\infty.$$

Lemma 5.12 applies. $\square$

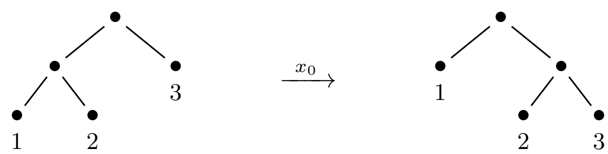


Figure 4. The tree table of x_0. Matching leaf labels give the prefix substitution: the source leaves $00, 01, 1$ map to the target leaves $0, 10, 11$, so $x_0 = \{00 \mapsto 0,\ 01 \mapsto 10,\ 1 \mapsto 11\}$. The element x_1 applies the same table inside the cylinder 1 and fixes its complement.

Remark 5.14 (The classical facts about V, and the classical route)**.** The group V is infinite, simple, and finitely presented [17, 8], and 2-generated [7]; it is perfect, since $[V, V]$ is a nontrivial normal subgroup of a simple group. These classical facts give a second proof of Proposition 5.13: a finitely presented LEF group is residually finite (Lemma 2.8), while an infinite simple group is not. Nothing in the main line of this paper consumes them—the proofs of Theorems A–D use only Lemma 5.12 and the explicit pair above—but they are the conceptual frame: the witness subgroup built in Section 6 is the image of the standard copy of Thompson's F inside the corner copy of V.

6. The compression scheme in adjacent ranks

This section presents the algebraic heart of the paper in its natural generality. The matrix identities below are verifiable by finite computation from the two Leavitt relations, over an arbitrary ambient ring. Beyond them, the section uses only the facts about Thompson's group V established in Section 5 and the two membership hypotheses isolated in Theorem 6.10, the latter discharged explicitly in Section 7 and K-theoretically in Section 8.

Throughout, A is a nontrivial associative unital ring carrying a binary Leavitt family (s_0, s_1, t_0, t_1) in the sense of Definition 5.1—in particular A may be, but need not contain a copy of, the Leavitt algebra $L = L_k(1, 2)$—and we retain the notation $p_0 = s_0 t_0$, $p_1 = s_1 t_1$. Fix $m \geq 2$ and put

$$n = m + 1, \qquad G = \mathrm{EL}_n(A), \qquad \Gamma = \{\hat{\gamma} = \mathrm{diag}(\gamma, 1) \mid \gamma \in \mathrm{EL}_m(A)\} \ \leq\ G.$$

Indeed $\Gamma \leq G$: an elementary matrix in the first m coordinates is an elementary matrix of rank n. The group $\Gamma \cong \mathrm{EL}_m(A)$ is infinite, since $a \mapsto x_{12}(a)$ is injective and A is infinite by Lemma 5.2.

We use the left comb (22) with n leaves:

$$\alpha_i = s_0^{i-1} s_1 \quad (1 \leq i \leq m), \qquad \alpha_n = s_0^m, \tag{25}$$

a complete leaf set, so that $\alpha_i^* \alpha_j = \delta_{ij}$ and $\sum_{i=1}^n \alpha_i \alpha_i^* = 1$ by Lemma 5.3.

6.1. **The comb compressor.** Define the column $b \in A^{m\times 1}$, the row $c \in A^{1\times m}$, and the matrices $u, u' \in M_n(A)$:

$$b = \begin{pmatrix} s_1\alpha_1^* \\ \vdots \\ s_1\alpha_m^* \end{pmatrix}, \qquad c = \begin{pmatrix} \alpha_1 t_1 & \cdots & \alpha_m t_1 \end{pmatrix},$$
$$u = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix}, \qquad u' = \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix}. \tag{26}$$

Leavitt/LeavittMatrixCompression
`matrixCompressionHom`
`matrixCompressionUnit`
`matrixCompression_recover`
Leavitt/RankFourCompressors
`compressor`
`compressor_val`

Lemma 6.1. *The following identities hold in A:*

$$bc = p_1 I_m, \quad cb = 1 - \alpha_n\alpha_n^*, \quad t_0 b = 0, \quad cs_0 = 0, \quad b\alpha_n = 0, \quad \alpha_n^* c = 0.$$

Consequently $uu' = u'u = I_n$; thus $u \in \mathrm{GL}_n(A)$ with $u^{-1} = u'$.

Proof. Entrywise: $(bc)_{ij} = s_1\alpha_i^*\alpha_j t_1 = \delta_{ij} s_1 t_1 = \delta_{ij} p_1$. Next, $cb = \sum_{i=1}^m \alpha_i t_1 s_1 \alpha_i^* = \sum_{i=1}^m \alpha_i\alpha_i^* = 1 - \alpha_n\alpha_n^*$ by Lemma 5.3. The entries of $t_0 b$ are $t_0 s_1 \alpha_i^* = 0$; those of cs_0 are $\alpha_i t_1 s_0 = 0$; those of $b\alpha_n$ are $s_1\alpha_i^*\alpha_n = 0$ and of $\alpha_n^* c$ are $\alpha_n^*\alpha_i t_1 = 0$, by orthogonality of distinct leaves. Then

$$uu' = \begin{pmatrix} s_0 t_0 I_m + bc & b\alpha_n \\ \alpha_n^* c & \alpha_n^*\alpha_n \end{pmatrix} = \begin{pmatrix} (p_0 + p_1) I_m & 0 \\ 0 & 1 \end{pmatrix} = I_n,$$

and likewise

$$u'u = \begin{pmatrix} t_0 s_0 I_m & t_0 b \\ c s_0 & cb + \alpha_n\alpha_n^* \end{pmatrix} = I_n. \qquad \square$$

Leavitt/RankFourCompressors
`compressor_conjugation`
`compressionEnd_injective`

Proposition 6.2 (Compression)**.** *For every $\gamma \in \mathrm{EL}_m(A)$,*

$$u\,\hat\gamma\,u^{-1} = p_1 \widehat{I_m + s_0}\gamma t_0\,, \tag{27}$$

and on elementary generators $u\,\hat x_{ij}(a)\,u^{-1} = \hat x_{ij}(s_0 a t_0)$ for $i \neq j \leq m$. Consequently

$$K := u\Gamma u^{-1} = \langle\, \hat x_{ij}(s_0 a t_0) \;:\; a \in A,\; i \neq j \leq m \,\rangle \;\leq\; \Gamma, \tag{28}$$

and conjugation by u is an isomorphism $\Gamma \xrightarrow{\cong} K$; in particular $K \cong \mathrm{EL}_m(A)$.

Proof. Block multiplication using Lemma 6.1:

$$u\,\hat\gamma\,u^{-1} = \begin{pmatrix} s_0\gamma & b \\ 0 & \alpha_n^* \end{pmatrix}\begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix} = \begin{pmatrix} s_0\gamma t_0 + bc & b\alpha_n \\ \alpha_n^* c & \alpha_n^*\alpha_n \end{pmatrix}$$
$$= \begin{pmatrix} s_0\gamma t_0 + p_1 I_m & 0 \\ 0 & 1 \end{pmatrix}.$$

For $\gamma = x_{ij}(a) = I_m + aE_{ij}$, the block is $p_1 I_m + s_0 t_0 I_m + s_0 a t_0 E_{ij} = I_m + s_0 a t_0 E_{ij} = x_{ij}(s_0 a t_0)$. Since Γ is generated by the $\hat x_{ij}(a)$ and conjugation is a group isomorphism onto its image, K is generated by the displayed elements, which lie in Γ. $\square$

The compressed copy K sits inside Γ with all matrix entries confined to the corner $s_0(\cdot)t_0 = p_0 A p_0$ of A; the complementary corner $p_1 A p_1$ is untouched, and this is the room in which the centralizing subgroup J will live.

6.2. **The involution z and generation.** Let $z \in M_n(A)$ agree with the identity matrix outside the coordinates 1 and n, where it is

(29)
$$z\big|_{\{1,n\}} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}; \quad \text{that is,} \quad z = I_n + (p_0 - 1)E_{11} + s_1 E_{1n} + t_1 E_{n1} - E_{nn}.$$

Lemma 6.3. *$z^2 = I_n$; in particular $z \in \mathrm{GL}_n(A)$. Moreover z centralizes K.*

Leavitt/Leavitt `z_sq` Leavitt/RankFourCompressors `involution_sq` Leavitt/DiagonalCornerCompression `matrixCompression_commutes_firstDiagonalCorner`

Proof. On the coordinates $\{1, n\}$,
$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = \begin{pmatrix} p_0 + p_1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$
using $p_0 s_1 = 0$, $t_1 p_0 = 0$, $t_1 s_1 = 1$; the remaining coordinates are fixed. For the centralizing statement, put $r = s_0 a t_0$ and note
$$p_0 r = r = r p_0, \qquad t_1 r = 0, \qquad r s_1 = 0.$$
Let $i \neq j \leq m$. If $1 \notin \{i, j\}$, then z acts as the identity on the coordinates i, j and $z\,\hat{x}_{ij}(r)\,z = \hat{x}_{ij}(r)$. If $i = 1$: the matrix rE_{1j} satisfies $z(rE_{1j})z = (p_0 r E_{1j} + t_1 r E_{nj})z = rE_{1j}z = rE_{1j}$, since the j-th row of z is e_j^{T} for $2 \leq j \leq m$. If $j = 1$: $z(rE_{i1})z = rE_{i1}z = rp_0 E_{i1} + rs_1 E_{in} = rE_{i1}$. In each case $z\,\hat{x}_{ij}(r)\,z^{-1} = \hat{x}_{ij}(r)$, and these elements generate K by (28). □

Define the second compressor

(30)
$$v = zu.$$

Then, by Proposition 6.2 and Lemma 6.3,

(31)
$$v\Gamma v^{-1} = z(u\Gamma u^{-1})z^{-1} = zKz^{-1} = K = u\Gamma u^{-1}:$$

the two compressors have one common image. The quotient $vu^{-1} = z$ is precisely what generates the missing last row and column:

Proposition 6.4 (Generation)**.** *The subgroup of $\mathrm{GL}_n(A)$ generated by Γ and z contains $\mathrm{EL}_n(A)$. Consequently, if u and z lie in $\mathrm{EL}_n(A)$, then*
$$\mathrm{EL}_n(A) = \langle \Gamma, z \rangle = \langle \Gamma, u, v \rangle.$$

Leavitt/RankFourCompressors `coreEmbedding_compressorSet_generate`

Proof. For $2 \leq j \leq m$ and $a \in A$, we claim
$$z\,\hat{x}_{1j}(s_1 a)\,z = x_{nj}(a), \qquad z\,\hat{x}_{j1}(a t_1)\,z = x_{jn}(a).$$
For the first: $z(s_1 a E_{1j})z = (p_0 s_1 a E_{1j} + t_1 s_1 a E_{nj})z = aE_{nj}z = aE_{nj}$, using $p_0 s_1 = 0$, $t_1 s_1 = 1$, and that the j-th row of z is e_j^{T}. For the second: $z(a t_1 E_{j1})z = a t_1 E_{j1} z = a t_1 p_0 E_{j1} + a t_1 s_1 E_{jn} = aE_{jn}$, using $t_1 p_0 = 0$, $t_1 s_1 = 1$, and that the j-th column of z is e_j. Thus $\langle \Gamma, z \rangle$ contains $x_{nj}(a)$ and $x_{jn}(a)$ for all $2 \leq j \leq m$ and $a \in A$, as well as all $\hat{x}_{ij}(a) = x_{ij}(a)$ with $i \neq j \leq m$. The remaining elementary matrices are supplied by the Steinberg relation, using $m \geq 2$ so that the coordinate $2 \notin \{1, n\}$ exists:
$$x_{1n}(a) = [\,x_{12}(a),\, x_{2n}(1)\,], \qquad x_{n1}(a) = [\,x_{n2}(a),\, x_{21}(1)\,].$$

Hence $\langle \Gamma, z\rangle \supseteq \mathrm{EL}_n(A)$. If moreover $u, z \in \mathrm{EL}_n(A)$, then $\langle \Gamma, z\rangle \subseteq \mathrm{EL}_n(A)$, giving the first equality; and since $z = vu^{-1}$, also $\langle \Gamma, u, v\rangle \supseteq \langle \Gamma, z\rangle = \mathrm{EL}_n(A) \supseteq \langle \Gamma, u, v\rangle$. □

6.3. **The commuting corner witness.** For $g \in V$, recall $\rho(g) = p_0 + s_1 U_g t_1 \in A^\times$ from Lemma 5.11, and define

$$j(g) = \mathrm{diag}\big(\rho(g), 1, \ldots, 1\big) \in \mathrm{GL}_n(A), \qquad J = \big\langle\, j(x_0),\ j(x_1)\,\big\rangle,$$

where $x_0, x_1 \in V$ are the two tree tables of Proposition 5.13. Thus J is a two-generator subgroup of the corner copy $j(V)$ of V; it is the image of the standard copy of Thompson's F. Two lemmas place its generators inside Γ by explicit commutator certificates.

For prefix-incomparable words v, w, write

$$T(v, w) = s_v t_w + s_w t_v + \big(1 - p_v - p_w\big) \in A^\times,$$

the *cylinder transposition*: it is the table unit U_g of the element of V exchanging the cylinders v and w and fixing everything else, and $T(v, w)^2 = 1$.

Leavitt/ThompsonWitness
`cylinderSwap_is_commutator`
`cylinderSwap_mem_commutator`

Lemma 6.5 (Transpositions are single commutators)**.** *Let v and w be prefix-incomparable. Then*

$$T(v, w) = \big[\, T(v0, v1)\, T(w0, w1),\ \ T(v0, w0)\,\big]$$

is a single commutator of units of A.

Proof. All five transpositions are table units, and multiplication of table units composes the underlying prefix maps (Proposition 5.10), so the identity may be verified at the level of maps of $\{0,1\}^{\mathbb{N}}$. Splitting the two cylinders into their children, $T(v, w)$ acts as the double transposition $(v0\ w0)(v1\ w1)$ of the four pairwise-incomparable child cylinders, fixing everything else; and in the symmetric group of these four cylinders, with $x = (v0\ v1)(w0\ w1)$ and $y = (v0\ w0)$,

$$xyx^{-1} = (v1\ w1), \qquad \text{so} \qquad [x, y] = (v1\ w1)(v0\ w0) = T(v, w)\text{'s action}.$$

All the maps involved fix the complement of the four child cylinders, so the identity holds outright. □

Leavitt/ThompsonWitness
`rootRotation`
`cornerWitnessSubgroup`
`cornerWitnessSubgroup_le_commutator`

Lemma 6.6 (Commutator certificates for the witness pair)**.** *The table units of x_0 and x_1 factor as*

$$U_{x_0} = T(0, 10)\ T(0, 11)\ T(0, 1), \qquad U_{x_1} = \kappa_{[1]}(U_{x_0}),$$

where $\kappa_{[1]}(u) = p_0 + s_1 u t_1$ is the corner insertion at the cylinder 1—multiplicative and unital by the computation of Lemma 5.11. Consequently U_{x_0} and U_{x_1} are products of three commutators of units of A each, and so are $\rho(x_0)$ and $\rho(x_1)$.

Proof. For the factorization, compose the three transpositions as prefix maps, rightmost first: a point $00v$ goes to $10v$, stays at $10v$, then goes to $0v$; a point $01v$ goes to $11v$, then to $0v$, then to $10v$; a point $1v$ goes to $0v$, then to $11v$, and stays. The composite is exactly the table $\{00 \mapsto 0,\ 01 \mapsto 10,\ 1 \mapsto 11\} = x_0$, and the displayed unit identity follows through Proposition 5.10. The identity $U_{x_1} = \kappa_{[1]}(U_{x_0})$ is the definition of x_1 (act as x_0 inside the cylinder 1, and note $1-p_1 = p_0$). Each transposition is a single commutator by Lemma 6.5, so U_{x_0} is a product of three commutators, and $\kappa_{[1]}$, being unital and multiplicative, carries each commutator certificate to a commutator certificate. Finally $\rho(g) = p_0 + s_1 U_g t_1 = \kappa_{[1]}(U_g)$ itself, so the same transport gives the claim for $\rho(x_0)$ and $\rho(x_1)$. ☐

Proposition 6.7. *$J \le \Gamma$: both $j(x_0)$ and $j(x_1)$ lie in $\mathrm{EL}_m(A)$, embedded in the first m coordinates. The subgroup J is generated by two elements, and it is not LEF.*

Leavitt/UniversalRankFour
`witnessEmbedding`
`witnessEmbedding_injective`
Leavitt/DiagonalElementary
`firstDiagonalUnit_mem_of_mem_commutator`
Leavitt/ThompsonWitness
`not_isLEF_cornerWitnessSubgroup`

Proof. By Lemma 6.6, $\rho(x_0)$ and $\rho(x_1)$ are products of commutators of units of A; the Whitehead commutator lemma (Lemma 5.5, with $m \ge 2$) places $\mathrm{diag}(\rho(x_i), 1, \dots, 1)$ in $\mathrm{EL}_m(A)$. Hence $J \le \Gamma$. The composite $g \mapsto j(g)$ is an injective homomorphism (Lemma 5.11), so $j(x_0), j(x_1)$ satisfy the two relators (24) and do not commute, exactly as x_0, x_1 do in the proof of Proposition 5.13; Lemma 5.12 shows J is not LEF. ☐

Remark 6.8 (The full corner copy of V)**.** Granting the classical perfectness of V (Remark 5.14), the same Whitehead argument places *every* $j(g)$, $g \in V$, in $\mathrm{EL}_m(A)$, so the full corner copy $j(V) \cong V$ is a subgroup of Γ, over every ring carrying a binary Leavitt family. The main theorems use only the two-generator witness J, whose certificates are explicit.

Proposition 6.9. $[K, J] = 1$ *and* $K \cap J = \{1\}$*. Consequently* $KJ \cong K \times J$*.*

Criterion/Scheme
`commute_compressed_corner`
`compressed_inf_corner`
Leavitt/GeneralCornerTheorem
`corner_not_isSofic`
`compressionSetup`
`coreEmbedding_compressorSet_generate`
`witnessEmbedding`
`corner_not_isSofic_of_charTwo`
Leavitt/GeneralScheme
`uUnit`
`zUnit`
`uMatrix_mul_uPrimeMatrix`
Criterion/CompressionSetup
`CompressionSetup`

Proof. Centralizing. Conjugating an elementary matrix by a diagonal matrix multiplies its coefficient: $j(g)\,\hat{x}_{il}(r)\,j(g)^{-1} = \hat{x}_{il}(\rho(g)^{[i=1]}\,r\,\rho(g^{-1})^{[l=1]})$, where the exponent $[i=1]$ is 1 if $i = 1$ and 0 otherwise. For $r = s_0 a t_0$, Lemma 5.11 gives $\rho(g) r = \rho(g) s_0\, a t_0 = s_0 a t_0 = r$ and $r\rho(g^{-1}) = s_0 a\, t_0 \rho(g^{-1}) = s_0 a t_0 = r$. Hence $j(g)$ commutes with every generator $\hat{x}_{il}(s_0 a t_0)$ of K from (28).

Trivial intersection. Suppose $j(g) \in K$. By (27), every element of K has the form $p_1 \widehat{I_m + s_0}\gamma t_0$ with $\gamma \in \mathrm{EL}_m(A)$; comparing $(1,1)$-entries, $\rho(g) = p_1 + s_0 \gamma_{11} t_0$. Sandwiching between t_1 and s_1: on one side, $t_1(\rho(g) - 1)s_1 = t_1(s_1 U_g t_1 - p_1)s_1 = U_g - 1$, using $t_1 p_0 = 0$; on the other, $t_1(p_1 + s_0\gamma_{11}t_0 - 1)s_1 = t_1(s_0\gamma_{11}t_0 - p_0)s_1 = 0$, using $t_1 s_0 = 0$ and $t_1 p_0 = 0$. Hence $U_g = 1$, so $g = 1$ by Proposition 5.10, and $j(g) = 1$.

The two computations quantify over every $g \in V$, so they prove the claims for the full corner copy $j(V)$; the witness $J = \langle j(x_0), j(x_1)\rangle \le j(V)$ inherits both. Since K and J commute and intersect trivially, the multiplication map $K \times J \to KJ$ is an isomorphism. ☐

6.4. **The Leavitt-corner theorem.**

Theorem 6.10 (Leavitt-corner theorem)**.** *Let A be a nontrivial associative unital ring carrying a binary Leavitt family, let $m \ge 2$, and put $n = m + 1$. Let $u, z \in \mathrm{GL}_n(A)$ be the matrices (26) and (29). Assume:*

Every adjacent pair $(m, m+1)$ with $m \ge 2$, at exactly the printed hypotheses; finite generation and countability derived from (T).

(i) $\mathrm{EL}_m(A)$ *and* $\mathrm{EL}_n(A)$ *have property* (T)*;*
(ii) $u, z \in \mathrm{EL}_n(A)$*; this holds, for instance, whenever* $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$*.*

Then $\mathrm{EL}_n(A)$ *is not sofic.*

Proof. Put $G = \mathrm{EL}_n(A)$ and let Γ, u, z, $v = zu$, K, J be as above. By hypothesis (i), G and $\Gamma \cong \mathrm{EL}_m(A)$ are property-(T) groups, hence finitely generated, and Γ is infinite. By hypothesis (ii) and Proposition 6.4, $G = \langle \Gamma, u, v\rangle = \langle \Gamma, Q\rangle$ with $Q = \{u, v\} \subseteq G$. By Proposition 6.2 and (31),

$$u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma,$$

so both elements of Q conjugate Γ into itself. Fix $q_0 = u$, so $K = q_0 \Gamma q_0^{-1}$. By Propositions 6.7 and 6.9, the subgroup $J \leq \Gamma$ is generated by two elements, satisfies $[K, J] = 1$ and $K \cap J = \{1\}$, and is not LEF. If G were sofic, Corollary 4.2 (Theorem D) would force J to be LEF, a contradiction. □

Endpoint/ManuscriptStatements
`cor_fgring_printed`
`cor_fgring_exact`
`cor_fgring_countableFree`
Criterion/CriterionAssembly
`not_isSofic_of_not_isLEF`
Leavitt/FamilyRankFour
`compressionSetup`
`ambient_not_isSofic`
`elementary_not_isSofic`
`units_not_isSofic`
`gl_not_isSofic`
Memberships discharged rather than assumed; every $r \geq 2$*,* $A^\times$*, and every* GL_r*.*

Corollary 6.11. *Let A be a nontrivial finite-type algebra over a finite field carrying a binary Leavitt family, and let $m \geq 2$, $n = m+1$. If $u, z \in \mathrm{EL}_n(A)$—for instance, if $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$—then $\mathrm{EL}_n(A)$ is an infinite, finitely generated, property-(T) group that is not sofic.*

Proof. Theorem 5.7 gives property (T) for $\mathrm{EL}_3(A)$, and the elementary rank equivalence (Proposition 5.8) transports it to $\mathrm{EL}_m(A)$ and $\mathrm{EL}_n(A)$ for every $m \geq 2$, supplying hypothesis (i) of Theorem 6.10; hence $\mathrm{EL}_n(A)$ is not sofic and has property (T). It is finitely generated by Lemma 5.6 (here $n \geq 3$, and a finite-type algebra over a finite field is a finitely generated ring), and infinite, since $a \mapsto x_{12}(a)$ embeds the ring A, which is infinite by Lemma 5.2. Over an arbitrary finitely generated unital ring and $m \geq 3$, the same conclusion holds verbatim, granting the full theorem of [13] in place of Theorem 5.7. □

The corollary's reach is wider than its statement, in the following precise sense. Nonsoficity spreads over the same algebra—by the self-similarity transport of Section 5, and, for $A^\times$ and $\mathrm{GL}_r(A)$, because a subgroup of a sofic group is sofic—to every elementary group of rank at least two, to the unit group, and to every positive-rank $\mathrm{GL}_r(A)$; the formal development records this at every target, with the countability of A derived from finite type over the finite field rather than assumed. The full four-property profile and the equality $\mathrm{GL}_r = \mathrm{EL}_r$ are finer statements, established for $L = L_k(1, 2)$ by the separate arguments of Section 8.

Remark 6.12 (The trust surface of the scheme)**.** The matrix identities of this section are verifiable by finite computation from the two Leavitt relations; beyond them, the section inputs only the Whitehead lemma, the two-relator obstruction and the explicit commutator certificates of Section 5, and hypotheses (i) and (ii) of Theorem 6.10. The witness is elementary for every ambient ring: Proposition 6.7 uses only Lemmas 6.5–6.6 and the Whitehead lemma, and no perfectness theorem; with the classical perfectness of V, the full corner copy of V is elementary as well (Remark 6.8). The entire membership burden therefore falls on hypothesis (ii), which concerns exactly two matrices, u and z: Section 7 discharges it by explicit elementary factorizations in characteristic two, and Section 8 discharges it over every finite field via $K_1(L) = 0$ and the GE property, with the K_1 input available in elementary form (Remark 8.4).

7. The explicit realization: no K-theory

In this section $k = \mathbb{F}_2$ and $R = L_{\mathbb{F}_2}(1, 2)$; the matrix identities and the argument extend without change to any finite field of characteristic two. Fix $m \geq 3$, put $n = m + 1$, and let $G = \mathrm{EL}_n(R)$, Γ, u, z, v, K, J be as in Section 6 with $A = R$. We now exhibit the two matrices u and z as explicit products of elementary matrices, discharging hypothesis (ii) of Theorem 6.10 with no K-theoretic input.

Proposition 7.1 (Explicit elementary factorization of the compressor)**.** *For $1 \leq i \leq m$, put*

$$U_i = x_{ni}(1 + t_0)\, x_{in}(1)\, x_{ni}(1 + s_0)\, x_{in}(t_0) \;\in\; \mathrm{EL}_n(R).$$

Then

$$u = U_m\, U_{m-1} \cdots U_1.$$

In particular $u \in \mathrm{EL}_n(R)$, as a product of $4m$ elementary matrices.

Proof. By Lemma 5.9(b), applied in the pair of coordinates (i, n), the matrix U_i agrees with the identity outside the coordinates i and n and equals $\left(\begin{smallmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{smallmatrix}\right)$ on them; explicitly,

$$U_i = I_n + (s_0 + 1)E_{ii} + s_1 t_1 E_{in} + (t_0 + 1)E_{nn},$$

signs being immaterial in characteristic two. Put $P_k = U_k U_{k-1} \cdots U_1$. We claim, by induction on k, that

$$P_k = \sum_{i \leq k} (s_0 E_{ii} + s_1 t_1 t_0^{i-1} E_{in}) + \sum_{k < i \leq m} E_{ii} + t_0^k E_{nn}.$$

For $k = 1$ this is the displayed form of U_1. For the inductive step, compute the rows of $P_{k+1} = U_{k+1} P_k$. For $i \notin \{k+1, n\}$, the i-th row of U_{k+1} is e_i^{T}, so the i-th row of P_{k+1} equals that of P_k. The $(k+1)$-st row of U_{k+1} is $s_0 e_{k+1}^{\mathsf{T}} + s_1 t_1 e_n^{\mathsf{T}}$; since the $(k+1)$-st row of P_k is e_{k+1}^{T} and the n-th row of P_k is $t_0^k e_n^{\mathsf{T}}$, the $(k+1)$-st row of P_{k+1} is $s_0 e_{k+1}^{\mathsf{T}} + s_1 t_1 t_0^k e_n^{\mathsf{T}}$. The n-th row of U_{k+1} is $t_0 e_n^{\mathsf{T}}$, so the n-th row of P_{k+1} is $t_0^{k+1} e_n^{\mathsf{T}}$. This is the claimed form for $k + 1$.

Finally, for the left comb (25), $s_1 t_1 t_0^{i-1} = s_1 (s_0^{i-1} s_1)^* = s_1 \alpha_i^*$ for $1 \leq i \leq m$, and $t_0^m = \alpha_n^*$. Hence

$$P_m = \sum_{i=1}^{m} (s_0 E_{ii} + s_1 \alpha_i^* E_{in}) + \alpha_n^* E_{nn} = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix} = u,$$

by (26). $\square$

Leavitt/GeneralRankWords
piece
piece_val
comb
comb_mem
comb_val
combMatrix
partialComb_val
Leavitt/RankFourCompressors
compressor
compressorMatrix
compressorPiece
compressorPiece_val
compressor_val
Every rank; the rank-four instance, valid in every characteristic, is kept for the spine.
Leavitt/GeneralRankWords
involutionWord
involutionWord_mem
involutionWord_val
involutionWord_sq
Leavitt/RankFourCompressors
involution
involutionMatrix
involution_sq
Leavitt/RawSwapCompressors
rawInvolutionWord_val
Every rank; the rank-four instance is valid in every characteristic.

Lemma 7.2 (Explicit elementary factorization of the involution)**.**

$$z = x_{1n}(s_1)\, x_{n1}(t_1)\, x_{1n}(s_1) \;\in\; \mathrm{EL}_n(R).$$

Proof. Lemma 5.9(a), applied in the pair of coordinates $(1, n)$, produces exactly the matrix z of (29). □

Endpoint/MainResults
`universalLeavittEL4_not_isSofic`
`universalLeavitt_profile`

Theorem 7.3 (The explicit spine)**.** *For every $m \geq 3$, the group $\mathrm{EL}_{m+1}(R)$ over $R = L_{\mathbb{F}_2}(1, 2)$ is infinite, finitely generated, has property (T), and is not sofic.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is finitely generated (by s_0, s_1, t_0, t_1 over the prime field). Proposition 7.1 and Lemma 7.2 verify $u, z \in \mathrm{EL}_{m+1}(R)$, and Corollary 6.11 applies. □

Proof of Theorem A. Theorem 7.3 gives every $m \geq 3$; the case $m = 3$ is the group $G = \mathrm{EL}_4(R)$. For $m \in \{1, 2\}$, the elementary rank equivalence of Proposition 5.8 is an explicit isomorphism $\mathrm{EL}_{m+1}(R) \cong \mathrm{EL}_4(R)$, and all four properties transport along it; in particular $\mathrm{EL}_2(R)$ and $\mathrm{EL}_3(R)$ have property (T) although Theorem 5.7 says nothing at rank two. The two auxiliary matrices are exhibited as explicit products of elementary matrices: u by Proposition 7.1 and z (hence $v = zu$) by Lemma 7.2. The remaining memberships, of the two witness matrices $j(x_0)$ and $j(x_1)$, follow from the explicit commutator certificates of Lemmas 6.5 and 6.6 together with the Whitehead identity of Lemma 5.5, which converts each certificate into an explicit elementary word. No K-theory is used. □

Remark 7.4 (Scope of the explicit spine)**.** The factorizations of this section are identities in characteristic two and hold over any field of characteristic two; finiteness of the field is used only through finite generation of the ring in Theorem 5.7. Thus $\mathrm{EL}_{m+1}(L_k(1, 2))$ is infinite, finitely generated, Kazhdan, and nonsofic, with all memberships explicit, for every finite field k of characteristic two and every $m \geq 3$—and then for every $m \geq 1$ by the elementary rank equivalence (Proposition 5.8). For rank $n = 3$ (that is, $m = 2$), hypothesis (i) of Theorem 6.10 for EL_2 is not supplied by Theorem 5.7; it is exactly here that the rank equivalence substitutes. Section 8 reaches the remaining groups of Theorem B—the unit group and every GL_r, over every finite field—through self-similarity.

8. Rank two and the full panorama

The compression–centralizer mechanism closes already in rank two, over a different and smaller group Γ: the full diagonal unit group, with property (T) transported through self-similarity rather than supplied by Theorem 5.7. Since all the groups in question turn out to be isomorphic, the value of the rank-two realization is not a new group but a map of the mechanism's boundary—it shows how little room the criterion actually needs. Combined with matrix self-similarity and the vanishing of K_1 (Proposition 8.2, with the elementary proof of Appendix A), this yields Theorem B.

8.1. The two-by-two realization.

Theorem 8.1 (Two-by-two Leavitt-corner theorem)**.** *Let A be a nontrivial associative unital ring carrying a binary Leavitt family. Suppose that $A^\times$ and $\mathrm{EL}_2(A)$ have property (T) and that $\mathrm{GL}_2(A) = \mathrm{EL}_2(A)$. Then $\mathrm{EL}_2(A)$ is not sofic.*

Leavitt/RankTwoCompression
rankTwo_not_isSofic
rankTwoInvolution_conj_upperNil
rankTwoInvolution_conj_lowerNil

Proof. Put $G = \mathrm{EL}_2(A) = \mathrm{GL}_2(A)$ and

$$\Gamma = \{\mathrm{diag}(a, 1) \,|\, a \in A^\times\} \le G.$$

Then $\Gamma \cong A^\times$ is a property-(T) group, hence finitely generated, and it is infinite because $\rho : V \hookrightarrow A^\times$ is injective (Lemma 5.11).

The compressor. Put

$$u = \begin{pmatrix} s_0 & p_1 \\ 0 & t_0 \end{pmatrix}, \qquad u' = \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix}.$$

Using $p_1 s_0 = s_1(t_1 s_0) = 0$, $t_0 p_1 = (t_0 s_1)t_1 = 0$, $t_0 s_0 = 1$, and $p_0 + p_1 = 1$:

$$uu' = \begin{pmatrix} s_0 t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = I_2, \qquad u'u = \begin{pmatrix} t_0 s_0 & t_0 p_1 \\ p_1 s_0 & p_1 + s_0 t_0 \end{pmatrix} = I_2,$$

so $u \in \mathrm{GL}_2(A) = G$ with $u^{-1} = u'$. For $a \in A$, put

$$\kappa(a) = p_1 + s_0 a t_0.$$

Then $\kappa(1) = 1$ and $\kappa(a)\kappa(a') = p_1 + s_0 a a' t_0 = \kappa(aa')$ by the same identities, so κ is unital and multiplicative and restricts to a group homomorphism $A^\times \to A^\times$, injective because $t_0 \kappa(a) s_0 = t_0 p_1 s_0 + t_0 s_0 a t_0 s_0 = a$; and

$$\begin{aligned} u\,\mathrm{diag}(a, 1)\,u^{-1} &= \begin{pmatrix} s_0 a & p_1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix} \\ &= \begin{pmatrix} s_0 a t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = \mathrm{diag}(\kappa(a), 1). \end{aligned}$$

Hence

$$K := u\Gamma u^{-1} = \{\mathrm{diag}(\kappa(a), 1) \,|\, a \in A^\times\} \le \Gamma.$$

The involution. Put

$$z = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}, \qquad \text{so} \qquad z^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = I_2,$$

using $p_0 s_1 = 0$, $t_1 p_0 = 0$, $t_1 s_1 = 1$; thus $z \in \mathrm{GL}_2(A) = G$ and $z^{-1} = z$. For any $c \in A$,

$$z\,\mathrm{diag}(c, 1)\,z = \begin{pmatrix} p_0 c p_0 + p_1 & p_0 c s_1 \\ t_1 c p_0 & t_1 c s_1 \end{pmatrix}. \tag{32}$$

The element $\kappa(a)$ satisfies $p_0 \kappa(a) = \kappa(a) p_0 = s_0 a t_0$, $t_1 \kappa(a) = t_1$, and $\kappa(a) s_1 = s_1$; substituting into (32) gives $z\,\mathrm{diag}(\kappa(a), 1)\,z = \mathrm{diag}(s_0 a t_0 + p_1, 1) = \mathrm{diag}(\kappa(a), 1)$. Hence z centralizes K, and $v := zu \in G$ satisfies $v\Gamma v^{-1} = zKz^{-1} = K$.

Generation. For any $a \in A$, the elements $1 + p_0 a t_1$ and $1 + s_1 a p_0$ are units of A, since $(p_0 a t_1)^2 = 0$ and $(s_1 a p_0)^2 = 0$ (using $t_1 p_0 = 0$ and $p_0 s_1 = 0$);

so the diagonal matrices $\operatorname{diag}(1+p_0at_1,1)$ and $\operatorname{diag}(1+s_1ap_0,1)$ lie in Γ. Substituting $c=1+p_0at_1$ and $c=1+s_1ap_0$ into (32) and using $t_1p_0=0$, $p_0s_1=0$, $t_1s_1=1$, $p_0^2=p_0$:

$$z\ \operatorname{diag}(1+p_0at_1,\,1)\,z=\begin{pmatrix}1 & p_0a\\ 0 & 1\end{pmatrix}=x_{12}(p_0a), \tag{33}$$

$$z\ \operatorname{diag}(1+s_1ap_0,\,1)\,z=\begin{pmatrix}1 & 0\\ ap_0 & 1\end{pmatrix}=x_{21}(ap_0). \tag{34}$$

Next, the *corner swap*

$$d=s_1t_0+s_0t_1\in A^\times,\qquad d^2=s_1t_1+s_0t_0=1,$$

gives $D=\operatorname{diag}(d,1)\in\Gamma$ with $D^{-1}=D$. Conjugation by D multiplies an upper coefficient on the left by d and a lower coefficient on the right by d; since $d\,p_0s_0t_1=d\,s_0t_1=p_1$ and $s_1t_0p_0\,d=s_1t_0\,d=p_1$,

$$D\,x_{12}\big(p_0(s_0t_1a)\big)\,D=x_{12}(p_1a),\qquad D\,x_{21}\big((as_1t_0)p_0\big)\,D=x_{21}(ap_1),$$

with the inner matrices supplied by (33) and (34). Finally

$$x_{12}(a)=x_{12}(p_0a)\,x_{12}(p_1a),\qquad x_{21}(a)=x_{21}(ap_0)\,x_{21}(ap_1),$$

so $\langle\Gamma,z\rangle$ contains every elementary matrix; hence $\langle\Gamma,z\rangle=G$, and since $z=vu^{-1}$,

$$G=\langle\Gamma,u,v\rangle,\qquad u\Gamma u^{-1}=v\Gamma v^{-1}=K\le\Gamma.$$

The commuting corner. Put $J=\big\langle\operatorname{diag}(\rho(x_0),1),\ \operatorname{diag}(\rho(x_1),1)\big\rangle\le\Gamma$; membership is automatic here, since $\rho(g)\in A^\times$. By Lemma 5.11 the pair inherits the relators (24) and the noncommutation of x_0,x_1, so J is a two-generator non-LEF subgroup by Lemma 5.12. The subgroups K and J commute—indeed K commutes with the whole diagonal corner copy:

$$\begin{aligned}\kappa(a)\rho(g)&=(p_1+s_0at_0)(p_0+s_1U_gt_1)=s_1U_gt_1+s_0at_0\\&=(p_0+s_1U_gt_1)(p_1+s_0at_0)=\rho(g)\kappa(a),\end{aligned}$$

using $p_1s_1=s_1$, $t_1p_1=t_1$, $t_0p_0=t_0$, $p_0s_0=s_0$, $t_0s_1=0$, $t_1s_0=0$. And $K\cap J$ is trivial: if $\kappa(a)=\rho(g)$, then sandwiching between t_0 and s_0 gives $a=t_0\kappa(a)s_0=t_0\rho(g)s_0=1$, so $\rho(g)=\kappa(1)=1$ and $g=1$ by injectivity of ρ.

Conclusion. The data G,Γ, $Q=\{u,v\}$, $q_0=u$, K, J satisfy all hypotheses of Corollary 4.2: G and Γ are finitely generated Kazhdan groups, Γ is infinite, $G=\langle\Gamma,Q\rangle$, each $q\in Q$ conjugates Γ into itself with $q_0\Gamma q_0^{-1}=K$, and $J\le\Gamma$ is finitely generated with $[K,J]=1$ and $K\cap J=\{1\}$. If G were sofic, J would be LEF, contradicting Lemma 5.12. $\square$

8.2. K_1 **vanishes and** $\mathrm{GL}=\mathrm{EL}$**.** Every ring-theoretic fact this subsection needs is proved in Appendix A; for context we recall the theorem of Ara–Goodearl–Pardo [5], of which the facts below are the $L_k(1,2)$-instances: a purely infinite simple unital ring R is a GE-ring (every matrix in $\mathrm{GL}_r(R)$ is a product of elementary matrices and one $\operatorname{diag}(a,1,\dots,1)$), and

$R^\times/[R^\times, R^\times] \cong K_1(R)$. Neither statement is invoked in any proof of this paper.

Proposition 8.2. *Let k be any field and $L = L_k(1,2)$. Then:*
(i) $K_1(L) = 0$*;*
(ii) *the unit group $L^\times$ is perfect;*
(iii) $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ *for every $r \geq 2$.*

KOne/RefineLoopDischarge
`K1_trivial`
`stableUnits_eq_top_holds`
KOne/AllRanksElementary
`elementaryGroup_eq_top`
`glAll_eq_elementary`
`binaryLeavittUnits_perfect`
KOne/WhiteheadQuotient
`BinaryLeavittWhiteheadK1`
`binaryLeavittWhiteheadK1_subsingleton`
KOne/ClassicalKOne
`BinaryLeavittClassicalK1`
`binaryLeavittClassicalK1_subsingleton`
`binaryLeavittElementaryColim_eq_top`
(i) in both forms, each a constructed group proved trivial; (ii) and (iii) as stated.

Proof. Parts (i) and (iii) are proved, elementarily and in full, in Appendix A: Theorem A.2 establishes $K_1(L) = 0$ in Whitehead form—every unit u has $\mathrm{diag}(u, 1) \in \mathrm{EL}_2(L)$—and Corollary A.23 deduces $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for every $r \geq 2$, by rank-two elimination and flattening transport. The only input of the appendix beyond the Leavitt relations is strong division (Lemma A.3), which holds because L is purely infinite simple: the graph criteria of [1] are immediate for the rose R_2 with one vertex and two loops—the single vertex admits no nontrivial hereditary saturated subset, every cycle has an exit, and the vertex lies on a cycle.

Part (ii) is Corollary A.23 as well: $\Theta_{\mathcal{C}_3}$ identifies $L^\times$ with $\mathrm{GL}_3(L) = \mathrm{EL}_3(L)$, and every elementary generator is a commutator by the Steinberg relation, so $L^\times$ is its own commutator subgroup. (Classically, (ii) also follows from (i) and the Ara–Goodearl–Pardo identification of K_1 with the abelianized unit group.)

For an independent confirmation of (i), the K-theory of Leavitt path algebras [4] gives the exact sequence

$$K_1(k) \xrightarrow{1-N^\mathsf{T}} K_1(k) \longrightarrow K_1(L) \longrightarrow K_0(k) \xrightarrow{1-N^\mathsf{T}} K_0(k),$$

where N is the adjacency matrix—here the 1×1 matrix (2), so $1 - N^\mathsf{T} = -1$, an isomorphism on both $K_0(k) \cong \mathbb{Z}$ and $K_1(k) \cong k^\times$, whence $K_1(L) = 0$. $\square$

Remark 8.3. For an independent recent account of generation phenomena for unit groups and matrix groups over Leavitt algebras, consistent with Proposition 8.2 and with the self-similarity of Proposition 5.4, see [19].

Remark 8.4 (An elementary route to $K_1(L) = 0$)**.** The classical computation of $K_1(L)$ goes through the localization sequence of [4]. Appendix A replaces this input by a complete elementary proof—a Gaussian elimination over the binary tree, in the spirit of a Wiener–Hopf factorization whose "symbol" is a matrix pencil with two shift variables. In outline: a width reduction narrows every unit, modulo the Whitehead class group H of Definition A.1, to one whose value lies in the degree window $[-1, 1]$; at a sufficiently deep leaf set such a narrow unit becomes a matrix pencil with scalar coefficient matrices (A_0, A_1, C, B_0, B_1) over the five-dimensional space $kt_0 \oplus kt_1 \oplus k \oplus ks_0 \oplus ks_1$, indexed by a pair of complete leaf sets; and a single loop with two exits eliminates the pencil, splitting one column per pass (the index shift of the Wiener–Hopf analogy) until either a reshape by two tree tables lands the value in a one-sided degree window, or the stacked matrix $[B_0; B_1]$ acquires a scalar left inverse and the *inverse* unit lands there instead; one-sided windows

KOne/RefineLoopDischarge
`K1_trivial`
`narrowReduction_holds`
`scalarReduction_holds`

die by a block-move induction whose base is the keystone Lemma A.16. Every step is carried out in Appendix A, so the K_1-theoretic input of [4] is eliminated from the trust surface of Theorem B; the citation is retained as an independent confirmation.

KOne/FactorizationCertificate
`exists_elementaryCertificate`
`exists_moveList`
`Move.eval_mem_stableUnits`
No run-time bound is claimed or formalized.

Remark 8.5 (The Wiener–Hopf picture)**.** The proof in Appendix A moves every unit of $L = L_k(1,2)$ through exactly four kinds of steps: cylinder unipotents $1 + s_v x t_w$ (Lemma A.6), code-scalar units—matrices over k transported along a complete leaf set (Lemma A.9)—tree-table units, elements of Thompson's group V (Lemma A.8), and one window of nonpositive degree (Proposition A.17); the loop's termination is Theorem A.22. Unwinding the closure memberships makes the first of these explicit: for every unit u, the matrix $\mathrm{diag}(u,1)$ is an explicit finite product of elementary matrices of $M_2(L)$, and, transported along Θ at the two-atom leaf set, the corner insertion $s_0ut_0 + s_1t_1$ is an explicit finite word in cylinder unipotents alone. In the classical Wiener–Hopf picture the loop is index normalization: each refinement is the index shift that trades a constant diagonal for a shift atom, and the two exits are the two triangular factorizations. As of August 2026 we are not aware of a prior proof of $K_1(L_k(1,2)) = 0$ that is elementary in this sense—finite words and window inductions from the Leavitt relations, with strong division as the only imported input; the localization-sequence computation of [4] is not elementary.

Leavitt/RankTwoCompression
`rankTwo_not_isSofic`
KOne/AllRanksElementary
`binaryLeavittRankTwo_not_isSofic`
Leavitt/UnitsGLProfile
`binaryLeavittUnits_profile`
`binaryLeavittGL_profile`
Over $L_k(1,2)$ for every finite field.

8.3. **Finite fields: all ranks and the unit group.**

Endpoint/ManuscriptStatements
`theoremB_exact`
Endpoint/MainResults
`binaryLeavitt_finiteField_profile`
`binaryLeavittUnits_not_isSofic`
`binaryLeavittGL_not_isSofic`
`universalLeavittEL3_not_isSofic`

Theorem 8.6. *Let k be a finite field and $L = L_k(1,2)$. Then $\mathrm{EL}_2(L) = \mathrm{GL}_2(L)$ is infinite, finitely generated, has property (T), and is not sofic.*

Proof. The ring L is finitely generated, by s_0, s_1, t_0, t_1 and the finitely many scalars. By Theorem 5.7, Proposition 5.8, and Lemma 5.6, $\mathrm{EL}_4(L)$ is a finitely generated Kazhdan group; by Proposition 8.2(iii), $\mathrm{EL}_4(L) = \mathrm{GL}_4(L)$; and by Proposition 5.4 with the left comb $\mathcal{C}_4$, $\mathrm{GL}_4(L) \cong L^\times$. Hence $L^\times$ is a finitely generated Kazhdan group. Similarly, $\mathrm{GL}_2(L) = \mathrm{EL}_2(L)$ by Proposition 8.2(iii), and $\mathrm{GL}_2(L) \cong L^\times$ by Proposition 5.4 with $\mathcal{C}_2 = (s_1, s_0)$; so $\mathrm{EL}_2(L)$ is Kazhdan as well. Theorem 8.1 with $A = L$ now shows that $\mathrm{EL}_2(L)$ is not sofic. It is infinite, containing $\mathrm{diag}(\rho(V), 1) \cong V$, and finitely generated, being Kazhdan. □

Leavitt/PrefixCode
`prefixRingEquiv`
`prefixRingEquiv_apply`
`prefixUnitsEquiv`
KOne/AllRanksElementary
`elementaryGroup_eq_top`
`glAll_eq_elementary`
Internal throughout; no external K-theory.

Theorem 8.7 (Panorama)**.** *Let k be a finite field and $L = L_k(1,2)$. Then the groups*

$$L^\times = \mathrm{GL}_1(L), \qquad \mathrm{GL}_r(L)\ (r \geq 1), \qquad \mathrm{EL}_r(L)\ (r \geq 2)$$

are mutually isomorphic—explicitly, $\Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \to L^\times$ is an isomorphism for every ordered complete leaf set $\mathcal{C}$ of size r, and $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for $r \geq 2$—and each is an infinite, finitely generated, property-(T) group that is not sofic.

Proof. The isomorphisms are Proposition 5.4, and the equalities $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ are Proposition 8.2(iii). Each of the listed groups is therefore isomorphic to $\mathrm{EL}_2(L)$, which is infinite, finitely generated, Kazhdan, and nonsofic by Theorem 8.6; all four properties are isomorphism-invariant. □

Proof of Theorem B. Part (i) and part (ii) are exactly Theorem 8.7, with the explicit isomorphism (2) given by Proposition 5.4. □

Remark 8.8 (Two routes to rank four)**.** For a finite field of characteristic two and rank $m + 1 \geq 4$, the membership hypotheses are discharged twice over: explicitly, with no K-theory, in Section 7; and via Proposition 8.2 here. Theorem B in its full strength—all ranks down to two, all finite fields, and the unit group itself—uses the second route. The step that makes the panorama characteristic-free is Proposition 6.7: the witness is elementary because its memberships go through single-commutator certificates and the Whitehead lemma, identities valid over every ring, rather than through transvection factorizations of transpositions, which require characteristic two.

9. Finitely presented nonsofic covers

Nonsoficity of a finitely generated group is witnessed by a finite fragment of its multiplication table; imposing that fragment as a presentation produces finitely presented nonsofic covers. The covers are proved nonsofic directly, from the forbidden table itself; quotient closure of soficity—which in fact fails, by Remark 10.16—plays no role.

Definition 9.1. Let G be a group, $F \subseteq G$ finite with $1 \in F$, and $\varepsilon > 0$. An (F, ε)-*model* of G is a map $\varphi : F \cup F{\cdot}F \to \mathrm{Sym}(Y)$, for some nonempty finite set Y, such that *Covers/TableCover* `TableModel`

$$d_{\mathrm{H}}(\varphi(g)\varphi(h),\, \varphi(gh)) \leq \varepsilon \qquad (g, h \in F), \tag{35}$$

$$d_{\mathrm{H}}(\varphi(g),\, \varphi(h)) \geq 1 - \varepsilon \qquad (g, h \in F,\ g \neq h). \tag{36}$$

Lemma 9.2 (Finite-table characterization)**.** *A countable group G is sofic if and only if it admits an (F, ε)-model for every finite $F \ni 1$ and every $\varepsilon > 0$.* *Covers/TableCover* `tableModel_of_isSofic` `exists_table_obstruction`

Proof. If σ_n is a sofic approximation, then for fixed (F, ε) the restriction of σ_n to $F \cup F{\cdot}F$ is an (F, ε)-model for all large n: (35) is Definition 2.1(a), and (36) follows from (a) and (b) since $d_{\mathrm{H}}(\sigma_n(g), \sigma_n(h)) \geq d_{\mathrm{H}}(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(h)) - d_{\mathrm{H}}(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(g)) = d_{\mathrm{H}}(\sigma_n(gh^{-1}), \mathrm{id}) - o(1)$.

Conversely, exhaust G by finite symmetric sets $F_1 \subseteq F_2 \subseteq \cdots$ containing 1, and let φ_n be an $(F_n, 1/n)$-model on Y_n; replacing Y_n by many disjoint copies of itself we may assume $|Y_n| \to \infty$. Define $\sigma_n(g) = \varphi_n(g)$ for $g \in F_n$ and $\sigma_n(g) = \mathrm{id}$ otherwise. Fix $g, h \in G$; for n large, $g, h, gh \in F_n$, and (35) gives Definition 2.1(a). For faithfulness, first note that $p_n = \varphi_n(1)$ satisfies $d_{\mathrm{H}}(p_n^2, p_n) \leq 1/n$ by (35) with $g = h = 1$, and $d_{\mathrm{H}}(p_n^2, p_n) = d_{\mathrm{H}}(p_n, \mathrm{id})$ by right invariance of d_{H}; so $d_{\mathrm{H}}(\varphi_n(1), \mathrm{id}) \leq 1/n$. Then for $g \neq 1$ and n large,

$$d_{\mathrm{H}}(\sigma_n(g), \mathrm{id}) \geq d_{\mathrm{H}}(\varphi_n(g), \varphi_n(1)) - d_{\mathrm{H}}(\varphi_n(1), \mathrm{id}) \geq 1 - \tfrac{2}{n},$$

by (36). Hence σ_n is a sofic approximation. □

Sofic/SoficAmplification
`powerPerm`
`hammingDistance_powerPerm`
`IsSoficWeak`
`IsSoficWeakLocal`
`isSofic_of_isSoficWeak`
`isSofic_iff_weak`
`isSofic_iff_weakLocal`
Sofic/SoficUltraproduct
`hammingLength`
`IsNullSeq`
`nullSubgroup`
`UniversalSofic`
`isSofic_of_soficEmbedding`
`no_soficEmbedding_of_not_isSofic`
`exists_soficEmbedding_of_soficApproximation`
Endpoint/StructuralProfile
`universalLeavittEL4_no_soficEmbedding`
`exists_soficEmbedding_of_isSofic`

Remark 9.3 (What nonsoficity refutes)**.** Condition (36) ties the separation $1-\varepsilon$ to the multiplicative accuracy ε. A common variant fixes the separation at a constant $\delta \in (0,1)$—often $1/4$—and shrinks only the multiplicative error. Tensor powers identify the two. Let $\varphi^{\otimes k}$ act on Y^k coordinatewise. Two such powers disagree at a tuple exactly when their factors disagree at some coordinate, so the agreement set is a full product:

$$d_{\mathrm{H}}(\varphi(g)^{\otimes k}, \varphi(h)^{\otimes k}) = 1 - \big(1 - d_{\mathrm{H}}(\varphi(g), \varphi(h))\big)^k.$$

A power lifts a fixed separation δ to $1-(1-\delta)^k$, past any prescribed $1-\varepsilon$ once k is large, and inflates a multiplicative error d only to $1-(1-d)^k \le kd$, which asking for accuracy ε/k at the outset absorbs. The exponent depends on δ and ε alone. Letting δ vary with the test set removes the last side condition.

That freedom reaches the third standard formulation, the one the operator-algebraic literature uses. There a countable group is sofic when it embeds into a *universal sofic group*: the metric ultraproduct $\prod_{\mathcal{U}} \mathrm{Sym}(Y_n)$, the quotient of $\prod_n \mathrm{Sym}(Y_n)$ by the sequences whose normalized Hamming length vanishes along $\mathcal{U}$. Lifting a finite test set makes multiplicativity as accurate as one likes on a $\mathcal{U}$-large set; injectivity, however, separates two elements only by an amount depending on the pair, never by $1-\varepsilon$. That is the hypothesis amplified above, so an embedding into a universal sofic group implies Definition 2.1. The converse holds for countable groups: a sofic approximation, read along a nonprincipal ultrafilter on $\mathbb{N}$, *is* such an embedding, both of its estimates transferring because a nonprincipal ultrafilter contains every cofinite set. The two formulations agree, and the group of Theorem A admits no injective homomorphism into any metric ultraproduct of finite symmetric groups—over any index set, along any ultrafilter.

Covers/TableCover
`tableGroup_no_model`
`exists_finitelyPresented_obstruction`

Theorem 9.4 (Finite multiplication-table covers)**.** *Let G be a finitely generated group that is not sofic. Then there is an infinite finitely presented group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

Proof. Finite groups are sofic, so G is infinite. By Lemma 9.2 there are a finite set $F \subseteq G$ with $1 \in F$ and an $\varepsilon_0 > 0$ such that G admits no (F, ε_0)-model; enlarging F, we may assume F is symmetric and generates G. Define the finitely presented group

$$H = H_F = \big\langle\, x_g \ (g \in F \cup F\!\cdot\! F) \;\big|\; x_1 = 1,\ \ x_g x_h = x_{gh}\ (g, h \in F) \,\big\rangle. \tag{37}$$

The assignments $x_g \mapsto g$ satisfy the relations in G, and F generates G, so there is a surjection

$$\pi : H \twoheadrightarrow G, \qquad \pi(x_g) = g. \tag{38}$$

In particular H is infinite, and $\pi(x_g) = g \ne h = \pi(x_h)$ shows $x_g \ne x_h$ in H for distinct $g, h \in F \cup F\!\cdot\! F$.

Suppose H were sofic, with sofic approximation $\sigma_n : H \to \mathrm{Sym}(Y_n)$. Define $\varphi_n : F \cup F\cdot F \to \mathrm{Sym}(Y_n)$ by $\varphi_n(g) = \sigma_n(x_g)$. For $g, h \in F$, the relation $x_g x_h = x_{gh}$ holds in H, so

$$d_{\mathrm{H}}(\varphi_n(g)\varphi_n(h), \varphi_n(gh)) = d_{\mathrm{H}}(\sigma_n(x_g)\sigma_n(x_h), \sigma_n(x_g x_h)) \to 0.$$

For distinct $g, h \in F$, the element $x_g x_h^{-1} \neq 1$ of H gives, as in the proof of Lemma 9.2,

$$d_{\mathrm{H}}(\varphi_n(g), \varphi_n(h)) \geq d_{\mathrm{H}}(\sigma_n(x_g x_h^{-1}), \mathrm{id}) - o(1) \to 1.$$

Hence for n large, φ_n is an (F, ε_0)-model of G—note that (35) and (36) refer only to the multiplication table of F inside G, which is faithfully copied by the generators of H. This contradicts the choice of (F, ε_0). Therefore H is not sofic. ☐

Theorem 9.5 (Kazhdan covers)**.** *Let G be a nonsofic group with property (T). Then there is an infinite finitely presented property-(T) group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

Kazhdan/ShalomFinitePresentation
`exists_presented_kazhdan_cover`
`exists_finitelyPresented_kazhdan_cover`
Covers/KazhdanCover
`exists_kazhdan_finitelyPresented_cover_of_not_isSofic`
Shalom's theorem is proved internally, not cited.

Proof. Property-(T) groups are finitely generated, and G is infinite as before. By Shalom's theorem [30], G is a quotient of a finitely presented property-(T) group: fix a surjection $\theta : P \twoheadrightarrow G$ with P finitely presented and Kazhdan. Choose (F, ε_0) as in the proof of Theorem 9.4, and for each $g \in F \cup F\cdot F$ choose a lift $\tilde{g} \in P$ with $\theta(\tilde{g}) = g$, taking $\tilde{1} = 1$. Define

$$H = P \,/\, \langle\!\langle\, \tilde{g}\,\tilde{h}\,\widetilde{gh}^{-1} \;:\; g, h \in F \,\rangle\!\rangle. \tag{39}$$

Then H is finitely presented (finitely many new relators over the finitely presented P) and has property (T) (a quotient of P). Each relator lies in $\ker\theta$, since $\theta(\tilde{g}\tilde{h}\widetilde{gh}^{-1}) = gh(gh)^{-1} = 1$; hence θ descends to a surjection $H \twoheadrightarrow G$, and H is infinite. Writing $y_g \in H$ for the image of $\tilde{g}$, we have $y_1 = 1$, $y_g y_h = y_{gh}$ for $g, h \in F$, and $y_g \neq y_h$ for distinct $g, h \in F$ (their images in G differ). If H were sofic, then $\varphi_n(g) = \sigma_n(y_g)$ would be an (F, ε_0)-model of G for large n, exactly as in the proof of Theorem 9.4, a contradiction. ☐

Proof of Theorem C. The first two assertions are Theorems 9.4 and 9.5. For the final assertion, apply Theorem 9.5 to the infinite finitely generated property-(T) nonsofic group $L^\times$ of Theorem B, over any fixed finite field; the resulting H surjects onto $L^\times$, and composing with the inverses of the isomorphisms $\Theta_{\mathcal{C}}$ of Theorem B(ii) exhibits a quotient of H isomorphic to each group listed there. ☐

Remark 9.6 (Effectivity)**.** The covers are effective in the logical sense: nonsoficity of G is equivalent, by Lemma 9.2, to the existence of a finite witness (F, ε_0), and any witness yields H by (37) or (39). The proof establishes the existence of a witness without computing one—no algorithm or complexity bound is claimed—and no claim is made that $L^\times$ itself is finitely presented.

10. Concluding remarks

formalized in part

Leavitt/ElementaryNoFiniteQuotients
`elGen_mul`
`elGen_commutator`
`exists_ne_zero_mem_elementaryKernel`
`elementaryGroup_finite_quotient_trivial`
`exists_third_index`
Endpoint/StructuralProfile
`rankFour_exists_third_index`
`universalLeavittEL4_finite_quotient_trivial`
`universalLeavittEL4_no_finite_quotient`
`eq_top_of_permHom_trivial`
`universalLeavittEL4_no_properFiniteIndex`
Sofic/HyperlinearMetric
`permMatrix_row_sq`
`permMatrix_dist_sq_eq`
`permMatrix_normalized_dist_sq_eq`
Sofic/Hyperlinear
`hsDistSq`
`permMatrix_hsDistSq`
`hammingDistance_inv`
`HyperlinearModel`
`IsHyperlinear`
`permMatrix_mem_unitaryGroup`
`isHyperlinear_of_isSofic`
`isHyperlinear_of_finite`
`not_isSofic_of_not_isHyperlinear`

Remark 10.1 (The trust surface of Theorem A)**.** Theorem A is independent of $\mathrm{GL}_r = \mathrm{EL}_r$ and of $K_1(L) = 0$: the memberships of u and z are explicit factorizations, the remaining memberships follow from the Whitehead identity and the explicit commutator certificates of the witness pair, and the only analytic inequality proved in the paper is the co-area estimate of Lemma 2.7—the approximation-theoretic depth resides in the cited theorems of Kun and Kun–Thom. The argument concerns permutation models only; what it leaves open for unitary ones is the subject of Remark 10.2.

Remark 10.2 (The witness and hyperlinearity)**.** Soficity implies hyperlinearity [10], and whether the converse holds is Question 3.4 of [27]: *is every hyperlinear group sofic?* So long as no nonsofic group was known, the question had nothing to be tested against. Theorem A supplies one. If the witness is hyperlinear the two classes differ; if it is not, no approximation property of this kind survives it. Hyperlinearity should not be confused with the Connes embedding problem for tracial von Neumann algebras, which was resolved in the negative by the $\mathrm{MIP}^* = \mathrm{RE}$ theorem [18].

The witness is not a neutral test object. It contains the corner copy of V of Remark 6.8, hence the standard copy of F, and hyperlinearity passes to subgroups; so a proof that the witness is hyperlinear would in particular prove F hyperlinear, and even the soficity of F is a long-standing open question.

One direction is elementary and is recorded here because it isolates what the open direction is not about. Soficity implies hyperlinearity because the two metrics are the same metric: for permutations σ, τ of a finite set Y, the permutation matrices satisfy $\|P_\sigma - P_\tau\|_{\mathrm{HS}}^2 = 2|\{y \mid \sigma y \neq \tau y\}|$, so after normalizing by $|Y|$ the squared Hilbert–Schmidt distance is exactly $2d_{\mathrm{H}}(\sigma,\tau)$. Every estimate of Definition 2.1 transports through that identity with the error doubled and square-rooted, and $\sigma \mapsto P_\sigma$ is a genuine homomorphism, so the passage itself creates no defect. The content of Question 3.4 is entirely in the converse.

The heuristic offered in [27] for a positive answer runs through Malcev's theorem—a finitely generated subgroup of $U(n)$ is residually finite, hence sofic—and so covers the groups that are linear. The witness is as far from linear as a group can be. *Every homomorphism from* $\mathrm{EL}_r(L)$ *to a finite group is trivial*, whenever L is infinite, every nonzero element of L divides 1 on both sides, and $r \geq 3$. The proof is three commutators and a division. For $i \neq j$ the map $a \mapsto x_{ij}(a)$ carries addition to multiplication, so a finite target must kill some $a \neq 0$; the Steinberg identity of Lemma 5.5 moves that a into a second root subgroup as ab and into a third as cab; and strong division writes an arbitrary d as $(du)\,a\,v$, so the third root subgroup dies entirely. The generators die with it.

Formalized: the triviality of every finite quotient, for the general ring hypotheses and for the witness, and the absence of a proper subgroup of finite index. Not formalized: the profinite completion and the Malcev step, both classical.

Consequently the witness has no proper subgroup of finite index, no nontrivial action on a finite set, and trivial profinite completion; and by

Malcev's theorem again, no nontrivial finite-dimensional linear representation over any field, since the image would be a finitely generated linear group inheriting the absence of finite quotients. Nonsoficity here is therefore not the shadow of some coarser failure of residual structure—there is no residual structure at all. Whatever settles Question 3.4 for this group, a linear model will not be what does it.

Remark 10.3 (The separation constant is a convention on one side only)**.** Remark 9.3 made the sofic separation constant a convention by taking tensor powers. The unitary computation is the same computation and is if anything cleaner. Write τ for the trace normalized by $|Y|$. For unitaries A, B the squared normalized Hilbert–Schmidt distance is a single trace,

$$\|A - B\|_{\mathrm{HS}}^2 = 2 - 2\Re\tau(AB^*),$$

because each of $\tau(AA^*)$ and $\tau(BB^*)$ is 1; and the k-fold tensor power multiplies traces, $\tau\big(A^{\otimes k}(B^{\otimes k})^*\big) = \tau(AB^*)^k$, so

$$\|A^{\otimes k} - B^{\otimes k}\|_{\mathrm{HS}}^2 = 2 - 2\Re\big(\tau(AB^*)^k\big). \tag{40}$$

And yet the amplification fails. A scalar enters the tensor power as its k-th power, so the k-th power identifies any two unitaries differing by a k-th root of unity. Take $A = 1$ and $B = i \cdot 1$ on any nonempty model: both are unitary, $\tau(AB^*) = -i$, and (40) gives the maximal separation 2 at $k = 1$ and the value 0 at $k = 4$, where the two tensor powers are literally equal. Permutation models cannot exhibit this, because $P_\sigma P_\tau^{-1}$ is again a permutation matrix and its normalized trace is the proportion of fixed points—a real number in $[0, 1]$. In Hamming a small trace means few agreements; over the unitaries a trace can be small because the agreements *cancel*, and tensoring recombines the phases.

Sofic/HyperlinearAmplification
`tensorPow`
`tensorPow_mul`
`tensorPow_smul`
`trace_tensorPow`
Sofic/HyperlinearUltraproduct
`hsLengthSq_conjTranspose_mul`
`UniversalHyperlinear`
`hsLengthSq_ge_of_separated`
`exists_hyperlinearApproximation_of_isHyperlinear`
`exists_hyperlinearEmbedding_of_approximation`
Sofic/HyperlinearScalar
`hsNormSq`
`ofReal_sum_normSq`
`ofReal_hsNormSq`
`hsNormSq_sub_smul_one`

The repair is to tensor with the conjugate rather than with a further copy. The map $A \mapsto A \otimes \bar{A}$ is again multiplicative and unitary, and $\tau(A \otimes \bar{A}) = |\tau(A)|^2$ is nonnegative, so

$$\|(A \otimes \bar{A})^{\otimes k} - (B \otimes \bar{B})^{\otimes k}\|_{\mathrm{HS}}^2 = 2 - 2|\tau(AB^*)|^{2k},$$

which runs to the maximum 2 as soon as $|\tau(AB^*)| < 1$. That hypothesis is not removable, and the reason is an identity rather than an estimate: for a unitary U and any scalar c,

$$\|U - c\cdot 1\|_{\mathrm{HS}}^2 = 1 - 2\Re\big(\bar{c}\,\tau(U)\big) + |c|^2, \qquad \text{so} \qquad \|U - \tau(U)\cdot 1\|_{\mathrm{HS}}^2 = 1 - |\tau(U)|^2.$$

The trace defect *is* the squared distance to the scalars. Hence $|\tau(U)| \le 1$ always, with equality exactly when $U = \tau(U) \cdot 1$—the collapse again, now with a modulus of continuity: a unitary with $|\tau| \ge 1 - \delta$ sits within $\sqrt{\delta}$ of a scalar, so a phase can be propagated through products and constrained by separation at every power. The transport that propagation needs is elementary and is recorded alongside: the norm is unchanged by unitary multiplication on either side and by conjugate transposition, it scales by $|c|^2$, it obeys $\|A + B\|^2 \le 2\|A\|^2 + 2\|B\|^2$, and—Cauchy–Schwarz against the identity—$|\tau(C)|^2 \leq \|C\|^2$, so a bound on a matrix difference is a bound on the difference of traces. In model terms the bad case is explicit—AB^* is a scalar precisely when the model sends some element of $F \cdot F^{-1}$ to a scalar.

So the separation constant of Definition 2.1 is a convention outright, while its unitary counterpart is a convention relative to a scalar-freeness hypothesis on the model. Nothing here decides whether the unitary conventions agree; what it does is remove the standard proof from the table and name what a replacement must supply. The definition used in Remark 10.2 is the strong one, which is the one soficity implies, so the implication recorded there and the reading ¬hyperlinear ⇒ ¬sofic are unaffected.

Remark 10.4 (Why the unitary separation constant is harder than the permutation one)**.** The hypothesis isolated in Remark 10.3 is a property a model either has or lacks, so it names a class. Call a unitary model *non-scalar* with constant δ when $|\tau(u_g u_h^*)|^2 \leq 1-\delta$ for distinct g, h in the test set, and call a group non-scalar hyperlinear when every finite test set admits such models of every multiplicative accuracy, with δ depending on the test set alone. This asks more than Definition 2.1's unitary counterpart, which bounds only $\Re\tau$ and so tolerates $u_g u_h^* \approx \pm i$. Both inclusions hold:

$$\text{sofic} \implies \text{non-scalar hyperlinear} \implies \text{hyperlinear}.$$

The first holds because a permutation matrix cannot hide a phase: $\tau(P_\pi)$ is the proportion of fixed points of π, so transporting a sofic model gives $\tau(u_g u_h^*) = 1 - d_{\mathrm{H}}(\varphi h, \varphi g)$, a real number in $[0,1]$ that separation forces below ε. The second is the conjugate-double amplification of Remark 10.3: k tensored copies of $A \otimes \bar{A}$ carry the bound $1-\delta$ to separation $2-2(1-\delta)^k$, and inflate the multiplicative error only by $2k$, which asking for accuracy $\varepsilon/2(k+1)$ at the outset absorbs.

Which scalars a strong model can produce is settled by arithmetic on the unit circle. Suppose $u_g u_h^*$ is close to a scalar ζ, $|\zeta|=1$. Writing $w = gh^{-1}$, approximate multiplicativity carries the closeness to the powers, $u_{w^l} \approx \zeta^l$, and separation confines every ζ^l with $w^l \neq 1$ to the left half plane. No point of the unit circle survives that for four powers:

$$\max(\Re\zeta,\ \Re\zeta^2,\ \Re\zeta^3,\ \Re\zeta^4) \ \geq\ \tfrac{3}{10}.$$

Writing $x = \Re\zeta$, the four real parts are the Chebyshev polynomials x, $2x^2-1$, $4x^3-3x$, $8x^4-8x^2+1$, and the minimum of their maximum is $(\sqrt{5}-1)/4 \approx 0.30902$, attained where the second and third cross. That crossing is exact: with φ the golden ratio and $x = -\varphi/2$, the identity $4x^3 - 2x^2 - 3x + 1 = 0$ follows from $\varphi^2 = \varphi + 1$, and both polynomials take the value $(\sqrt{5}-1)/4$ there while the first and fourth take $-(1+\sqrt{5})/4$. So the constant $3/10$ is within three percent of optimal and cannot be replaced by $1/3$. Four powers are needed and no fewer, and the exceptions are exactly the roots of unity of order at most four: if $\zeta^3 = 1$ or $\zeta^4 = 1$ with $\zeta \neq 1$ then $\Re\zeta \leq 0$, so -1, $\pm i$ and the primitive cube roots keep all their nontrivial powers in the closed left half plane. The propagation itself is uniform, with an absolute constant. Write $\tau = \tau(U)$. The identity $U^{l+1} - \tau^{l+1} = U^l(U-\tau) + \tau(U^l - \tau^l)$, together with the fact that left multiplication by the unitary U^l and scaling by τ (of modulus at most one) are norm-nonincreasing, gives $a_{l+1} \leq 2a_1 + 2a_l$ for $a_l = \|U^l - \tau^l \cdot 1\|^2$, hence

$$\|U^4 - \tau^4 \cdot 1\|^2 \ \leq\ 22\,(1 - |\tau|^2).$$

Four steps is a fixed number, chosen before any accuracy is, so the factor 2 per step is harmless and no square roots enter. Feeding this into the circle bound—which survives a modulus merely close to one—yields: *if* $|\tau(U)|^2 \geq 1 - 10^{-6}$ *then* $\Re\tau(U^l) \geq 1/4$ *for some* $1 \leq l \leq 4$. In a model

formalized in part
Sofic/HyperlinearNonScalar
`permMatrixC_conjTranspose`
`normTrace_permMatrix`
`NonScalarModel`
`IsHyperlinearNonScalar`
Sofic/PhaseOrder
`re_pow_max_ge`
`re_pow_max_sharp`
`re_nonpos_of_pow_three_eq_one`
`re_nonpos_of_pow_four_eq_one`
Sofic/PhasePropagation
`normTrace_smul_one`
`normTrace_sub`
`hsNormSq_pow_succ_sub_le`
`hsNormSq_pow_four_sub_le`
Formalized: both inclusions, and the arithmetic on the unit circle. Not formalized: Radulescu's theorem, quoted from the literature, which is what makes the two classes coincide.

separated as in Definition 2.1's unitary counterpart every u_{w^l} with $w^l \neq 1$ has $\Re\tau \le \varepsilon/2$, so once $\varepsilon < 1/2$ no element with w, w^2, w^3, w^4 all nontrivial can be sent within 10^{-6} of a scalar. The last comparison, between u_{w^l} and u_w^l, is the same recursion with the model's own defect in place of the scalar one: $u_{w^{l+1}} - u_w^{l+1} = (u_{w^l w} - u_{w^l} u_w) + (u_{w^l} - u_w^l)u_w$, so the deficit is at most 14ε by the fourth step, and u_1 itself is within ε of the identity because $u_1 - u_1 u_1 = u_1(1 - u_1)$. Assembling: *if $\varepsilon \le 10^{-4}$ and w, w^2, w^3, w^4 all lie in the test set and are all nontrivial, then $|\tau(u_w)|^2 < 1 - 10^{-6}$.* A separated unitary model cannot put a phase on an element of order five or more, and by the root-of-unity computation above the scalars that remain available are exactly those of order at most four. The obstruction of Remark 10.3 is therefore confined to small torsion—which is precisely where the lamp groups of Theorem A's wreath-product relatives live.

The exception at order two is not available twice. Let g, h be nontrivial involutions with g, h, gh in the test set and $gh \neq 1$. If both were sent within 10^{-6} of a scalar then each scalar would be a near-square-root of one with nonpositive real part, hence near -1; their product would be near $+1$; and $\tau(u_{gh})$ is near that product, contradicting the separation of gh from the identity. So *a separated model cannot put a phase on two involutions at once*. In an elementary abelian 2-group—where every nontrivial element is an involution and the product of two distinct ones is again nontrivial—at most one element can carry a phase. The escape hatch that order two leaves open is therefore closed as soon as the group in question is a lamp group, which is the only case in which it was ever going to matter.

The two classes in fact coincide, but not for an elementary reason. By Radulescu's theorem—Theorem 8.5 of [27], from [28]—a countable group is hyperlinear exactly when its group von Neumann algebra embeds into R^ω, and restricting a trace-preserving embedding to the group unitaries gives $\tau(u_g) = \delta_{g,e}$ on the nose, hence non-scalar models with δ as close to 1 as one likes. What the two inclusions above record is the proof-theoretic asymmetry this hides. On the permutation side the corresponding normalization is Theorem 3.5 of [27], whose separation constant $1/4$ is made arbitrary by the amplification of Remark 9.3. Remark 3.7 of [27] asserts the same refinement for the unitary characterization of its Theorem 3.6, without proof—and Remark 10.3 shows that the amplification cannot supply it, since the tensor power identifies unitaries differing by a root of unity. The unitary refinement rests on the von Neumann algebra theorem; the elementary route available on the permutation side provably does not exist on the unitary side. The asymmetry is visible at the level of the ultraproducts themselves. The unitary metric ultraproduct $\prod_{\mathcal{U}} U(Y_i)$ is built exactly as its permutation counterpart is—null sequences are those whose squared Hilbert–Schmidt length vanishes along $\mathcal{U}$, and the three closure properties follow from $\|A+B\|^2 \le 2\|A\|^2 + 2\|B\|^2$ together with unitary invariance on both sides, a factor 2 per step being harmless for a limit. Conjugation invariance makes the null sequences normal, so the quotient is a group—the universal hyperlinear group over the family—and the metric it carries is the one the models are measured in, since $\|w^*v - 1\|^2 = \|v - w\|^2$ for unitary w, the exact counterpart of the identity $|q^{-1}p| = d_{\mathrm{H}}(p, q)$ that plays the same role for permutations. One direction of the bridge transfers intact, and it applies to hyperlinear groups rather than to a hypothesis: for a countable group, exhausting by finite test sets while shrinking the accuracy turns the local definition into a sequence of models, and such a sequence defines an injective homomorphism into the quotient: the multiplicative defect of a product is exactly the length of the corresponding sequence, by the identity above, and injectivity is the estimate that in a separated model every nontrivial test element has length at least $2 - \varepsilon - 1/50$. Neither step uses amplification, which is why both survive. What does not transfer is the converse: the permutation version turns an embedding back into a model of full separation by amplification, and that is the step Remark 10.3 closes here. What is elementary is exactly the two inclusions: soficity gives the trace normalization directly, because a permutation matrix cannot hide a phase, and the normalization amplifies back to full separation.

formalized in part
Sofic/DivisibleInvisible
`invPowSubgroup`
`pruferSubgroup_divisible`
`prufer_map_eq_zero`
`pruferSubgroup_nontrivial`
Sofic/MonomialModel
`monomialMatrix`
`monomialMatrix_one`
`hammingDistance_le_hsDistSq_monomial`
Formalized: the Prüfer centre, its divisibility and invisibility to finite groups, and the two monomial inequalities. Not formalized: Thom's construction and its properties, quoted from the literature.

Remark 10.5 (A candidate on the other side)**.** Theorem A supplies a nonsofic group to test Question 3.4 against. A candidate on the other side has been available since 2008 and deserves to be named here, because the distinction it turns on is the one of Remark 10.3. Let $K_0(\mathbb{Z}[1/p])$ be de Cornulier's group of block-unipotent matrices in $\mathrm{SL}_8(\mathbb{Z}[1/p])$, and $K = K_0(\mathbb{Z}[1/p])/\mathbb{Z}$ its quotient by the copy of $\mathbb{Z}$ inside its centre $\mathbb{Z}[1/p]$. Then K is finitely presented, Kazhdan and non-Hopfian, hence not residually finite [9]; and K is *hyperlinear* [31, Prop. 3.2], because $K_0(\mathbb{Z}[1/p])$ is residually finite and hyperlinearity passes to quotients by central subgroups. Its centre contains the Prüfer group $\mathbb{Z}[1/p]/\mathbb{Z}$, which is divisible, and a divisible group has no nontrivial homomorphism to a finite group: write $d = e^{|B|}$ and apply Lagrange. Note that $\mathbb{Z}[1/p]$ itself is *not* divisible—only p has been inverted—so the divisibility is a property of the quotient and is proved by two moves. Dividing by p stays inside $\mathbb{Z}[1/p]$ exactly, since $c/p^N = p \cdot (c/p^{N+1})$; dividing by an n prime to p is possible only modulo $\mathbb{Z}$, by Bézout: choose u, v with $un + vp^N = 1$, and then $c/p^N - n \cdot (uc/p^N) = vc \in \mathbb{Z}$. Writing $n = p^k m$ with $\gcd(m, p) = 1$ and iterating the first move k times reduces the general case to these two. The same argument applies to $\mathbb{Q}/\mathbb{Z}$, of which the Prüfer group is the p-primary part. So every finite quotient of K kills that centre, and by Malcev no finite-dimensional representation sees it either: the approximations are forced to be projective. Its soficity is open—in Thom's words, "we were not able to show that the group K is sofic". A proof that K is not sofic would answer Question 3.4 in the negative.

What blocks the passage is exactly the phase. Thom's microstates for K are not unitary in a shapeless way: they lie in the monomial group $T_n \rtimes S_n \subseteq U(n)$, permutation matrices carrying diagonal phases, assembled from the twisted group algebras $\mathbb{C}_{\beta\circ\alpha}K_1(\mathbb{Z}/q\mathbb{Z})$ attached to the central extension. Their traces are near zero for nontrivial elements because the phases *cancel*. Soficity would require the same smallness with the torus removed—and there, by the identity recorded in Remark 10.4, the normalized trace of a permutation matrix is its proportion of fixed points, a real number in $[0, 1]$ that can only be small by genuine scarcity of agreements. So the gap between what is proved about K and what Question 3.4 asks of it is precisely the gap between a trace that vanishes by cancellation and one that vanishes by disagreement, and Remark 10.3 shows that tensor amplification cannot close it.

Two inequalities measure that gap exactly. Write $A = A(d, \sigma)$ for the monomial matrix with permutation σ and unimodular phases d. First, the permutation part inherits the whole multiplicative defect: two monomial matrices whose permutations disagree at i differ in two entries of row i, each of modulus one, so

$$2\, d_{\mathrm{H}}(\sigma, \tau) \;\le\; \|A(d, \sigma) - A(e, \tau)\|_{\mathrm{HS}}^2.$$

The underlying permutations of a monomial model are therefore an approximately multiplicative map into $\mathrm{Sym}(Y)$, with no loss at all. Second, the trace is bounded by the fixed-point fraction of that permutation:

$$|\tau(A(d,\sigma))|^2 \ \le\ \big(1 - d_{\mathrm{H}}(\sigma,1)\big)^2.$$

So a small trace is *permitted* by many fixed points and never *caused* by them. Together: a monomial model is a sofic model exactly when its underlying permutations are already almost fixed-point free, and what a monomial model may do instead is let the phases cancel over the fixed points. The purest instance is a two-point model with the identity permutation and phases $1, -1$: the trace is 0 while nothing whatever is moved. A permutation model cannot imitate it, because there the normalized trace *is* the fixed-point fraction. For K the missing step is exactly the removal of that torus, and what removal requires can be said exactly: a monomial model whose data are multiplicative *combinatorially*—permutation and phase agreeing outside density ε—and separated by scarcity of points where permutation and phase both agree, *is* a permutation model, on m times as many points. Those two conditions are precisely the two quantities the untwisting identities below compute, so the criterion is not a reformulation but a statement of what a model must satisfy. Its converse holds trivially—a permutation model is monomial data with a single phase—so

$$\text{sofic} \quad \Longleftrightarrow \quad \text{combinatorial monomial approximability.}$$

Requiring the phases to be multiplicative *combinatorially*, agreeing as elements of $\mathbb{Z}/m$ outside a small set, therefore adds nothing at all to soficity. What the open question is about is the *metric* alternative, in which nearby phases count as equal; and the two prices of a scalar below say exactly how far apart those two demands are.

Removal is always available as a *construction*, and its price is exact. Phases valued in $\mathbb{Z}/m$ are absorbed by a second coordinate: $(y, j) \mapsto (\sigma y,\ j + d(y))$ is an honest permutation of $Y \times \mathbb{Z}/m$, the standard embedding of $(\mathbb{Z}/m)^Y \rtimes \mathrm{Sym}(Y)$ into $\mathrm{Sym}(Y \times \mathbb{Z}/m)$. Under it,

$$d_{\mathrm{H}}\big(\widetilde{(d,\sigma)}, \widetilde{(e,\tau)}\big) = \frac{|\{y \mid \sigma y \ne \tau y \text{ or } d(y) \ne e(y)\}|}{|Y|},$$
$$d_{\mathrm{H}}\big(\widetilde{(d,\sigma)}, 1\big) = 1 - \frac{|\{y \mid \sigma y = y \text{ and } d(y) = 0\}|}{|Y|}.$$

The second identity is what a sofic model needs, and it says the separation is scarcity of *trivially phased* fixed points: cancellation among the phases, which is what makes a monomial trace small while moving nothing, buys exactly nothing after untwisting. The first is the price: a phase difference invisible to the Hilbert–Schmidt metric—two nearby roots of unity—is fully visible in Hamming. So untwisting trades a metric defect for a combinatorial one, and the two identities say precisely what must be traded. A scalar factor is the extreme case: it leaves the permutation part untouched and changes every phase, so it costs the whole of Y—which is Remark 10.3 again, in the monomial category.

Sofic/ScalarCocycle

`pow_card_eq_one_of_scalarCommute`

`hsDistSq_monomial_const`

`hammingDistance_wreathPerm_blockConst`

Sofic/CharacterCount

`card_annihilator`

`hammingDistance_characterUntwist`

`hammingDistance_characterUntwist_generator`

Remark 10.6 (Exact infeasibility is a determinant, not a gap)**.** It is tempting to read emptiness of an *exact* finite-dimensional solution locus, at every dimension, as half of a non-closure profile, with only robustness left to establish. The scalar commutation relation shows that reading is unsound, and shows it with the cheapest witness available. For unitaries U, V on a model of size n, the relation $UV = c\,(VU)$ forces $c^n = 1$: take determinants, $\det U \det V = c^n \det V \det U$, and the determinants are nonzero. So for c not a root of unity the exact locus is empty at every dimension—one line, no rigidity. Yet the relation is exactly solvable whenever c is a q-th root of unity, on a model of size q: the shift $U_{ij} = [j = i+1]$ and the clock $V_{ij} = [j = i]\,\zeta^{\,i}$ on $\mathbb{Z}/q$ satisfy $UV = \zeta\,(VU)$ on the nose. Since the roots of unity are dense in the circle, every c is a limit of exactly solvable instances. Emptiness of every exact locus therefore carries no dimension-free gap for the relaxed loci.

Two features of the witness are worth recording. It is a *monomial* pair—a permutation carrying a diagonal of phases—so the entire phenomenon lives in the group of Remark 10.5; and what it evades is a *scalar* cocycle, which is exactly the configuration Remark 10.3 shows tensor amplification cannot remove and Remark 10.5 prices at the whole of the model when one tries to untwist it. The same object appears in all three places, which is some evidence that it is the right object.

The two prices of a scalar can be stated exactly, and their mismatch is the whole difficulty. Multiplying every phase of a monomial matrix by ζ moves it in Hilbert–Schmidt by precisely $|1-\zeta|^2$, which is arbitrarily small for ζ near one. Shifting every phase by a nonzero constant leaves the permutation part untouched and changes the second coordinate at every point, so the untwisted permutations of Remark 10.5 disagree *everywhere*: the Hamming distance is exactly 1. A scalar cocycle is therefore invisible to the metric that hyperlinearity uses and maximal in the metric that soficity uses. That is why a projective model cannot simply be untwisted, and it is the precise sense in which the two approximation categories part company.

A projective model assembled from a family of characters pays a graded version of the same price. If the phase difference is constant on each block of a partition, the untwisted permutations disagree exactly on the blocks where that constant is nonzero, so the multiplicative defect after untwisting is the *proportion of characters that see the cocycle*. The global scalar above is the one-block case, costing everything; a cocycle invisible to almost every character costs almost nothing. That is the quantity a construction of the kind described in Remark 10.5 has to control, and it is a finite count rather than an analytic estimate—indeed it is a greatest common divisor. The characters of $\mathbb{Z}/n$ are $\beta_l(x) = e^{2\pi i l x/n}$ for $l \in \mathbb{Z}/n$, and β_l kills α exactly when $l\alpha = 0$; the map $l \mapsto l\alpha$ has image of order $n/\gcd(n,\alpha)$, so by Lagrange its kernel has order $\gcd(n,\alpha)$ and

$$d_{\mathrm{H}}\big(\widetilde{(d,\sigma)}, (d+\widetilde{\beta_\bullet(\alpha)},\sigma)\big) \;=\; 1 - \frac{\gcd(n,\alpha)}{n}$$

on the model indexed by all n characters. The two ends of that formula are the two readings above. A cocycle datum invisible to almost every character has $\gcd(n, \alpha)$ close to n and costs almost nothing. A datum that *generates*—which is the case for Thom's microstates at level p^k, where the cocycle takes values generating the p^k-torsion of the Prüfer centre—has $\gcd = 1$: exactly one character out of p^k kills it, the defect is $1 - p^{-k}$, and untwisting destroys the model as k grows. So the obvious passage from Thom's projective microstates to a sofic approximation does not merely fail to be obvious. It fails, with a computed defect.

The extreme case is worth stating on its own, because it is the obstruction in a single line. On any nonempty model take the monomial matrices with the same permutation part σ and constant phases 1 and -1. They sit at the *maximal* Hilbert–Schmidt distance 4, while their permutation parts are equal, so their untwisted permutations coincide and no model derived from them separates the pair at all. Metric separation, which is what hyperlinearity supplies, therefore gives no combinatorial separation whatever, which is what soficity demands—and by the characterization above that passage is the whole of the difference between the two classes.

Sofic/NoRounding
`circle_map_eq_zero`
`additiveRounding_no_separation`
`roundMod_not_additive`
`no_pointwise_rounding`

Remark 10.7 (No pointwise rounding of phases)**.** The characterization of Remark 10.5 says that soficity is *combinatorial* monomial approximability, while hyperlinearity supplies only *metric* agreement of phases. The obvious way to close that gap is to round: replace each phase, which lives in the circle, by a nearby m-th root of unity, and hope that multiplicativity survives. It cannot, and the obstruction is a fork whose two horns exhaust the pointwise rules.

If the rounding is *multiplicative*, it is trivial. The circle is divisible and $\mathbb{Z}/m$ is finite, so every homomorphism between them is zero—the two lines of Remark 10.5, applied one level up, to the group the phases live in rather than to the group being modelled. A multiplicative rounding therefore sends every phase to 1; two monomial data differing only in their phases untwist to the *same* permutation, and the separation such a rounding provides is exactly 0, for every pair of phase systems and every permutation part.

If the rounding is *nearest-point*, it is not multiplicative, at every m and however fine. The witness is uniform in m: $3/10m$ rounds down to 0 while its double rounds up to $1/m$. Refining does not help, because the failure is scale-invariant—it is the statement that the sawtooth $x \mapsto \mathrm{round}(x) - x$ is not additive, and rounding to μ_m is that sawtooth read at scale $1/m$.

A pointwise rule either respects the group law or it does not, so the two horns are exhaustive. The passage from a metric phase system to a combinatorial one, if it exists at all, therefore cannot be pointwise: it has to use the group being modelled and the model itself. This is the shape of Remark 10.3 again—not a difficulty in carrying out a strategy, but a proof that the strategy does not exist—and it is why the gap priced in Remark 10.6 is not an artifact of a bad choice of m.

Remark 10.8 (A lamp chart is a coordinate chart)**.** A referee audit of this development observed that the action-level route to hyperlinearity of a generalized wreath product cannot help a candidate whose coordinate action is not already sofic, because the lamp data *contains* the coordinate data. The observation is written there against two papers' definitions, but its content depends on neither. Encode a coordinate x as the lamp $\delta_x(k_0)$ for a fixed $k_0 \neq 1$. That encoding is injective, and it is equivariant; consequently charts for the lamp action restrict along it to charts for the coordinate action, with the covariance identity transporting verbatim and nothing used about where the charts take their values.

Sofic/CoordinateTransfer
`OrbitChartData`
`trivialOrbitChartData`
`OrbitChartData.comap`
`lampEmbedding_injective`

Both hypotheses are isolated as such here: $x \mapsto \delta_x(k_0)$ is injective for every nonzero lamp value, and it intertwines the coordinate action with the site action. The transfer is then a pullback of chart data along an injective equivariant map, valid for charts into any class whatever. The reading is the audit's: if the coordinate action admits no chart system, neither does the lamp action. This closes a route to hyperlinearity; it says nothing about whether such a wreath product is hyperlinear by other means, and nothing here says it is.

Sofic/ImplementerCocycle
`implementerCocycle`
`implementerCocycle_twisted`
`correctedImplementer_defect`
`hsDistSq_orthogonal_rankOne`

Remark 10.9 (Two obstructions beyond elementwise approximation)**.** Two cautions, both due to an external referee audit of this development and formalized here because they are exactly the kind of statement that should not travel as prose. Neither bears on Theorem A; both bear on how a unitary model of a wreath-type candidate could be built.

Implementer coherence is a cocycle problem. Suppose an action is implemented elementwise: for each g a unitary v_g with $\mathrm{Ad}(v_g)$ the required automorphism. Doing this for every g separately is not the same as having a representation, and the failure is measured by $c(g,h) = v_g v_h v_{gh}^{-1}$, which satisfies the twisted identity

$$c(g,h)\, c(gh,k) = \big(v_g\, c(h,k)\, v_g^{-1}\big)\, c(g,hk),$$

while correcting the implementers by b_g replaces c by a coboundary: $g \mapsto b_g v_g$ is multiplicative at (g,h) exactly when $b_g\,(v_g b_h v_g^{-1})\, c(g,h)\, b_{gh}^{-1} = 1$. So "each automorphism is approximately inner" is incomplete data: a construction choosing implementers one element at a time still owes the trivialization of a class. The calibration is the Pauli pair, where $\mathrm{Ad}\,X$ and $\mathrm{Ad}\,Z$ commute on $M_2(\mathbb{C})$ because $XZ = -ZX$, both are inner, and no commuting implementers exist. Everything above is a group identity and is stated at that level.

Normalized rank blindness. Two *orthogonal* rank-one projections on a model of size n sit at normalized squared Hilbert–Schmidt distance $2/n$, which tends to zero although the projections are as different as two projections can be. Any argument that tracks a distinguished vector, transports a Kazhdan fixed vector, or preserves an orbit chart must therefore first bound the *relative trace* of the support it tracks; normalized-rank information alone will not do it.

Remark 10.10 (Two compressors, one image)**.** The pair $Q = \{u, v\}$ resolves a genuine tension: generation of G requires elements that move Γ, while compression requires elements that map Γ into itself. The present realization divides the two roles. The compressor u seats $K = u\Gamma u^{-1}$; the involution $z = vu^{-1}$ generates the missing last row and column of EL_n (Proposition 6.4); and the two compressors share the single image K, so the criterion sees one coherent compression while z carries the generation. The criterion itself is indifferent to the size of Q; the algebra here simply supplies two natural compressors rather than one.

Criterion/FiniteQuotientBlindness
`compressedImage_le`
`compressedImage_eq`
`compressorImage_normalizes`
`compressorImage_normalizes_inv`

Remark 10.11 (The median is canonical up to scale)**.** The normalization $f = s/(s+m)$ of Section 4 pins at $1/2$, and the value $1/2$ is not a trace or a spectral constant: it is the fixed point of the exchange of source and target sizes, and it is a vertex-weighted median by construction. Median pinning is exactly the statement that a permutation compressing one expander decomposition into another must asymptotically preserve the size profile—the conservative core of the argument, with exact conservation (Lemma 3.2) converting a one-sided drift estimate into a two-sided one at no cost.

Remark 10.12 (Why the criterion needs approximations at all)**.** Hypothesis (4) is a *strict* compression: $u\Gamma u^{-1}$ is a proper subgroup of Γ, and everything in Section 4 is arranged to exploit that strictness. No finite quotient records it. Let $\varphi : G \to Q$ be any homomorphism into a finite group. Then $\varphi(u)\varphi(\Gamma)\varphi(u)^{-1}$ is *conjugate* to $\varphi(\Gamma)$, hence of the same cardinality, while $u\Gamma u^{-1} \leq \Gamma$ makes it a subgroup of $\varphi(\Gamma)$; a subgroup of a finite group with the cardinality of the whole is the whole. So $\varphi(u)$ normalizes $\varphi(\Gamma)$, and since Γ and the compressors generate G, the image of Γ is *normal* in every finite quotient of G.

This is co-Hopfianity of finite groups, and it is the reason the criterion cannot be a residual argument. The compression data—the only thing distinguishing the hypothesis from normality—has been quotiented away before any finite construction begins. It becomes visible only in an approximation, where the maps are not homomorphisms, and property (T) enters exactly to make the approximate structure rigid enough to exploit: Kun's theorem supplies components, and the matching of Section 3 compares them. Remark 10.13 is the same statement one category up, with cardinality replaced by dimension.

Remark 10.13 (Exact compressions normalize for free)**.** Theorem 4.1 spends its length turning the one-sided hypothesis (4) into a two-sided comparison, and it is worth recording how little of that survives if the action is *exact*. Let $\varphi : G \to \mathrm{Sym}(Y)$ be a genuine homomorphism with Y finite, and write $\mathrm{Fix}(\Gamma)$ for the points fixed by every element of Γ. The points fixed by $t\Gamma t^{-1}$ are exactly $\varphi(t)\,\mathrm{Fix}(\Gamma)$, while $t\Gamma t^{-1} \leq \Gamma$ makes $\mathrm{Fix}(\Gamma)$ a subset of them; a finite set contained in its own bijective image equals it, so $\varphi(t)$ carries $\mathrm{Fix}(\Gamma)$ *onto* itself. As Γ and the compressors generate G, the fixed set is invariant under all of $\varphi(G)$—no expansion, no median, no property (T). The whole difficulty of Theorem 4.1 is thus that a sofic approximation is not exact. It is worth naming what the exact argument actually consumes, since it is a single property and it is sharp: a finite set contained in its own injective image equals it, and so does a finite-dimensional subspace. Both statements say that the ambient category is co-Hopfian, and that is the whole content of the collapse. The same argument therefore runs verbatim for genuine finite-dimensional unitary representations, with dimension in place of cardinality, and stops dead one step further out—a II_1 factor is not co-Hopfian, so an automorphism may carry a subalgebra strictly into itself, and the trace, being preserved, counts nothing. What the approximation-theoretic sections of this paper buy is precisely a substitute for co-Hopfianity where none is available.

Criterion/ExactCompression
`fixedSet`
`fixedSet_conj`
`fixedSet_image_eq`
`fixedSet_smul_eq_of_closure`
`fixedSubmodule`
`fixedSubmodule_map_eq`
`fixedSubmodule_compressed_eq`
Criterion/CommutantRigidity
`adjointRep`
`commutant_no_growth`

The commutant form is the same statement about a different representation, and it is the one an operator-algebraic reader wants. For a finite-dimensional π, the commutant of $\pi(\Gamma)$ in $\mathrm{End}\,V$ is the fixed space of the adjoint representation $g \mapsto (x \mapsto \pi(g)x\pi(g)^{-1})$; so the displayed collapse, applied to $\mathrm{Ad}\,\pi$ instead of π, says that an endomorphism commuting with

$\pi(t\Gamma t^{-1})$ already commutes with $\pi(\Gamma)$. *In finite dimension the relative commutant does not grow under compression.* Whether it grows in a II_1 factor is precisely what a co-Hopfian argument can no longer decide.

Criterion/CriterionAssembly
`isLEF_of_ambientDecomposition`
`not_isSofic_of_ambientDecomposition`
The ambient Kazhdan hypothesis is removed outright; the hypothesis on Γ remains and is used twice.

Remark 10.14 (What property (T) is used for)**.** Property (T) enters three times, in three different roles: (a) through Theorem 2.9 applied to Γ, producing the source decomposition that is compressed; (b) through Theorem 2.9 applied to G, producing the ambient decomposition carrying the co-area inequality—and by Remark 4.3 this use is replaceable by any componentwise ℓ^1 Poincaré decomposition; (c) through the Kazhdan factor hypothesis of Theorem 2.10. The local Theorem 4.1 quantifies over the first two uses and leaves only the third to the black box.

Use (b) is not merely replaceable in principle; it can be deleted. Taking the ambient decomposition as a hypothesis rather than deriving it, the criterion proves the same conclusion with *no Kazhdan assumption on G at all*: if every sofic approximation of G admits an ambient expander decomposition on S_G, and the witness J is not LEF, then G is not sofic. Only Γ need be Kazhdan, and its hypothesis is genuinely used twice—once for its own decomposition, once for the Kazhdan pair that Theorem 2.10 consumes. So the ambient hypothesis of Theorem D measures a property of the *approximations*, not of G; property (T) is one sufficient condition for it, by Theorem 2.9, and the criterion never asks for more. How much weaker the hypothesis is we do not determine: the per-approximation form is genuinely weaker, asking for a decomposition of the single approximation at hand, while the nonsoficity form quantifies over all approximations, and whether that property is strictly weaker than (T) is not settled here. What is settled is the direction of dependence, so any future sufficient condition for the decomposition can replace (T) without touching the proof.

Endpoint/QuotientNonclosure
`exists_sofic_group_with_nonsofic_quotient`
`exists_isLEF_group_with_non_isLEF_quotient`
`exists_residuallyFinite_group_with_non_residuallyFinite_quotient`
Sofic/FreeGroupResiduallyFinite
`freeGroup_residuallyFinite`
`isSofic_freeGroup`
Residual finiteness of free groups is proved from scratch.

Remark 10.15 (Scope)**.** The Leavitt-corner theorem (Theorem 6.10) applies to an arbitrary nontrivial unital ring carrying a binary Leavitt family, whenever the two adjacent elementary groups are Kazhdan and the two matrices u, z are elementary; for finite-type algebras over a finite field the Kazhdan hypothesis is automatic (Corollary 6.11), and over an arbitrary finitely generated ring with $m \geq 3$ it is likewise automatic granting the full theorem of [13]. Finite fields are where all hypotheses are verified simultaneously (Theorem 8.7); characteristic two is where they are verified explicitly (Theorem 7.3); and the construction can already be carried out in rank two (Theorem 8.1).

Remark 10.16 (Soficity is not closed under quotients)**.** Theorem A settles a companion closure question in the negative. Every group is a quotient of a free group, and free groups are residually finite—hence LEF, hence sofic; presenting the nonsofic group of Theorem A as such a quotient exhibits a residually finite group with a nonsofic quotient. Consequently none of the three classes—residually finite, LEF, sofic—is closed under quotients. This is also why the covers of Section 9 are proved nonsofic directly, from the forbidden table: a quotient-closure argument would have sufficed there, and none is valid.

Sofic/SoficTransfer
`isSofic_of_injective`
`SoficApproximation.restrict`
Endpoint/MainResults
`exists_finitelyPresented_nonsofic_group`
Sofic/SoficPositiveControl
`isSofic_of_finite`
Markov's two conditions are formalized; Adian–Rabin and Hall's embedding theorem are cited.

Remark 10.17 (Soficity is undecidable)**.** Soficity passes to subgroups, immediately from Definition 2.1: a model for the ambient group, read along an injective homomorphism, is a model for the source—multiplicativity transports along the homomorphism, separation along the injection. Every group into which a nonsofic group embeds is therefore nonsofic.

With Theorem C this makes soficity a *Markov property* of finitely presented groups. The trivial group is sofic; the finitely presented nonsofic group of Theorem C embeds in no sofic group at all, hence in no finitely presented one. By the Adian–Rabin theorem [2, 29] no algorithm decides, from a finite presentation, whether the group it defines is sofic. Markov's second condition asks for a *finitely presented* negative witness; that is why Theorem C, and not Theorem A alone, is what this rests on. Neither hyperlinearity nor weak soficity has such a witness, and both questions stay open.

Subgroup closure also carries the witness upward: every countable group embeds into a finitely generated simple group [16], so there is a finitely generated infinite simple group that is not sofic.

Remark 10.18 (Related work)**.** Three preprints from the first week of August 2026 bear on the material here.

Fournier-Facio [15] applies the criterion of Theorem D to construct finitely presented torsion-free nonsofic groups, and observes that the bare existence of a finitely presented nonsofic group follows from the existence of any nonsofic group, by openness of nonsoficity in the space of marked groups. Theorem C gives more than existence: an explicit finite obstruction, a surjection onto the original group, and the Kazhdan refinement.

Kun and Thom [23] carry the mechanism of Section 4 into ergodic theory and operator algebras. They call $\Gamma \leq G$ *infranormal* when $\{g \in G \mid g\Gamma g^{-1} \leq \Gamma\}$ generates G—the compression hypothesis (4) read as a condition on a subgroup—and prove that for an infranormal, non-normal Kazhdan pair the generalized wreath product $\left(\bigoplus_{G/\Gamma} \mathbb{Z}/2\mathbb{Z}\right) \rtimes G$ is not sofic; that in a sofic Kazhdan group the centralizer of an infranormal Kazhdan subgroup is normal; and that the Γ-fixed algebra of a sofic probability-measure-preserving action is G-invariant, so that generalized Bernoulli actions over G/Γ are not sofic—which answers a question of Păunescu in the negative and yields nonsofic ergodic equivalence relations generated by free actions of residually finite groups. They also exhibit explicit residually finite Kazhdan pairs $\mathrm{EL}_r(\mathbb{F}_q[x_1, \dots, x_d]) < \mathrm{EL}_r(\mathbb{F}_q[x_1^{\pm 1}, \dots, x_d^{\pm 1}]) \rtimes \mathrm{SL}_d(\mathbb{Z})$ to which all of this applies.

Alekseev and Thom [3], in work independent of this paper and completed before it, reach the same phenomenon from the other side. They show that the centralizer of a sofic embedding of a Kazhdan group into a metric ultraproduct of symmetric groups is again a metric ultraproduct of finite permutation groups, and deduce that when that centralizer acts ergodically on the Loeb space the group is LEF, hence residually finite if finitely presented. Where Theorem D uses compression to force a commuting subgroup to be LEF, they use ergodicity of the whole centralizer; both are readings of Kun's decomposition to the effect that a Kazhdan group cannot carry a sofic approximation whose centralizer is large. Theorem A is the extreme case, in which no sofic approximation exists at all.

Appendix A. An elementary proof of $K_1(L_k(1,2)) = 0$

This appendix proves the vanishing of K_1 announced in Remark 8.4, in the Whitehead form in which Section 8 uses it, by elementary means: no localization sequence, no spectra, and no input beyond the two Leavitt relations and one classical fact about L recorded as Lemma A.3 below. The proof is a Gaussian elimination over the binary tree, carried out by finitely many explicit kinds of moves; Remark 8.5 names them. The appendix is written to be read and checked entirely on paper.

Throughout, k is a field and $L = L_k(1,2)$. Recall from Section 5 the word notation s_w, s_w^* (we write $t_w = s_w^*$ for the word w, so $t_w s_{w'} = \delta_{w,w'}$ for $|w| = |w'|$), the cylinder idempotents $p_w = s_w t_w$, and complete leaf sets. A *complete code of depth* $\le r$ (resp. $\ge r$) is a complete leaf set all of whose words have length $\le r$ (resp. $\ge r$).

A.1. The Whitehead class group.

Leavitt/DiagonalClassGroup
`stableUnits`
`ScalarReduction`

Definition A.1. Let

$$H \;=\; \{u \in L^\times \,|\, \mathrm{diag}(u,1) \in \mathrm{EL}_2(L)\}\,.$$

Since $\mathrm{diag}(uv,1) = \mathrm{diag}(u,1)\,\mathrm{diag}(v,1)$, H is a subgroup of $L^\times$. The main theorem of this appendix is:

KOne/RefineLoopDischarge
`K1_trivial`
`narrowReduction_holds`
`scalarReduction_holds`
`stableUnits_eq_top_holds`
KOne/AllRanksElementary
`glAll_eq_elementary`

Theorem A.2. *$H = L^\times$. Equivalently, for every $u \in L^\times$ and every $r \ge 2$, the matrix* $\mathrm{diag}(u,1,\ldots,1)$ *is a product of elementary matrices in* $\mathrm{EL}_r(L)$.

(The two forms are equivalent because $\mathrm{EL}_2 \hookrightarrow \mathrm{EL}_r$ in the first two coordinates.) The single input beyond the Leavitt relations is the following division property, a standard consequence of pure infiniteness and simplicity of L—external to this appendix, which cites it from [1], and internal to the formal development, which proves it from the relations.

Leavitt/LeavittSimplicity
`exists_mul_mul_eq_one`

Lemma A.3 (Strong division)**.** *For every nonzero $x \in L$ there are $p, q \in L$ with $pxq = 1$.*

Leavitt/MatrixDiagonalization
`exists_elementary_mul_diag`
`binaryLeavitt_exists_elementary_mul_diag`

The first consequence is a Gaussian elimination for invertible two-by-two matrices; it is the rank-two case of the GE property of purely infinite simple rings [5] in elementary form, and rank two is all the sequel needs—every higher rank is reached by flattening (Corollary A.23).

Lemma A.4 (Elementary diagonalization)**.** *Let A be a nontrivial unital ring in which every nonzero element x admits p, q with $pxq = 1$, and let $X \in \mathrm{GL}_2(A)$. Then there are $E, F \in \mathrm{EL}_2(A)$ and $u \in A^\times$ with*

$$E\,X\,F = \mathrm{diag}(u, 1).$$

Proof. Write $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Assume first $a \neq 0$, and pick p, q with $paq = 1$. Three elementary operations plant a literal 1 in the corner: the column operation $C_2 \mathrel{+}= C_1 \cdot \rho$ with $\rho = q(1 - pb)$ replaces b by $b' = a\rho + b$, which satisfies $pb' = paq(1 - pb) + pb = 1$; the row operation $R_2 \mathrel{+}= wR_1$ with $w = (1 - d')p$, where $d' = c\rho + d$, replaces d' by $wb' + d' = (1 - d')pb' + d' = 1$; and with a 1 at position $(2, 2)$, the operations $R_1 \mathrel{+}= (-b')R_2$ and $C_1 \mathrel{+}= C_2 \cdot (-c'')$ (where c'' is the current $(2, 1)$ entry) clear the off-diagonal entries. The result is $\mathrm{diag}(v, 1)$ with $v \in A^\times$, since the product of invertible matrices is invertible and a diagonal matrix is invertible exactly when its diagonal entries are. If $a = 0$, then some entry of X is nonzero (as X is invertible and $A \neq 0$), and one elementary row or column addition moves a nonzero element into position $(1, 1)$; this reduces to the previous case. ☐

Lemma A.5. $\mathrm{EL}_2(L)$ *is normal in* $\mathrm{GL}_2(L)$*, and H is normal in $L^\times$.*

Leavitt/DiagonalClassGroup
`conj_mem_elementaryGroup_of_division`
`stableUnits_normal`

Proof. Conjugation by an elementary matrix preserves EL_2; conjugation by $\mathrm{diag}(u, v)$ sends $x_{12}(a)$ to $x_{12}(uav^{-1})$ and $x_{21}(a)$ to $x_{21}(vau^{-1})$, hence also preserves EL_2; and by Lemma A.4 every element of $\mathrm{GL}_2(L)$ is a product of elementary matrices and one matrix $\mathrm{diag}(u, 1)$. For the second claim, if $u \in H$ and $g \in L^\times$, then $\mathrm{diag}(gug^{-1}, 1) = \mathrm{diag}(g, 1)\,\mathrm{diag}(u, 1)\,\mathrm{diag}(g, 1)^{-1} \in \mathrm{EL}_2(L)$ by the first claim. ☐

By Lemma A.5 the quotient $L^\times/H$ is a group; we write $[u]$ for the class of u, so that Theorem A.2 says $[u] = 1$ for every unit. Three families of moves generate equalities in $L^\times/H$.

Lemma A.6 (Unipotent moves)**.** *Let $a, b \in L$ with $ba = 0$. Then $1 + ab \in H$. In particular, for prefix-incomparable words v, w and any $x \in L$, the* cylinder *unipotent $1 + s_v x t_w$ lies in H.*

KOne/StableUnitsGenerators
`mem_stableUnits_of_val_unipotent`
KOne/IncomparableUnipotents
`incomparableUnit_mem`

Proof. From $ba = 0$, $(ab)^2 = a(ba)b = 0$, so $1 + ab$ is a unit with inverse $1 - ab$, and the four-term identity

$$\mathrm{diag}(1 + ab,\ 1) = x_{12}(a)\,x_{21}(b)\,x_{12}(-a)\,x_{21}(-b)$$

holds by direct block computation: $x_{12}(a)x_{21}(b) = \begin{pmatrix} 1+ab & a \\ b & 1 \end{pmatrix}$, multiplying by $x_{12}(-a)$ clears the $(1, 2)$ entry using $-(1 + ab)a + a = -a(ba) = 0$, and $x_{21}(-b)$ clears the $(2, 1)$ entry. For the particular case take $a = s_v x$, $b = t_w$: then $ba = t_w s_v\, x = 0$ since v, w are incomparable. ☐

Lemma A.7 (The flip)**.** *Let $x, y \in L$ and suppose $1 + xy \in L^\times$. Then $1 + yx \in L^\times$, with inverse $1 - y(1 + xy)^{-1}x$, and*

$$[\,1 + xy\,] = [\,1 + yx\,] \qquad \text{in } L^\times/H.$$

Leavitt/WhiteheadFlip
`flipUnit`
`diagPair_flipUnit_inv_mem`
`mul_flipUnit_inv_mem_stableUnits`

Proof. The stated inverse is checked directly: $(1+yx)(1-y(1+xy)^{-1}x) = 1+yx-y(1+xy)(1+xy)^{-1}x = 1$, and likewise on the other side. The matrix $\mathrm{diag}(1+xy,\,(1+yx)^{-1})$ factors into four elementary matrices,

$$x_{12}(x)\,x_{21}(y)\,x_{12}(-(1+xy)^{-1}x)\,x_{21}(-(1+yx)y),$$

again by direct computation. Hence

$$\mathrm{diag}((1+xy)(1+yx)^{-1},\,1) = \mathrm{diag}(1+xy,\,(1+yx)^{-1})\cdot\mathrm{diag}((1+yx)^{-1},\,1+yx),$$

and the second factor is a Whitehead diagonal (Lemma 5.5); both factors lie in $\mathrm{EL}_2(L)$, so $(1+xy)(1+yx)^{-1} \in H$. □

Leavitt/LeavittDiagonalClass
`kappaUnit_mul_inv_mem_stableUnits`
`central_mem_stableUnits`
`stableUnits_eq_top`

Lemma A.8 (Corner insertion)**.** *For a binary word w and $u \in L^\times$, the corner insertion $\kappa_w(u) = s_w u t_w + (1-p_w)$ is a unit, and $[\kappa_w(u)] = [u]$. Consequently every central unit lies in H.*

Proof. The matrix $X_w = \begin{pmatrix} s_w & 1-p_w \\ 0 & t_w \end{pmatrix}$ is invertible, with inverse $\begin{pmatrix} t_w & 0 \\ 1-p_w & s_w \end{pmatrix}$ (using $t_w s_w = 1$ and $(1-p_w)s_w = 0$), and it intertwines:

$$X_w\,\mathrm{diag}(u,1)\,X_w^{-1} = \mathrm{diag}(\kappa_w(u),\,1).$$

Thus $\mathrm{diag}(\kappa_w(u)u^{-1}, 1) = [X_w, \mathrm{diag}(u,1)]$, a commutator of invertible matrices, one of which is elementary-diagonalizable; by Lemmas A.4 and 5.5 such a commutator lies in $\mathrm{EL}_2(L)$, so $\kappa_w(u)u^{-1} \in H$.

For a central unit c, centrality gives $s_0ct_0 + s_1ct_1 = c(s_0t_0+s_1t_1) = c$; on the other hand a direct computation using $t_1s_0 = 0$ shows

$$\kappa_0(c)\,\kappa_1(c) = s_0ct_0 + s_1ct_1.$$

Hence $[c] = [\kappa_0(c)][\kappa_1(c)] = [c]^2$ in $L^\times/H$, and $[c] = 1$. □

A.2. Code moves. Complete leaf sets play the role of block structures. Two further families of units lie in H.

KOne/CodeScalarMoves
`codeScalar_unit_mem`

Lemma A.9 (Code scalars)**.** *Let $\mathcal{D} = (w_1,\dots,w_\ell)$ be a complete leaf set and $G \in \mathrm{GL}_\ell(k)$ an invertible scalar matrix. Then*

$$\Sigma_{\mathcal{D}}(G) \;=\; \sum_{i,j} s_{w_i}\,G_{ij}\,t_{w_j}$$

is a unit of L lying in H.

Proof. The assignment $G \mapsto \Sigma_{\mathcal{D}}(G)$ is unital and multiplicative by the leaf identities (Lemma 5.3), so $\Sigma_{\mathcal{D}}(G)$ is a unit with inverse $\Sigma_{\mathcal{D}}(G^{-1})$. Over the field k, G is a product of transvections and one diagonal matrix (Gaussian elimination). A transvection $I + \lambda E_{ij}$, $i \neq j$, transports to $1 + \lambda\, s_{w_i}t_{w_j}$, a cylinder unipotent in H by Lemma A.6 (leaves of a complete leaf set are pairwise incomparable). A diagonal matrix $\mathrm{diag}(d_1,\dots,d_\ell)$ transports to $\prod_i \kappa_{w_i}(d_i)$ with central scalars $d_i \in k^\times$; each factor lies in H by Lemma A.8. □

KOne/CodeChangeUnits
`codeChange_mem_stableUnits`
KOne/CodeChangeGlue
`codeBijection_mem_stableUnits`

Lemma A.10 (Code changes)**.** *Let* $(\tau_1, \dots, \tau_\ell)$ *and* $(\sigma_1, \dots, \sigma_\ell)$ *be complete leaf sets of the same size. Then the* code change

$$\Omega \;=\; \sum_{i=1}^{\ell} s_{\tau_i} t_{\sigma_i}$$

is a unit of L *lying in* H*.*

Proof. Ω is a unit with inverse $\sum_i s_{\sigma_i} t_{\tau_i}$, by the leaf identities. Note that the Ω are precisely the images U_g of Thompson's group V under the embedding (23), together with their compositions with index bijections; the lemma is a generation theorem for the Higman–Thompson groupoid. Induct on ℓ. For $\ell = 1$ both sets are the empty word and $\Omega = 1$. For $\ell \geq 2$, the source set, being the leaf set of a finite full binary tree, contains a sibling pair $\sigma_a = w0$, $\sigma_b = w1$. Composing Ω with at most two *transpositions* aligns the pairing so that the targets of σ_a, σ_b are again a sibling pair $v0, v1$ in the same order; here a transposition is a code change that swaps two leaves x, y of one complete leaf set and fixes the rest, i.e. $T(x,y) = s_x t_y + s_y t_x + (1 - p_x - p_y)$, and $T(x,y) \in H$ because it is $\Sigma_{\mathcal{D}}(P)$ for the transposition matrix P on the complete leaf set $\mathcal{D}$ of which x and y are leaves—a code scalar (Lemma A.9). After alignment, the two sibling terms merge through the completeness relation, $s_{v0}t_{w0} + s_{v1}t_{w1} = s_v(s_0t_0 + s_1t_1)t_w = s_v t_w$, producing a code change between complete leaf sets of size $\ell - 1$; the inductive hypothesis and normality of H conclude. ☐

Corollary A.11 (Reshaping)**.** *Let* (R_i)*,* (C_j)*,* (P_i)*,* (Q_j) *be complete leaf sets, with* $|R| = |P|$ *and* $|C| = |Q|$*, and let* $E_{ij} \in L$*. If* *KOne/PencilReshape* `reshaped_pencil_mem_iff` `exists_reshaped_pencil` `pencilVal_window_mem`

$$u = \sum_{i,j} s_{R_i}\, E_{ij}\, t_{C_j} \quad \text{and} \quad v = \sum_{i,j} s_{P_i}\, E_{ij}\, t_{Q_j}$$

are both units, then $v = \Omega_1 u\, \Omega_2$ *for code changes* $\Omega_1 = \sum_i s_{P_i} t_{R_i}$ *and* $\Omega_2 = \sum_j s_{C_j} t_{Q_j}$*, and hence* $u \in H \iff v \in H$*.*

Proof. $\Omega_1 u \Omega_2 \;=\; \sum s_{P_i}(t_{R_i} s_{R_a}) E_{ab} (t_{C_b} s_{C_j}) t_{Q_j} \;=\; v$ by the leaf identities; apply Lemma A.10 and normality. ☐

A.3. The degree filtration and width reduction. The defining relations of L are homogeneous for $\deg s_i = +1$, $\deg t_i = -1$, so the free-algebra grading descends: $L = \bigoplus_{d \in \mathbb{Z}} L_d$ is $\mathbb{Z}$-graded, with L_d spanned by the monomials $s_\alpha t_\beta$, $|\alpha| - |\beta| = d$. Write

$$W[a,b] = \operatorname{span}_k \{ s_\alpha t_\beta \mid a \leq |\alpha| - |\beta| \leq b \},$$

and, for $x \in L$, write $x = \sum_d x^{(d)}$ for its graded components. Elements of $W[0,0]$ are called *balanced*; by the leaf identities, the balanced elements supported at depth $\leq n$ are exactly the matrices $\sum_{\gamma,\delta} G_{\gamma\delta}\, s_\gamma t_\delta$ over the full code $\{0,1\}^n$ with scalar G, i.e. the image of $M_{2^n}(k)$ under Σ-transport—so

Lemma A.9 says: *balanced units are exactly the transported invertible scalar matrices, and they lie in* H.[1]

Leavitt/LeavittWindowReduction
exists_window_reduction

Lemma A.12 (Width reduction). *Every* $u \in L^\times$ *satisfies* $[u] = [u']$ *for some unit* u' *with value in* $W[-1, 1]$ *(a* narrow *unit).*

Leavitt/LeavittGradingSpans
exists_corner_move
Leavitt/BinaryLeavittWindow
exists_narrow_representative

Proof. Every element of L lies in some window $W[-M, N]$. We show: if $N \geq 2$, then $[u] = [u'']$ for a unit u'' with value in $W[-M, \max(N - 1, 1)]$; the mirror statement for $M \geq 2$ follows by the same argument applied to the anti-automorphism θ of Lemma A.13 below, and iterating both lands in $W[-1, 1]$.

The engine is the *corner move*: for any $v, w \in L$,

$$\Big[\, u \,\Big] = \Big[\, s_0(u + vw)t_0 + s_0vt_1 + s_1wt_0 + s_1t_1 \,\Big],$$

obtained by transporting the 2×2 identity $\left(\begin{smallmatrix} 1 & v \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} u & 0 \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ w & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} u+vw & v \\ w & 1 \end{smallmatrix}\right)$ along $\Theta_{\mathcal{C}_2}$ of Proposition 5.4: the images of the two unipotent factors are the cylinder-type unipotents $1+s_0vt_1$ and $1+s_1wt_0$, which lie in H by Lemma A.6, and the image of $\operatorname{diag}(u, 1)$ is $\kappa_0(u)$, congruent to u by Lemma A.8.

Let x be the value of u, with top component $x^{(N)}$, $N \geq 2$. Since $y\, t_0 s_0 = y$ for every y, the top component factors as $x^{(N)} = b\, s_0$ with $b = x^{(N)}t_0 \in W[N - 1, N - 1]$. Apply the corner move with $v = -b$, $w = s_0$: the new value is

$$s_0(x - x^{(N)})t_0 \;-\; s_0bt_1 \;+\; s_1s_0t_0 \;+\; s_1t_1.$$

The first summand lies in $W[-M, N - 1]$ (conjugation by $s_0(\cdot)t_0$ preserves degree), the second has degree $N - 1$, and the last two have degrees 1 and 0. Hence the new unit has value in $W[-M, \max(N - 1, 1)]$, as claimed. ☐

A.4. Windows die. The next stage shows that units confined to a *one-sided* window already lie in H. This is where the analytic heart of the argument sits.

KOne/ThetaStable
thetaMatUnit_mem_elementaryGroup
thetaMatUnit_diagUnit

Lemma A.13 (Transpose). *The assignment* $s_i \mapsto t_i$, $t_i \mapsto s_i$ *extends to an anti-automorphism* θ *of* L *with* $\theta^2 = \mathrm{id}$*; it maps* $W[a, b]$ *onto* $W[-b, -a]$*, and* $u \in H \iff \theta(u) \in H$*.*

Proof. The relations are symmetric under the assignment composed with order reversal, so θ exists; it negates degree. Applying θ entrywise to the transpose of matrices sends $x_{ij}(a)$ to $x_{ji}(\theta(a))$ and $\operatorname{diag}(u, 1)$ to $\operatorname{diag}(\theta(u), 1)$, so it preserves EL_2 and hence membership in H. ☐

KOne/PureTailNilpotency
pure_tail_nilpotent

Lemma A.14 (Pure tails are nilpotent). *Let* $1 + \eta \in L^\times$ *with* $\eta \in W[1, 1]$*. Then* η *is nilpotent.*

[1] A balanced element that is a unit of L has invertible scalar matrix at its own depth—a singular matrix over a field is an annihilator of a nonzero matrix, hence a zero divisor—and therefore has balanced inverse.

Proof. Let $y = (1+\eta)^{-1}$, with graded components $y^{(d)}$ supported in a finite window. The equation $(1+\eta)y = 1$ reads degreewise: $y^{(d)} + \eta\, y^{(d-1)} = \delta_{d,0}$. From the bottom of the window upward, $y^{(d)} = 0$ for d below the window; then $y^{(d)} = -\eta y^{(d-1)}$ for $d < 0$ forces all negative components to vanish, $y^{(0)} = 1$, and $y^{(d)} = (-\eta)^d$ for $d > 0$. Since the window is finite, $(-\eta)^d = 0$ for large d. □

Lemma A.15 (Nilpotent tails die)**.** *Let $z \in L$ be balanced with $s_1 z$ nilpotent, and let $u \in L^\times$ have value $1 + s_1 z$. Then $u \in H$.*

KOne/NilpotentTailKill
`nilpotent_tail_mem_stableUnits`

Proof. Induct on the nilpotency index D of $s_1 z$; for $D = 1$, $z = t_1(s_1 z) = 0$ and $u = 1$. For $D \geq 2$, put $r = t_1(s_1 z)^{D-1}$, of pure degree $D-2$, so that $(s_1 z)^{D-1} = s_1 r$ and $r\,(s_1 z) = t_1(s_1 z)^D = 0$. Factor $r = R s_0^{D-2}$ with $R = r t_0^{D-2}$ balanced, take a balanced pseudo-inverse Ξ of R—balanced elements are scalar matrices along a full code, and every scalar matrix R admits Ξ with $R \Xi R = R$—and set

Leavitt/RankNormalForm
`exists_rank_normal_form`

$$e = t_0^{D-2}\, \Xi\, r,$$

a balanced element with the two properties that drive the induction: $e\,(s_1 z) = t_0^{D-2} \Xi\, (r(s_1 z)) = 0$, and $re = (r t_0^{D-2}) \Xi r = R \Xi R\, s_0^{D-2} = R s_0^{D-2} = r$, whence $(s_1 z)^{D-1} e = s_1(re) = s_1 r = (s_1 z)^{D-1}$. The *mover* $1 - s_1(ze) = 1 - (s_1 z)e$ lies in H by Lemma A.6 with $(a, b) = (s_1 z, -e)$, since $ba = -e(s_1 z) = 0$; and for the same reason

$$(1 - s_1(ze))\,(1 + s_1 z) = 1 + s_1(z - ze),$$

the cross term being $-s_1(ze)(s_1 z) = -s_1 z\,(e(s_1 z)) = 0$. The new tail satisfies $(s_1(z - ze))^{D-1} = (s_1 z)^{D-1}(1 - e) = 0$, using $(1-e)(s_1 z) = s_1 z$ to telescope and then the second property of e; so its index is at most $D - 1$, and the inductive hypothesis applies to u times the mover. □

Lemma A.16 (The keystone)**.** *Let $u = c + \zeta \in L^\times$ with c balanced and $\zeta \in W[1,1]$. Then $c \in L^\times$.*

KOne/ZeroKOne
`balanced_component_isUnit`

Proof. Suppose not. Balanced elements at depth n are scalar matrices $M_{2^n}(k)$ along the full code; by the rank normal form over the field k there are balanced units g, h with $gch = e := \sum_{\gamma \in S} p_\gamma$, a cylinder idempotent over a subset S of the depth-n code, and c is invertible exactly when S is everything. So assume S proper, and let $f = 1 - e = \sum_{\gamma \in T} p_\gamma$ with $T \neq \emptyset$. Replace u by $w = guh = e + \zeta'$ (with $\zeta' = g\zeta h \in W[1,1]$); invertibility of c is not affected. Let $y = w^{-1}$, with graded components $y^{(d)}$, $d \in [\mathrm{lo}, \mathrm{hi}]$, and let $M = -\mathrm{lo}$.

The equation $wy = 1$ reads degreewise $e\, y^{(d)} + \zeta' y^{(d-1)} = \delta_{d,0}$. Multiplying the $d = 0$ equation by f on both sides ($fe = 0$, $f^2 = f$) gives $f = f\zeta'\, y^{(-1)} f$, and substituting the relation $y^{(d)} = f y^{(d)} - \zeta' y^{(d-1)}$ (valid for $d \leq -1$, from $e y^{(d)} = -\zeta' y^{(d-1)}$) repeatedly yields, after M steps (the components below the window vanish),

$$f \;=\; \sum_{j=0}^{M-1} \underbrace{f\zeta'(-\zeta')^j}_{\deg=j+1} \cdot \underbrace{f\,y^{(-1-j)}f}_{\deg=-1-j}. \tag{41}$$

Now count ranks at a deep uniform interface. For ℓ large, every factor in (41) is represented by a scalar matrix between the full codes of appropriate depths: an element of $W[d,d]$ supported at bounded depth is, for all large q, a $2^{q+d} \times 2^q$ scalar matrix. Indeed, let $x \in W[d,d]$ be a combination of monomials $s_\alpha t_\beta$ with $|\alpha| - |\beta| = d$ and $|\alpha|, |\beta| \leq m$. For $q \geq m$, inserting the completeness relation on the right of each monomial gives

$$s_\alpha t_\beta = \sum_{|\nu|=q-|\beta|} s_\alpha s_\nu t_\nu t_\beta = \sum_{|\nu|=q-|\beta|} s_{\alpha\nu} t_{\beta\nu}, \qquad |\alpha\nu| = q+d, \quad |\beta\nu| = q,$$

so $x = \sum_{|\gamma|=q+d,\,|\delta|=q} X_{\gamma\delta}\, s_\gamma t_\delta$ for a scalar matrix X, unique because the $s_\gamma t_\delta$ over a pair of full codes are the transported matrix units and hence linearly independent. Since $t_\delta s_{\delta'} = \delta_{\delta\delta'}$ for words of equal length, presentations meeting at a common interface depth compose by matrix multiplication, which is what licenses the rank count. Representing the j-th summand at interface depths $\ell \to \ell - 1 - j \to \ell$, the middle factor passes through the matrix of f at depth $\ell - 1 - j$, whose rank is exactly $|T|\,2^{\ell-1-j-n}$. Hence the matrix of the right-hand side of (41) at depth ℓ has rank at most $\sum_{j<M} |T|\,2^{\ell-1-j-n} < |T|\,2^{\ell-n}$—a geometric series short of its limit—while the matrix of f at depth ℓ has rank exactly $|T|\,2^{\ell-n}$. This contradiction proves the lemma. □

KOne/WindowNonnegReduction
`window_nonneg_mem_stableUnits`
KOne/WindowNonposReduction
`window_nonpos_mem_stableUnits`
KOne/WidthTwoReduction
`window_zero_one_mem_stableUnits`

Proposition A.17 (Windows die)**.** *A unit of L whose value lies in $W[0,N]$ for some N, or in $W[-N,0]$, lies in H.*

Proof. By the transpose (Lemma A.13) it suffices to treat $W[0,N]$; induct on N.

Base, $N \leq 1$. Let $u = c + \zeta$, c balanced, $\zeta \in W[1,1]$. By the keystone (Lemma A.16) c is a balanced unit, so $c \in H$ with balanced inverse, and $u = c(1+\eta)$ with $\eta = c^{-1}\zeta \in W[1,1]$. By Lemma A.14 η is nilpotent. The corner insertion $\kappa_1(1+\eta) = 1 + s_1(\eta t_1)$ is congruent to $1+\eta$ (Lemma A.8), has balanced tail coefficient $z = \eta t_1$, and $s_1 z = s_1 \eta t_1$ inherits nilpotency from η (indeed $(s_1\eta t_1)^j = s_1 \eta^j t_1$); Lemma A.15 finishes.

Step, $N \geq 2$. Write the value as $1 + (a + \eta)$ with $a \in W[0, N-1]$—subtracting 1 costs nothing since 1 is balanced—and η the top component, of degree N. Split $\eta = s_0 q_0 + s_1 q_1$ with $q_z = t_z \eta$ of degree $N-1$. Perform the stable Whitehead block move along the four depth-two cylinders: with

$$X_z = s_{00}\, s_z\, t_{w_z}, \qquad Y_z = s_{w_z}\, q_z\, t_{00}, \qquad (w_0, w_1) = (01, 10),$$

set $m_2 = (1 - X_0)(1 - X_1)$ and $m_1 = (1 + Y_0)(1 + Y_1)$, products of cylinder unipotents in H (Lemma A.6), and let $\kappa = \kappa_{00}(u) = 1 + s_{00}(a+\eta)t_{00}$, congruent to u. In the product $m_2\,\kappa\,m_1$ all cross terms vanish by incomparability of the three cylinders $00, 01, 10$, except

$$X_0Y_0 + X_1Y_1 = s_{00}\,(s_0q_0 + s_1q_1)\,t_{00} = s_{00}\,\eta\,t_{00},$$

which cancels the top component inside κ. The value of $m_2\kappa m_1$ is therefore

$$1 + \big(s_{00}\,a\,t_{00}\ -\ X_0 - X_1\ +\ Y_0 + Y_1\big),$$

whose components have degrees in $[0, N-1]$: the a-part keeps its degrees, $\deg X_z = 1$, and $\deg Y_z = N-1$. By induction $m_2\kappa m_1 \in H$, hence $u \in H$. $\square$

A.5. The pencil and the elimination loop.

Lemma A.18 (Pencil form)**.** *Let $x \in W[-1,1]$. Then there are m and scalar matrices A_0, A_1, C, B_0, B_1, indexed by the full code $\mathcal{F} = \{0,1\}^{m+1}$, with* KOne/PencilForm exists_pencil_form KOne/RefineLoopDischarge pencilEntry_mem_window codePair_expansion

$$x = \sum_{i,j\in\mathcal{F}} s_i\,E_{ij}\,t_j, \qquad E_{ij} = A_{0,ij}\,t_0 + A_{1,ij}\,t_1 + C_{ij}\cdot 1 + B_{0,ij}\,s_0 + B_{1,ij}\,s_1.$$

Proof. Choose m with all monomials of x of depth $\le m$, and expand $x = \sum_{i,j} s_i(t_ixs_j)t_j$ along the full code of depth $m+1$ (completeness relation on both sides). For a monomial $s_\alpha t_\beta$ of x and code words i, j of length $m+1$: t_is_α vanishes unless α is a prefix of i, in which case it is $t_{i'}$ with $|i'| = m+1-|\alpha| \ge 1$, and symmetrically $t_\beta s_j = s_{j'}$ with $|j'| = m+1-|\beta| \ge 1$; the product $t_{i'}s_{j'}$ collapses, by the prefix calculus, to 0, a scalar, or a single letter s_z or t_z, according to the degree $|\alpha|-|\beta| \in \{-1,0,1\}$. Hence each corner entry t_ixs_j lies in the five-dimensional scalar pencil space displayed. $\square$

We call a unit in this form a *pencil unit* over the code pair (rows, columns); the data is the quintuple of scalar matrices, and the *stack* is the vertical block matrix $\binom{B_0}{B_1}$.

Lemma A.19 (Strict negativity)**.** *Let u be a pencil unit over complete leaf sets (R_i), (C_j), and suppose the stack admits a scalar left inverse: matrices G_0, G_1 with $G_0B_0 + G_1B_1 = I$. Then there is N such that every corner entry of the inverse,* KOne/StrictNegativePencil entry_window_negative_of_B_full

$$y_{ji} = t_{C_j}\,u^{-1}\,s_{R_i},$$

lies in $W[-N,-1]$.

Proof. All y_{ji} lie in a common finite window; let d^* be the largest degree in which any y_{ji} has a nonzero component, and suppose $d^* \ge 0$. From $u\,u^{-1} = 1$, sandwiching and inserting the completeness relation for (C_j),

$$\sum_j E_{ij}\,y_{ji'} = \delta_{ii'} \qquad (\text{all } i, i').$$

Take the component of degree $d^*+1 \ge 1$ on both sides. On the left only the degree-$+1$ part of E can contribute against $y^{(d^*)}$—components of y above d^* vanish—so

$$\sum_j \big(B_{0,ij}s_0 + B_{1,ij}s_1\big)\,y^{(d^*)}_{ji'} = 0.$$

Multiplying on the left by t_0 and by t_1 separates the two shifts:

$$\sum_j B_{0,ij} y^{(d^*)}_{ji'} = 0 \qquad \text{and} \qquad \sum_j B_{1,ij} y^{(d^*)}_{ji'} = 0;$$

that is, the stack annihilates the matrix $Y = (y^{(d^*)}_{ji'})$ on the left. Applying the scalar left inverse, $Y = (G_0B_0 + G_1B_1)Y = 0$, contradicting the choice of d^*. Hence all components have degree ≤ -1. □

KOne/RefineStep
`refine_column`
`pencilEntry_mul_s`
KOne/GLVectorNormalization
`exists_isUnit_matrix_col`

Lemma A.20 (Refinement step)**.** *Let u be a pencil unit over (R, C) whose column j_0 is* stack-free*: $B_{0,ij_0} = B_{1,ij_0} = 0$ for all i. Then u is also a pencil unit over the code pair (R, C'), where C' replaces the word C_{j_0} by its two children $C_{j_0}0$, $C_{j_0}1$, with data given columnwise by*

$$(A_0, A_1, C, B_0, B_1)_{\cdot\, j_0 z} = \big(0,\ 0,\ A_{z,\cdot j_0},\ [z{=}0]\, C_{\cdot j_0},\ [z{=}1]\, C_{\cdot j_0}\big),$$

all other columns unchanged.

Proof. Insert $1 = s_0t_0 + s_1t_1$ before $t_{C_{j_0}}$: each term $s_{R_i} E_{ij_0} t_{C_{j_0}}$ becomes $\sum_z s_{R_i} (E_{ij_0} s_z)\, t_{C_{j_0} z}$, and for a stack-free entry the split identity

$$(a_0t_0 + a_1t_1 + c)\, s_z = a_z + c\, s_z$$

shows $E_{ij_0} s_z$ is again a pencil entry, with the A-data falling to the constant slot and the constant to the B-slot—the index shift of the Wiener–Hopf analogy. □

KOne/CodeShapeSupply
`exists_shallow_code`
`exists_deep_code`

Lemma A.21 (Code supply)**.** *For $1 \leq \kappa \leq 2^r$ there is a complete leaf set of size κ and depth $\leq r$; for $\kappa \geq 2^M$ there is one of size κ and depth $\geq M$.*

Proof. Kraft counting. Shallow: start from the full code of depth $\lceil \log_2 \kappa \rceil \leq r$ and merge sibling pairs until κ leaves remain (each merge reduces the count by one and cannot increase depth). Deep: start from the full code of depth M, of size $2^M \leq \kappa$, and split leaves (each split increases the count by one and produces words of length $\geq M$). □

Figure 5 charts the elimination.

KOne/RefineLoopDischarge
`pencil_free_exit`
`pencil_full_exit`
`pencil_unit_mem_pow`

Theorem A.22 (The elimination loop)**.** *Every pencil unit over a row code of 2^r words lies in H.*

Proof. Let u be a pencil unit over (R, C), $|R| = 2^r$, $|C| = \kappa$. We induct on $2^{r+1} - \kappa$ (when this is ≤ 0, the first case below applies).

Free exit ($\kappa \geq 2^{r+1}$). By Lemma A.21 choose a shallow complete code P of size 2^r and depth $\leq r$, and a deep complete code Q of size κ and depth $\geq r + 1$. Reshape u to the pair (P, Q) (Corollary A.11). Each pencil entry lies in $W[-1, 1]$, so each monomial of the reshaped value has degree at most $|P_i| + 1 - |Q_j| \leq r + 1 - (r + 1) = 0$: the reshaped unit lies in $W[-N', 0]$ for some N', hence in H by Proposition A.17, hence $u \in H$.

Full stack. If the stack has a scalar left inverse, Lemma A.19 puts every corner entry y_{ji} of u^{-1} in $W[-N, -1]$. Expand $u^{-1} = \sum_{j,i} s_{C_j} y_{ji} t_{R_i}$ and reshape to a shallow column code of depth $\leq r+1$ (possible: $\kappa \leq 2^{r+1}$ in this

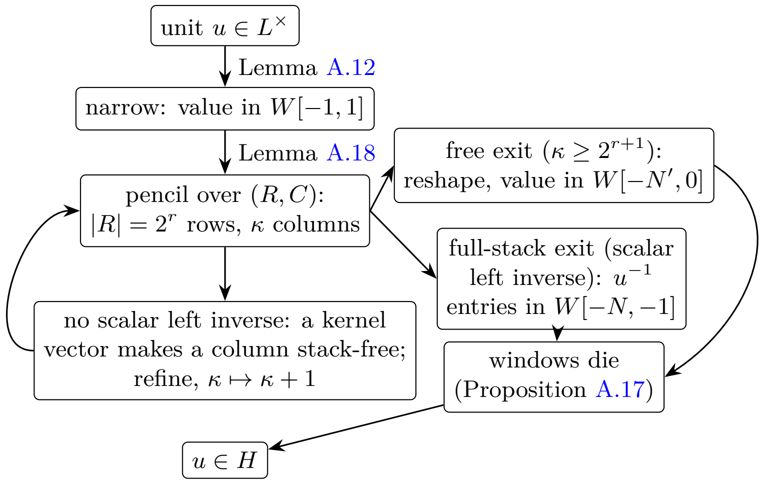


FIGURE 5. The elimination loop of Theorem A.22. Each pass either exits—shallow-versus-deep reshaping lands the unit, or its inverse, in a one-sided window—or splits one stack-free column, raising the column count; the counter $2^{r+1} - \kappa$ bounds the number of passes.

branch, or the free exit applies) and a deep row code of depth $\geq r$. Monomial degrees are at most $(r+1) + (-1) - r = 0$, so $u^{-1} \in H$ by Proposition A.17, and $u \in H$.

Refine. Otherwise the stack, having no scalar left inverse, has a nonzero kernel vector v (over the field k, a matrix with trivial kernel has a left inverse). Choose an invertible scalar matrix G whose j_0-th column is v (extend v to a basis). The code scalar $\Sigma_C(G)$ lies in H (Lemma A.9), and $u \cdot \Sigma_C(G)$ is a pencil unit over (R, C) with data multiplied columnwise by G; its column j_0 is stack-free because v kills both B_0 and B_1. Apply the refinement step (Lemma A.20): a pencil unit over (R, C') with $\kappa + 1$ columns. The counter $2^{r+1} - \kappa$ has dropped; the inductive hypothesis gives $u \cdot \Sigma_C(G) \in H$, hence $u \in H$. □

KOne/RefineLoopDischarge
`glTwo_eq_elementary_holds`
`glFour_eq_elementary_holds`
KOne/AllRanksElementary
`elementaryGroup_eq_top`
`glAll_eq_elementary`

Proof of Theorem A.2. Let $u \in L^\times$. By width reduction (Lemma A.12), $[u] = [u']$ with u' narrow; by the pencil form (Lemma A.18), u' is a pencil unit over the full code pair of depth $m+1$, a row code of 2^{m+1} words; by the elimination loop (Theorem A.22), $u' \in H$. Hence $u \in H$. □

`binaryLeavittUnits_perfect`
KOne/WhiteheadQuotient
`BinaryLeavittWhiteheadK1`
`binaryLeavittWhiteheadK1_subsingleton`

A.6. Consequences and effectivity.

Corollary A.23. *For every field k and $L = L_k(1,2)$: $K_1(L) = 0$; $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for every $r \geq 2$; and $L^\times$ is perfect.*

KOne/ClassicalKOne
`BinaryLeavittClassicalK1`
`binaryLeavittClassicalK1_subsingleton`

Proof. Rank two. For $X \in \mathrm{GL}_2(L)$, Lemma A.4 writes $EXF = \mathrm{diag}(u, 1)$, and Theorem A.2 places the diagonal factor in $\mathrm{EL}_2(L)$; hence $\mathrm{GL}_2(L) =$

Both forms, over every field, each a constructed group proved trivial.

$\mathrm{EL}_2(L)$. (This is the only use of Gaussian elimination; every higher rank now follows by transport.)

Every rank, by flattening. The coefficient isomorphism $\Theta_{\mathcal{C}}^{-1} : L \cong M_n(L)$ of Proposition 5.4 carries EL_p onto EL_p entrywise, and block flattening identifies $\mathrm{EL}_p(M_q(L)) = \mathrm{EL}_{pq}(L)$ inside $\mathrm{GL}_{pq}(L)$ for $p \geq 2$ (the two inclusions proved in Proposition 5.8); both maps are restrictions of ring isomorphisms of the ambient matrix rings, so each transports the *equality* $\mathrm{GL} = \mathrm{EL}$. Starting from rank two: $\mathrm{GL}_2(L) = \mathrm{EL}_2(L)$ transports through $L \cong M_n(L)$ and flattening to $\mathrm{GL}_{2n}(L) = \mathrm{EL}_{2n}(L)$ for every n; and for arbitrary $r \geq 2$, the same two maps with the roles of r and 2 exchanged transport $\mathrm{GL}_{2r} = \mathrm{EL}_{2r}$, already established, back down to $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$.

K_1 and perfectness. Since $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ at every rank, every class of the stable quotient defining $K_1(L)$ is trivial: $K_1(L) = 0$. For perfectness, $\Theta_{\mathcal{C}_3}$ restricts to a group isomorphism $\mathrm{GL}_3(L) \cong L^\times$, and $\mathrm{GL}_3(L) = \mathrm{EL}_3(L)$ is perfect: by the Steinberg relation, every generator $x_{ij}(a) = [\, x_{il}(a), x_{lj}(1)\,]$ with $l \notin \{i, j\}$ is a commutator. Hence $L^\times = [L^\times, L^\times]$. $\square$

The proof of Theorem A.2 is explicit throughout: the width reduction processes one graded component per corner move; the pencil form is a change of basis at depth $m+1$; each loop pass computes a kernel vector or a left inverse of the stack; and both exits are reshapes along explicitly constructed codes followed by the window inductions.

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